

Generic Protocols for Natural Farming

Natural Farming involves of set of holistic and comprehensive practices from tillage to selection of crops & varieties, raising of crops including management of on-farm inputs for realizing optimum productivity, economic advantage and improving soil health. The generic protocols for evaluation and demonstration of Natural Farming are described below.

1. Selection of Crops and Varieties: Natural farming promotes crop diversity having multiple crops grown simultaneously. Combination of cereals, legumes, fruits, vegetables along with trap/indicator crops such as marigold, okra, sunflower etc on bunds are preferred choice. Monocropping should be strictly avoided. Proven intercropping methods recommended for the region can be adopted. Generally indigenous/traditional varieties are preferred; however, improved varieties of crops can also be used under natural farming.

2. Tillage and sowing: Reduced / no tillage needs to be adopted in all the seasons. During summer, no tillage is preferred for raising green manures or multi-variate crops. Shallow tillage of 10-15 cm depth is preferred during *kharif* and *rabi* seasons. Deep tillage practices should be avoided. Sowing can be done manual or mechanical methods. Shallow seed drills for sowing of crops are preferred.

3. Cropping during summer/fallow period

3a. Pre-Monsoon Dry Seeding (PMDS) (rainfed areas): PMDS is a system of sowing, tilling and tending the land wherein the farmer grows crops in non-farming seasons or whenever there is no crop cover on the land. This can be practiced before the advent of monsoon, during summer, after *Kharif* and before beginning of *Rabi* season. The PMDS involves raising of 8 to 15 diverse crops, as a mixed crops, creates some special conditions for the seed germination through the beejamrit seed coating and plant survival during the dry seasons. The mulching material, which would be spread across the field, as a part of PMDS, acts as the catalyst to harness the water vapour from the atmosphere that drops to the land surface in the form of early morning dew. The mulching material facilitates the percolation of the dew into the soil and prevents its evaporation again. Therefore, PMDS needs to be practiced during March-May/June, followed by *Kharif* crops, Pre-Rabi Dry Sowing (PRDS) and *Rabi* crops. The crops grown in PMDS and PRDS are used, ultimately, as green manure, after obtaining intermittently some cash income and food items and green fodder to animals.

3b. Multi-variate cropping (irrigated areas): In general, the field is left fallow, or green manure crops such as *Sesbania aculeata*, *Sesbania rostrata*, sun hemp, etc., are grown during the summer period with conventional deep tillage operations. In natural farming, either green manuring crops such as *Sesbania aculeata*, *Sesbania rostrata*, sunhemp, moong etc. or multi-variate cropping during the fallow period in irrigated areas should be practiced under no-till/reduced till to keep the soil covered, enhance the soil microbial activities through root exudates, and incorporate the biomass having varied C: N ratio to ensure a longer period of decomposition thereby extending the period of nutrient release. Crops of different duration and nature (cereals, pulses, oilseeds, fibre, green manures) are to be grown as mixed under multi-variate cropping. Typically, a minimum of 09 crop seeds are to be mixed and broadcasted during the summer or fallow period. About 25% of the seed rate of the sole crop is to be adopted for each crop. Before the main crop is sown or planted, the biomass from multivariate cropping needs to be incorporated into the soil after harvesting of economic produce if any from the system. The multivariate cropping helps to reduce soil-borne pathogens through annihilation and allelopathy besides the other benefits.

4. Poly/Inter cropping: During the main seasons such as *kharif*, *rabi* and other seasons also poly or intercropping are advocated under natural farming. Proven intercropping practices can be adopted with recommended row ratios. Legumes are generally preferred as intercrops. Apart from this, trap crops such as marigold, okra, sorghum and indicator crops such as sunflower should also be grown in bunds and borders. Rice fields are not normally amenable to intercropping or mulching due to anaerobic conditions created through water inundation. In rice, under natural farming, practices such as green manuring of dhaincha, moong or any legume crop, growing trap crops and legumes on bunds, and also mulching with *Azolla* should be adopted. Intercropping is practiced by simultaneously growing *Azolla* with rice and continuously harvesting as and when required, providing an additional source of income to the farmer. *Sesbania aculeata* and *Sesbania rostrate* can be intercropped in rice and incorporated after 30 to 35 days using cono weeders. Direct seeding machines for simultaneous sowing rice and green manures are available. Use of green manure as intercrops in rice helps for reducing the weed infestation and contributes for nitrogen fixation. Addition of green biomass also helps for improving soil organic matter.

5. Concoctions (Beejamrit, Jeevamrit and Ghanjeevamrit) for active soil biology

Although many local and on-farm-based natural formulations or concoctions (mixtures) are used in chemical free farming approaches such as organic farming, *Panchagavya* farming,

biodynamic farming, nateco farming, etc., the natural farming relies on on-farm prepared formulations or concoctions among which the important ones used are seed invigoration through *Beejamrit*, re-invigorating the soil through *Jeevamrit* and *Ghanjeevamrit*. The preparation technique and detailed guidelines for the use of these formulations are described by Hari Om *et al.* (2024).

5a. *Beejamrit* (*Beejamrutha* or *Bheejamrutham*) is a microbial paste made using 5 kg of fresh dung from desi cows, 5 liters of *desi* cow urine, 50-100 g of virgin soil (preferably from forests, banyan trees), 50 g of calcium chloride, and 20 liters of water. The mixture is thoroughly stirred and fermented for 24 hours. It is used for seed dressing or dipping seedling roots. Seed dressing involves coating seeds with *Beejamrit* before sowing and drying them in the shade. *Beejamrit* has been shown to protect seeds and young seedling roots from soil-borne and seed-borne pathogens, reduce germination time, and enhance seedling vigor, leading to better growth. The application of beneficial microbes to seeds effectively places microbial inocula in the soil, positioning them to colonize seedling roots and protect them against soil-borne diseases and pests. Microbes improve seed germination, shoot and root length, seedling vigor, and crop performance. *Beejamrit* promotes root dry weight and root proliferation, enhancing nitrogen, nutrient, and water uptake by crop plants.

5b. *Ghanjeevamrit*: Ghanjeevamrit is solid in form and is prepared using ingredients such as desi cow dung (150 kg), jaggery (1 kg), pulse flour (2 kg), virgin fertile soil for microbial culture (50-100g), native/desi cow's urine as required.

Method 1: Mix well cow dung, jaggery, pulse flour and soil. Add a little bit of cow urine to it as needed. Dry the mixture in shade for 2-4 days. Use in the field after drying and grinding well into small aggregates. It can be stored and used for 6 months.

Method 2: Take 1-1.5 quintal well rotten desi cow dung manure and sprinkle 8-10 liters of Jeevamrit on it. Mix well and use by broadcasting uniformly in the field. After preparing, keep it covered in the shade. It can be used for 6 months.

Method 3: Take 1.5 quintals of dry biogas slurry, a kilogram of jaggery, 2 kilograms of pulse flour and a handful of virgin soil. Mix these well and broadcast uniformly in the field. Ghanjeevamrit can also be prepared by sprinkling Jeevamrit solution on this slurry as per requirement. If Ghanjeevamrit is prepared by adding Jeevamrit solution, then there is no need to add jaggery, pulse flour and soil separately.

Method 4 (Best quality method): Make a pile of fresh cow dung partially dried up to 3-4 days up to a height of 1.5 feet. Size of the pile might be as per convenience (say 12x3 feet). Make holes with a wooden log at a distance of about 1 feet and fill up these holes with already prepared Jeevamrit. Turn the dung pile up and down after one week. Then again make the holes and fill up these holes with Jeevamrit and turn up the pile again after one week. This way this process is repeated 2-3 times. The Ghan Jeevamrit will be ready in 30-35 days. Then after pounding it in small sized particles, it can be broadcasted uniformly in the field or stored in shade after covering with gunny bags.

Generally, it is recommended to use 800 to 1000 kg per acre of land as basal dose. However, this can be applied as top dressing also as per crop duration.

5c. Jeevamrit (Jivamrita or Jeevamrita or Jeevamrutha or Jeevamrutham) is a fermented microbial culture made from on-farm ingredients: 10 kg of fresh dung from *desi* cows, 10 liters of desi cow urine, 1.5 kg of cane jaggery (sugar) or sweet fruit pulp, and 1.5 kg of pulse flour mixed in a barrel with 180 liters of water. A handful (50-100 g) of virgin soil from banyan tree areas, forests etc. is added as a local microflora inoculant. The mixture is stirred daily and fermented for 48 hours in the shade. During fermentation, aerobic and anaerobic bacteria in cow dung and urine multiply as they decompose the organic ingredients. Cane jaggery enhances fermentation, and pulse flour provides protein energy. *Jeevamrit* is ready after 3-4 days in summer and 6-7 days in winter and is applied to soil twice a month at a rate of 200 to 600 liters per acre depending upon crops and soil conditions, either through irrigation water or as a foliar spray. Application of Jeevamrit or Ghanjeevamrit through drip or sprinkler irrigation is preferred. *Jeevamrit* acts as a catalyst, promoting soil microorganism activity, including earthworms, and making nutrients available to crops. Indigenous cow dung and urine produce the best *Jeevamrit* due to higher beneficial microbial populations.

6. Acchadana (Mulching of soil surface)

Natural farming promotes many types of mulching including (i) Soil mulching (tillage of the soil as normal, but to a reduced depth of 10-15 cm), (ii) Straw mulching (spread dried crop residues or straw from the previous crop), and (iii) Live mulching (raising diverse legume intercrops with cereals for supplying phosphorus, potassium, and sulphur to monocotyledons, such as rice, wheat, ragi, etc.). Crop residue mulching should be applied immediately after sowing of the crops to reduce germination of weeds in the field. Depending upon the crop residues, generally 5 to 7 tonnes are required to cover one ha area for all the crops except rice

in which Azolla should be used as mulching material under natural farming. Mulching should be done in all the seasons. Some of the leguminous cover crops which can be introduced in natural farming are spreading type cowpea, horse gram and calapogonium. Application of *jeevamrut* through drip/micro irrigation has accelerating effect on multi/relay crops and polythene mulching may also be recommended depending upon the soil condition and requirement.

7. Whapasa

Whapasa refers to the soil condition where a balanced concentration of water and air molecules exists in the crop root zone. Natural farming promotes this state of soil condition with the application of minimal amount of irrigation, reduced tillage, and enhanced soil humus due to residue mulching and addition of concoctions. By maintaining *Whapasa*, water use by crops can be reduced significantly. *Whapasa* soil condition in the crop root zone under field conditions is feasible through application of *Jeevamrit* using precision technologies such as micro irrigation systems such as drip and sprinkler irrigation including sensors, controllers, etc., to deliver a measured quantity of water with *Jeevamrit* precisely in the vicinity of roots to achieve optimal soil-water-plant relations. Alternate furrow irrigation and surge irrigation can also be adopted. Flood irrigation should be avoided.

8. Botanical concoctions for insect Management

As such, natural farming promotes the use of various *kashaya* (decoctions) from locally sourced ingredients like cow urine, neem leaves, green chilies, garlic, tobacco, sour buttermilk, etc., to act as natural fungicides and pesticides. Some of the prominent ones used are *Agniastra*, *Bramhastra*, *Neemastra* and *Dashparni ark*. Prophylactic spray of these botanical concoctions in every 15 days or in a month depending upon the crops are recommended under natural farming.

8a. *Agniastra* is made by mixing 20 liters of cow urine with crushed tobacco powder (500 g), green chili (0.5 kg), garlic (0.25 kg), and neem leaves (5 kg). In a pot, add 20 liters of local cow urine, stir well, boil, cool, and ferment for 48 hours. After fermentation, filter the solution with cloth. *Agniastra* spray effectively controls pests like leaf roller, stem borer, fruit borer, and pod borer. Take 6-8 liters of filtered solution and dissolve it in 200 liters of water and spray it on one acre crop. It should be used within 3 months.

8b. *Brahmastra* uses 10 liters of desi cow urine and a paste of neem, Pongamia (karanj), custard apple, castor, and datura leaves etc. (2 kg each). The manually crushed paste is boiled until

frothing appears, filtered, and fermented for 48 hours in shade, stirred twice daily. *Brahmastra* is sprayed at 6–8 liters per 200 liters of water for one acre and effectively controls sucking pests, fruit borer, and pod borer. After preparation, it should be used within one month.

8c. *Neemastra* involves mixing 5 liters of desi cow urine, 1 kg of fresh cow dung, and 5 kg of neem leaf paste in 100 liters of water. Ferment the mixture for 48-96 hours, stirring twice daily, and filter with cloth. *Neemastra* is effective against sucking pests and leaf-eating caterpillars. Prepared material is sufficient for one acre.

8d. *Dashparni ark*: Dashparni ark preparation involves ingredients of water (200 liters), urine of native cow (20 liters), dung of native cow (2 kg), leaves of ten plants (Neem, Karanj, Castor, Sitaphal, Bael, Marigold, Tulsi, Dhatura, Mango, Aak etc.) 2.0 kg each, turmeric, asafoetida powder 500 g, ginger chutney 500g (heeng) powder 10 gm, dry ginger powder 200 gm, tobacco powder 1.0 kg, hot green chilli chutney 1.0 kg and desi garlic chutney 1.0 kg. Dissolve cow dung and cow urine well in water and keep it for 2 hours. Mix turmeric powder, ginger chutney and asafoetida powder well and keep it in the shade for 24 hours. Stir this mixture, mix dry ginger powder, tobacco powder, hot green chillies and deshi garlic chutney well and keep it for 24 hours. Crush the leaves of plants and press them in this mixture. Cover the mixture with a gunny bag and keep it for 30-40 days and stir it for 2-3 minutes in the morning and evening. Mix 6-8 liters of extract of this solution in 200 liters of water and spray it on one acre crop. Dashparni ark is used for control of borers, maggots/ caterpillars.

9. Botanical concoctions for disease management

9a. *Sonthastra*: Ingredients for using in one acre includes dry ginger 200 gm, desi cow's milk 5 liters and water 200 liters. Grind dry ginger and make powder. Mix it in 2 liters of water and boil it. Cool it when one liter is left. Boil 5 liters of indigenous cow's milk in a pot once. Cool the milk. Remove the cream of milk and use it domestically. Sieve the ginger water and boiled milk with the help of a cloth. After mixing in 200 liters of water, spray it on one acre crop. It is used for the control of fungus.

9b. *Khatti lassi (Sour butter milk)*: Ingredients for one acre include water 100 liters, khatti lassi (3-5 days old) 5 liters, a small piece of copper, turmeric powder 50 gm. Keep the small piece of copper dipped for 3-5 days in khatti lassi. Mix water and khatti lassi well and spray it on the crop after dissolving 50g turmeric powder in it. This is used to control fungal diseases. Also found to be virus killer.

9c. Use of cow urine: Apply 30-40 liters of cow urine along with pre-sowing irrigation for controlling soil and seed borne diseases. It also controls seedling and wilt diseases in plants. The 10-20% spray of cow urine helps in controlling canopy diseases of crops. While planting fruit crops, apply 10% urine (one liter cow urine in 10 liter water) in the pits filled with the mixture of soil and cow dung manure/Ghan Jeevamrit before planting orchard seedlings. After planting, occasional application of 10-20% cow urine protects the plant from bacterial and fungal diseases.

9d. Saptdhanyankur ark: It is basically advocated to increase the grain shine and weight. The ingredients include 100 gm seeds each of sesame (oilseed), moong, urad, cowpea, moth, gram (5 types of pulse seeds) and wheat. Soak sesame seeds in a little water overnight in a bowl. On the second day, mix sesame, moong, urad, cowpea, moth, wheat, gram seeds and soak them in water overnight. On the third day, after taking out all the seven types of seeds from the water, tie them in a bundle in a coarse cotton cloth and hang them inside the house and keep the water safe. Then after the seeds have sprouted, grind all the seeds and make them like a sauce (chutney). Prepare a mixture of 200 liters of water, 10 liters of cow urine and safe water of seeds. In this, with the help of hands, mix the chutney of all the seven types of seeds and stir it with the help of a wooden stick. After the solution is ready, cover it with a gunny bag and keep it for two hours. After that filter it with a cloth. In this way Saptdhanyakur extract (ark) is ready for use. Specially use this extract within 48 hours. This is to be used during the milk stage of grains, infancy of the fruit (small beans) and bud stage in flower crops. Use of Saptdhayankur ark extract brings shine on the grains and reduces the dropping of flowers and fruits.

In addition to the above, depending upon the requirement, mechanical traps, pheromone traps, yellow sticky traps, bio-control agents such as *Trichoderma*, *Pseudomonas*, *Trichogramma*, *Bacillus etc* can also use in the natural farming to manage insect and diseases in crops.

10. Harvesting: Sowing and harvesting can be done either manually or mechanically depending upon the crops and availability of machineries.

General Guidelines

Research

1. All the generic protocols given above should be mandatorily adopted for evaluation.
2. Minimum experimental plot size of 150 square meter should be maintained for plain areas while 75 square meters should be planned for hills.

3. Study observations should focus on soil physical, chemical and biological properties apart from yield and economics. Since experiment plots are not too big, studies related to earthworm contribution may be undertaken in farmer's field who are practising natural farming for many years.
4. Nutrient cycling under natural farming through active soil biology needs focussed attention of researchers.
5. Efforts should be made to take treatment of natural farming separately in one acre, if available, otherwise non experimental area should be kept.

Demonstrations by KVKs

1. All the generic protocols given above should be mandatorily adopted for demonstration.
2. Minimum demonstration area should be of one-acre size.
3. Observations should focus yield, economics and soil organic carbon.
4. Preference should be given to develop dairy unit in the premises.

Reference

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Natural Farming Package of Practices

1. Name of the state : Gujarat

2. Natural Farming Practices

a. Soil Health Management practices

i. Name of practice: Mulching and Intercropping

ii. Brief description of the practice: Mulching with straw enhances soil moisture and encourages microorganisms and earthworms as well as fostering seed germination. Comprising dried biomass waste or any dry organic material, it contributes to improve soil fertility by decomposing through microbial activity.

iii. Benefit of the practices:

- ✓ Sustain and improve the soil organic carbon
- ✓ Increase soil microbial population
- ✓ Regulate soil temperature
- ✓ Improve Water Holding Capacity (WHC) and sustain soil moisture
- ✓ Reduce weeds population

iv. Protocols to be followed: Keeping soil surface covered year around with the help of crop residues available in the farm.

Inter-cropping of cereal crops with pulses/oilseed [e.g. Wheat + Chickpea (4:2), Maize + Soyabean (2:3)].

b. Nutrient management practices

i. Name of practice: *Ghan jivamrut*, *Jivamrut* and Intercropping

ii. Brief description of the practice: *Jivamrut* and *Ghan jivamrut* rich in nutrients like carbon, nitrogen, phosphorus, calcium etc. It boosts soil microorganisms, enhancing soil fertility and crop yield. It's made from cow dung and urine, promoting nutrient availability and microbial activity. This leads to increasing earthworm population in soil. Soil microorganisms actively impact by cycling nutrients like carbon and nitrogen which is essential for plant growth.

iii. Benefit of the practices:

- ✓ Enriches the soil with nutrients
- ✓ Sustain and improve the soil organic carbon
- ✓ Addition and fulfil the nutrients requirement of different crops
- ✓ Increase soil microbial population
- ✓ Improve WHC

iv. Protocols to be followed:

- ✓ 2 t/ha FYM once in a year during *kharif* season (when 1st time starting of NF and then reduce FYM @ 25 % from subsequent years)
- ✓ 2 t/ha *Ghan jivamrut* before sowing of the crops in each season as soil application
- ✓ Soil application of 1250 L/ha *Jivamrut* at the time of sowing thereafter every 15 days apply in the soil.
- ✓ Foliar application of *jivamrut* 1.0 litre/10 litre of water at 20 days after sowing and *jivamrut* 2.5 litre/10 litre of water at 40 and 60 days after sowing.

c. Pest and disease management practices:

i. Name of practice: *Agniastra*, *Brahmastra* and *Neemastra*

ii. Brief description of the practice: To control the sucking and other pest

Recommendations for farmers: (For effective management of root-knot nematodes in tomato)

Farmers growing tomato in middle Gujarat are recommended to apply *Agniastra* 800 mL in 10 litres of water and then dip the seedling roots for six hours. Thereafter, drench 500 mL *Agniastra* solution per plant at the time of transplanting and at 15, 30 and 45 days after transplanting for effective management of root-knot nematodes. The method for preparation of *Agniastra* is mentioned below:

Ingredients	Required Quantity
Cow urine (<i>desi</i>)	20 L
Neem leaves paste	5 kg
Garlic paste	500 g
Green chillies paste	500 g
Tobacco dust	1 kg

- Mix all the ingredients together in vessel and boil it 4-5 times continuously at medium flame and keep it for 48 hrs. Filter this by cloth and then use it.

iii. Benefit of the practices:

To prevent/control the pest with nature based low-cost product

iv. Protocols to be followed:

- ✓ Spraying of liquid formulation (400 ml /10 litre of water) at pest incidence occur
- ✓ Sowing of yellow marigold flower in field as a trap crop
- ✓ Spraying of prepared solution (10 days old butter milk dip with copper road or wire for 10 days) @ 5 % at 45 and 60 DAS for disease control

d. Weed management practices

i. Name of practice: Mulching, Intercropping and stale seedbed technique

ii. Brief description of the practice: Mulch and intercropping is a highly effective tool for weed control in natural farming systems. It works by blocking light, preventing seed germination, regulating moisture and temperature and even enhancing soil health. This reduces the need for manual weeding, creating a more sustainable and eco-friendly farming system.

iii. Benefit of the practices:

Mulching blocks sunlight, inhibiting the germination of weed seeds by creating a physical barrier.

Intercropping creates a more diverse environment where weeds have to compete for light, water, and nutrients.

iv. Protocols to be followed:

- ✓ In stale seedbed technique, allowing weed seeds to germinate and then killing them before sowing of crop with the help of tillage
- ✓ Mulching with available on farm crop residues
- ✓ Inter-cropping in cereals crops with pulses/oilseeds
- ✓ Use weed free seeds
- ✓ Add mulching material 10 days after sowing so that germination is not affected

e. Seed selection, treatment and sowing practices

i. Name of practice:

- ✓ Use of deshi/improved varieties of seeds
- ✓ Seed treatment with *Beejamrut*
- ✓ Drilling, Broadcasting and Dibbling etc.

ii. Brief description of the practice: These traditional varieties are often better suited to the local environment, as they have evolved over generations to withstand local pests, diseases, climate and soil conditions. *Bijamrut* is an essential component of natural farming that promotes healthier crops. *Bijamrut* made from natural inputs like cow urine and cow dung, it improves seed germination, enhances plant growth and fosters a sustainable farming ecosystem.

iii. Benefit of the practices:

- ✓ Low infestation of pests and diseases
- ✓ High nutritive value
- ✓ Better germination
- ✓ The microbial activity in *Bijamrut* helps to reduce the incidence of soil-borne diseases

iv. Protocols to be followed:

- ✓ Use of deshi and improved varieties
- ✓ Treat the seeds with 300 mL *Bijamrut*/kg of seeds before 8-10 hrs of sowing

f. Cropping systems

i. Name of the practice: Intercropping

ii. Brief description of the practice: e.g. Maize + Soyabean (2:3) and Wheat + chickpea (4:2) intercropping pattern.

iii. Benefit of the practices:

Intercrops helps to restore soil fertility when legumes are included as component crops. Suppress the weeds as well as reduce soil erosion, as the soil remains covered with plants.

iv. Protocols to be followed:

- ✓ Inter-cropping of cereal crops with pulses/oilseed [e.g. Wheat + Chickpea (4:2), Maize + Soyabean (2:3)].

g. Water management practices

i. Name of the practice: Check basin

ii. Brief description of the practice: Due to good source of irrigation water in area, irrigation provided through check basin method.

iii. Benefit of the practices:

✓ Simplest method

✓ Uniform distribution of water is possible

iv. Protocols to be followed: Proper size of bund height is required. Length and width of the basin depending on the slop of the field.



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Natural Farming Package of Practices

1. Name of the state: Gujarat

2. Natural Farming practices:

a. Soil health management practices:

i. Name of the Practices:

- ☐ Minimum tillage,
- ☐ Mulching, Inter-cropping,
- ☐ 365 days soil cover

ii. Brief description of the practice:

- ☐ Minimum tillage
- ☐ Mulching or live soil mulch
- ☐ Inter-cropping / Mixed cropping
- ☐ Pre-monsoon dry sowing with different groups of crops

iii. Benefits of the practices (impacts):

- ☐ Sustain and improve the soil organic carbon,
- ☐ Increase soil microbial population,
- ☐ Maintain soil temperature
- ☐ Improve WHC and sustain soil moisture
- ☐ Reduce weeds population

iv. Protocols to be followed:

- ☐ Minimum tillage with indigenous implements or mini-tractor
- ☐ Mulching with available on farm crop residues
- ☐ Live mulch with green-manuring crops *i.e.* Sun hemp/pulses *etc.*
- ☐ Inter-cropping in cereals/oilseed crops with pulses [e.g. Wheat + gram (4:1), Wheat + Lucerne (mixed cropping), groundnut + sesame (4:2), Pearl millet + Pulses (1:1), Castor + Green gram (1:2), Castor + Groundnut (1:2), Pigeon pea + groundnut (1:2), Turmeric + cowpea (2:1), Sorghum+ Kalingda (6:1) *etc.*].

b. Nutrient management practices:

i. Name of the practices:

- ☐ Use of FYM,
- ☐ *Ghanjivamrut*,
- ☐ *Jivamrut*
- ☐ Intercropping

ii. Brief description of the practice:

Jivamrut and *Ghan jivamrut* rich in nutrients like carbon, nitrogen, phosphorus, calcium *etc.* It boosts soil microorganisms, enhancing soil fertility and crop yield. It's made from cow dung and urine, promoting nutrient availability and microbial activity. This leads to increasing earthworm population in soil. Soil microorganisms actively

impact by cycling nutrients like carbon and nitrogen which is essential for plant growth.

iii. **Benefits of the practices (impacts):**

- ☐ Sustain and improve the soil organic carbon,
- ☐ Addition and fulfil the nutrients requirement of different crops
- ☐ Increase soil microbial population,
- ☐ Improve WHC
- ☐ Improve soil structure

iv. **Protocols to be followed:**

- ☐ 2-4 t/ha FYM once in year during kharif season
- ☐ 2-4 t/ha *Ghanjivamrut* before sowing of the crops in each season
- ☐ Soil application of 1000-1500 l/ha *Jivamrut* at the time of sowing thereafter twice in month with irrigation.
- ✓ Foliar spray of 5-10 % of *Jivamrut* at 20 days after sowing jivamrut 2.5 litre/10 litre of water at 40 and 60 days after sowing.

c. **Pest and Disease management practices:**

i. **Name of the Practices:**

- ☐ Use of Neemastra,
- ☐ Bhramastra,
- ☐ Agniashtra,
- ☐ Dashparni Ark,
- ☐ Sunthastra,
- ☐ Jivamrut and Intercropping.

ii. **Brief description of the practice:**

- ☐ To control the sucking and other pest

iii. **Benefits of the practices (impacts):**

- ☐ To prevent the infestation of pest and disease

iv. **Protocols to be followed:**

- ☐ Spraying of Neemastra @ 10 % (1000 ml /10 litre of water) at 45 DAS followed by Brahmastra @10% followed by Agniashtra @10 % at 10 days interval of each spray.
- ☐ Erect 40 bird perches per hectare
- ☐ Install one light trap per hectare
- ☐ Sowing of yellow marigold flower in field as a trap crop.
- ☐ Soil application of Agniashtra @ 200 litres/ha or Neem cake @ 1 tone /hectare before sowing.
- ☐ Drenching of 10 days decomposed liquid material (10 kg pieces of green cactus + 10 kg green leaves of milkweed (*Calotropis procera*) and 20 litre of 10 days old butter milk) @ 200 litre/ha after 20-25 DAS (ITK).
- ☐ Apply small pieces or paste of calotropis plant with irrigation or apply neem cake @ 1 tone /hectare before sowing and maintain soil moisture by irrigation to avoid attack on root zone for termite control.

- ☐ Spraying of prepared solution (10 days old butter milk dip with copper road or wire for 10 days) @ 5 % at 45 and 60 DAS for disease control.

d. Weed management practices:

i. Name of the Practices:

- ☐ Stale seedbed,
- ☐ Mulching,
- ☐ Intercropping

ii. Brief description of the practice:

Mulch and intercropping is a highly effective tool for weed control in natural farming systems. It works by blocking light, preventing seed germination, regulating moisture and temperature and even enhancing soil health. This reduces the need for manual weeding, creating a more sustainable and eco-friendly farming system.

iii. Benefits of the practices (impacts):

- ☐ Suppress the weed population

iv. Protocols to be followed:

- ☐ Allowing weed seeds to germinate and then killing them before sowing of crop
- ☐ Mulching with available on farm crop residues
- ☐ Live mulch with green-manuring crops i.e. Sun hemp/pulses *etc.*
- ☐ Inter-cropping in cereals/oilseed crops with pulses

e. Seed selection, treatment and sowing practices:

i. Name of the practices:

- ☐ Use of deshi & improved varieties
- ☐ Seed treatment with Beejamrut
- ☐ Drilling, Broadcasting & Dibbling *etc*

ii. Brief description of the practice:

These traditional varieties are often better suited to the local environment, as they have evolved over generations to withstand local pests, diseases, climate and soil conditions. Bijamrut is an essential component of natural farming that promotes healthier crops. Bijamrut made from natural inputs like cow urine and cow dung, it improves seed germination, enhances plant growth and fosters a sustainable farming ecosystem.

iii. Benefits of the practices (impacts):

- ☐ Low infestation of pests and diseases
- ☐ High nutritive value
- ☐ Good germination

v. Protocols to be followed:

- ☐ Use of deshi & improved varieties [i.e. Bansi (wheat), local maize]
- ☐ Treat the seeds with 200 ml *Bijamrut*/kg of seeds before 8-10 hrs of sowing.

f. Cropping systems

i.Name of the practices: Intercropping, Mixed cropping

ii. Brief description of the practice:

Inter-cropping in cereals/oilseed crops with pulses,
Wheat + Lucerne (mixed cropping)

iii. Benefit of the practices:

Intercrops helps to restore soil fertility when legumes are included as component crops. Suppress the weeds as well as reduce soil erosion, as the soil remains covered with plants.

iv.Protocols to be followed:

Inter-cropping in cereals/oilseed crops with pulses [e.g. Wheat + gram (4:1), Wheat + Lucerne (mixed cropping), groundnut + sesame (4:2), Pearl millet + Pulses (1:1), Castor + Green gram (1:2), Castor + Groundnut (1:2), Pigeon pea + groundnut (1:2), Turmeric + cowpea (2:1), Sorghum+ Kalingda (6:1) etc.].

g. Water management practices

i. Name of the practice: Micro irrigation/Check basin

ii. Brief description of the practice: Due to shortage of irrigation water and loamy sand texture soil

Benefit of the practices:

Saving of water
Higher yield

ivProtocols to be followed:

Mostly irrigation on the basis of CPE

Natural Farming Package of Practices



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A. Soil health management practices

i. Name of the practice:

1. *Jeevamrit and Ghanjeevamrit*
2. Residue Mulching
3. Permanent (365 Days) Soil Cover/ Pre-Season Dry Sowing
4. Minimum Soil Disturbance
5. Crop Diversification

ii. Brief description of the practice:

1. *Jeevamrit* and *Ghanjeevamrit* are rich in beneficial microbes, such as bacteria, fungi and actinomycetes. These microbes improve the soil biological activity, leading to a dynamic and healthy soil ecosystem. Regular application **increases microbial population and interactions**, which are essential to **control plant pathogens, accelerate organic matter decomposition and nutrient cycling**.
2. Residue mulching is a vital practice for improving and maintaining soil health by creating a **protective organic layer** on the soil surface. This layer **reduces erosion of nutrients, prevents water loss through evaporation which also controls salinity**, and moderates soil temperature, providing a stable environment for soil biota (**Microbes and Earthworms**). As the mulch decomposes, it **adds organic matter to the soil, enriching it with nutrients and increasing soil organic carbon (SOC) content**. This process **stimulates microbial activity in the rhizosphere, enhancing nutrient cycling and the formation of humus**, a stable organic substance that **improves soil structure, nutrient availability and water-holding capacity**.
3. Maintaining a 365-day soil cover through plantations provide a continuous supply of root exudates, creating a dynamic interface between plants and soil microbes that significantly enhances soil organic carbon. **Additionally, plant diversity supports microbial diversity by providing different habitats and resources, which helps create a balanced soil microclimate**. The living roots exude compounds like sugars, amino acids, and organic acids, which act as energy sources for soil microorganisms, stimulating their activity and facilitating the decomposition of organic matter. This process contributes to nutrient cycling and the formation of stable SOC pools, essential for soil fertility and

carbon sequestration. Moreover, year-round soil cover minimizes erosion and organic matter loss, protects soil from temperature extremes, and supports a diverse microbial community, all of which synergistically enhance SOC, nutrient cycling and promote long-term soil health. Moreover, Various mycorrhizal fungi associate with nearly 90% of global plant species, relying on living roots which reciprocate by supplying nutrients and water from the soil to the roots.

4. Conventional tillage practices lead to the disruption of soil aggregates in surface layers and increases aeration, which reduces total carbon by microbial decomposition of soil organic carbon to CO₂ particularly in **macroaggregates**. Additionally, adoption of zero tillage **reduces the number of micro pores (15-150 µm) in the soil, which is crucial for microbial activity thereby improving the physical conservation of soil organic carbon**. Therefore, by minimizing soil disturbance through the adoption of zero tillage practices, it is anticipated that CO₂ emissions from the soil to the atmosphere can be reduced, contributing to the mitigation of global climate change and improve soil organic carbon status. Zero tillage seeding systems without disturbing the soil **surface maintains optimum proportions of gaseous exchange in the rooting-zone, reduces C losses as atmospheric CO₂, moderates organic matter oxidation, improves porosity for water movement and limits the germination of weed seeds**.
5. Crop diversification positively impacts soil health by **enhancing soil biodiversity, improving nutrient cycling, and reducing pest and disease pressures**. Different crops have **varying root structures and nutrient requirements**, which promote a **more balanced use of soil nutrients and prevent depletion**. **Deep-rooted crops can bring nutrients from deeper soil layers to the surface**, while including **legumes** in crop diversification significantly enhances soil health due to their unique **ability to fix atmospheric nitrogen through symbiotic relationships with soil bacteria**. This natural nitrogen enrichment **eliminates the need for synthetic fertilizers, improving soil fertility sustainably**.

Additionally, legume root systems contribute organic matter to the soil, **enhancing its structure, water-holding capacity, and microbial activity**. In mixed cropping systems, legumes help **balance nutrient dynamics**, support other crops' growth, and suppress weeds. By preventing soil exhaustion and fostering a diverse microbial ecosystem, legume-based diversification is a cornerstone for long-term soil health and productivity.

ii. Benefits of the practices (impacts):

1. The application of *Jeevamrit* and *GhanJeevamrit* enriches the soil with beneficial microbes, as demonstrated by **metagenomic findings in natural farming soils**, showcasing their crucial role in improving soil health and nutrient availability. These bio-inputs foster a diverse microbial ecosystem, including dominant groups like ***Proteobacteria*, *Actinobacteria*, *Firmicutes*, and *Cyanobacteria***, which are essential for biogeochemical processes such as nutrient cycling and organic matter decomposition. At the genus level, microorganisms like ***Streptomyces*, *Bacillus*, *Pseudomonas*, and *Rhizobium*** are particularly prominent in Natural Farming soil. *Streptomyces* and *Bacillus* enhance phosphorus solubilization, while *Pseudomonas* supports plant growth through the production of phytohormones and bioactive substances. *Rhizobium* is instrumental in nitrogen fixation, converting atmospheric nitrogen into bioavailable forms for plants.

Metagenomic analysis also reveals an increase in the relative **abundance of genes linked to critical metabolic functions, such as hydrolase, isomerase, and ligase**, reflecting a robust microbial community with enhanced functional capacity. These enzymes **accelerate the decomposition of organic matter, releasing vital nutrients like nitrogen, phosphorus, and trace elements into the soil**. This dynamic microbial activity underpins the improvement of soil health, ensuring sustainable nutrient availability and fostering resilient agricultural productivity.

2. The **Residue mulch** layer reduces the impact of rain and wind on the soil, significantly **minimizing erosion and nutrient runoff**. By keeping nutrients in place, it ensures their **availability for plant uptake**, a critical factor in nutrient-depleted soils often encountered in conventional farming.

Mulching reduces water loss through evaporation by providing a barrier between the soil and the atmosphere. This not only conserves water but also helps **maintain a balanced soil moisture level, crucial for microbial activity and plant growth**. By **preventing excess evaporation, mulching also mitigates salinity buildup**, which can occur due to rising salts in the absence of adequate soil moisture.

Mulch acts as an insulating layer, **moderating soil temperature by keeping it cooler during hot periods and warmer during colder seasons**. This stable environment is

conductive to the thriving of soil biota, including microbes and earthworms, which play a vital role in soil **fertility and nutrient cycling**.

As the mulch decomposes, **it adds organic matter to the soil**, enriching it with essential nutrients and **increase SOC levels**. This increase in SOC not only improves soil fertility but also contributes to **carbon sequestration**, a critical component of climate change mitigation strategies.

The decomposition of mulch fuels microbial activity in the rhizosphere, leading to enhanced nutrient cycling. **Microbes decompose organic residues into simpler compounds, making nutrients readily available to plants. This process also promotes the formation of humus**, a stable organic substance that further enriches the soil.

The addition of organic matter through **mulching enhances soil aggregation**, resulting in better soil structure. This improved structure **increases the soil's water-holding capacity, reducing the need for frequent irrigation** while providing a more consistent water supply to plants.

3. **Plant diversity within a soil cover supports microbial diversity** by providing a range of habitats and resources. **Different plants exude different compounds that attract specific microbial communities**, leading to a more varied and resilient soil ecosystem. **This diversity promotes a balanced soil microclimate**, which is key to long-term soil health. By fostering beneficial microorganisms, such as nitrogen-fixing bacteria, mycorrhizal fungi, and decomposers, the soil's nutrient cycling processes are enhanced. Pre-season sowing aids in nutrient cycling by facilitating the release of nutrients trapped in organic matter. Microbes play a central role in breaking down plant residues and releasing essential nutrients like nitrogen, phosphorus, potassium and micronutrients , which are then available for the next crop.

A **year-round soil cover**, provided by cover crops or **pre-season sowing, ensures a constant supply of root exudates**, such as sugars, amino acids, and organic acids. These exudates act as a continuous **energy source for soil microorganisms**, promoting their activity and **enhancing the decomposition of organic matter**. This results in improved SOC content, essential for maintaining soil fertility, enhancing microbial diversity, and supporting nutrient cycling.

By continuously covering the soil with plant roots, **organic matter is steadily added through root growth, decay, and the breakdown of plant residues**. This process significantly contributes to the **formation of stable SOC pools**, which improves soil fertility and provides a **natural means of carbon sequestration, crucial in the fight against climate change**. The **presence of living roots also helps to reduce soil erosion**. By keeping the soil covered throughout the year, the **roots of cover crops help bind the soil particles, reducing the risk of erosion caused by wind or water**.

Cover crops protecting the soil from extreme temperatures. **During hot weather, they reduce soil evaporation and heat stress on soil microbes and plant roots, while in colder months, they help retain warmth. This temperature moderation creates a more stable environment for soil biota, enhancing microbial activity and soil health.** A stable temperature range also encourages the continued breakdown of organic matter, supporting nutrient availability.

Cover crops with taproots can create macro pores and alleviate compaction. Fibrous-rooted cover crops can promote aggregation and stabilize the soil. Species of cover crops that host mycorrhizal fungi can sustain and increase the population of these beneficial fungi.

4. **Soil aggregates** are formed through the interaction of organic matter, clay minerals, microbial products, and soil fauna activity. The **physical protection of organic carbon** occurs when organic matter, such as **decomposed plant residues, is incorporated into these aggregates**. This **encapsulated carbon is shielded from rapid microbial degradation**, as the structure of aggregates limits the access of microbes to the organic matter contained within. Microaggregates, which form within macroaggregates, play a significant role in this process. These microaggregates provide a physical barrier that prevents microorganisms from breaking down the organic carbon quickly. As a result, the carbon within aggregates is more stable and is slowly mineralized over a much longer time frame.

Conventional tillage, a common practice in many agricultural systems, **disrupts the natural aggregation of soil particles, particularly in the surface layers**. This disruption increases **aeration in the soil**, making organic matter more accessible to soil microbes, which leads to **accelerated microbial decomposition of organic carbon**. As a

result, tillage practices often lead to the loss of organic carbon in the form of CO₂ emissions, contributing to climate change. The disruption of aggregates also reduces the soil's ability to physically protect organic carbon, as the organic matter is no longer encapsulated within stable aggregates.

In contrast, **zero tillage**, a core principle in natural farming, minimizes soil disturbance and preserves the integrity of soil aggregates. **Under zero tillage, the soil is left undisturbed, allowing for the natural formation and stabilization of aggregates.** This practice maintains the **physical protection of organic carbon** within aggregates by preventing the disruption of microaggregates and macroaggregates. The **organic carbon in these aggregates is shielded from rapid decomposition**, as microbes are unable to penetrate the dense structure of aggregates, where oxygen and water levels are low, inhibiting microbial activity.

Zero tillage also helps maintain the soil's natural porosity, including the formation of micropores, which are crucial for microbial activity and the efficient breakdown of organic matter. These micropores, which are reduced in conventional tillage, help create a favourable environment for soil biota, supporting the process of nutrient cycling and the stabilization of organic carbon.

5. Legumes are central to the principles of natural farming, especially because of their ability to **fix atmospheric nitrogen through a symbiotic relationship with nitrogen-fixing bacteria (such as *Rhizobium*) in their root nodules.** This process naturally enriches the soil with nitrogen, an essential nutrient for plant growth, without the need for synthetic fertilizers. The ability of legumes to supply their own nitrogen is particularly important in systems that aim for sustainability and minimal external inputs. **By incorporating legumes into crop diversification systems,** natural farmers can significantly reduce their reliance on chemical fertilizers, thus **minimizing environmental pollution and reducing the cost of production.** Moreover, legumes contribute organic matter to the soil, which improves soil structure, increases water retention, and promotes microbial activity—further supporting soil health and fertility.

Natural farming is rooted in the idea that farming should work in harmony with natural ecosystems, rather than relying on artificial inputs. **Crop diversification is a fundamental practice in natural farming that encourages biodiversity at both the**

plant and microbial levels. A variety of crops supports a broader range of soil organisms, such as **earthworms, fungi, and bacteria, which work together to maintain soil structure and health.** By planting different crops, natural farmers create an environment in which these organisms can thrive, ultimately enhancing soil fertility and its resilience to disturbances.

crop diversification improves **nutrient cycling by using plants with different nutrient requirements.** Some crops take up specific nutrients, while others can replenish or balance the nutrients in the soil. For instance, **deep-rooted crops (such as trees or certain vegetables) access nutrients from deeper soil layers that may not be available to shallow-rooted crops.** This helps prevent nutrient depletion in the upper soil layers. As crops like **legumes fix nitrogen from the air, they contribute directly to soil fertility without the need for external inputs like synthetic fertilizers.** This supports a more balanced and sustainable nutrient cycle that relies on the natural processes of soil organisms.

iii. Protocols to be followed:

1. Preparation of Jeevamrit

Sr. No.	Component	Quantity
1	Indigenous cow dung	10 kg
2	Indigenous cow urine	8 - 10 litres
3	Jaggery	1.5 - 2.0 kg
4	Gram flour	1.5 - 2.0 kg
5	Water	180 litres
6	Soil collected from area under the large canopy trees added as a native microflora inoculant	500 g

To prepare *Jeevamrit*, the ingredients are mixed in a 200-liter drum, stirred twice daily in a clockwise direction, and left to ferment for 48 hours in a shaded area.

During fermentation, aerobic and anaerobic bacteria from cow dung and urine actively multiply and decompose the organic matter. Cane jaggery acts as a energy source for microbes enhance fermentation, while pulse flour supplies protein energy.

In summer, *Jeevamrit* is ready within 3–4 days, while in winter, it requires 6–7 days to complete the fermentation process.

Soil Application of *Jeevamrit*

For application, *Jeevamrit* should be used at the rate of **400 liters per acre during each irrigation cycle. For better efficiency and uniform application, *Jeevamrit* can also be atomized through sprinkler or drip irrigation systems. This ensures optimal coverage and maximizes its beneficial effects. Adequate soil moisture is crucial when applying *Jeevamrit*.**

The canopy region (i.e., the shaded region of a tree at noon) should be sprayed once or twice a month in a circular pattern.

For crops with a duration of 60 to 90 days

First spray: Mix 5 litres of filtered *Jeevamrit* with 100 litres of water per acre and spray it 21 days after sowing.

Second spray: Mix 20 litres of filtered *Jeevamrit* with 200 litres of water per acre and spray it 21 days after first spraying.

Third spray: Mix 5 litres of sour buttermilk or lassi with 200 litres of water per acre and spray it 21 days after the second spraying.

For crops with a duration of 90 to 120 days

First spray: Mix 50 litres of filtered *Jeevamrit* with 100 litres of water per acre and spray it 21 days after sowing.

Second spray: Mix 10 litres of filtered *Jeevamrit* with 150 litres of water per acre and spray it 21 days after first spraying.

Third spray: Mix 20 litres of *Jeevamrit* with 200 litres of water per acre and spray it 21 days after the second spraying.

Fourth and last spray: Mix 5 litres of sour buttermilk or 2 litres of coconut water with 200 litres of water per acre and spray it at the milking stage of growing or fruit initiation stage of the crop.

For crops with a duration of 120 to 135 days

First spray: Mix 5 litres of filtered *Jeevamrit* with 200 litres of water per acre and spray it one month after sowing.

Second spray: Mix 10 litres of filtered *Jeevamrit* with 150 litres of water per acre and spray it 21 days after first spraying.

Third spray: Mix 5 litres of sour buttermilk or lassi with 200 litres of water per acre and spray it 21 days after the second spraying.

Fourth spray: Mix 20 litres of filtered *Jeevamrit* with 200 litres of water per acre and spray it 21 days after the third spraying.

Fifth and last spraying: Mix 5 litres of sour buttermilk or 2 litres of coconut water with 200 litres of water per acre and spray it at the milking stage of growing or fruit initiation stage of the crop.

For crops with a duration of 135 to 150 days

First spray: Mix 5 litres of filtered *Jeevamrit* with 100 litres of water per acre and spray it a month after sowing.

Second spray: Mix 10 litres of filtered *Jeevamrit* with 150 litres of water per acre and spray it 21 days after first spraying.

Third spray: Mix 5 litres of sour buttermilk or lassi with 200 litres of water per acre and spray it 21 days after the second spraying.

Fourth spray: Mix 20 litres of filtered *Jeevamrit* with 200 litres of water per acre and spray it 21 days after the third spraying.

Fifth spray: Mix 20 litres of filtered *Jeevamrit* with 200 litres of water per acre and spray it 21 days after the fourth spraying.

Last spraying: Mix 5 litres of sour buttermilk or 2 litres of coconut water with 200 litres of water per acre and spray it at the milking stage of growing or fruit initiation stage of the crop.

For crops with a duration of 165 to 180 days

First spray: Mix 5 litres of filtered *Jeevamrit* with 150 litres of water per acre and spray it a month after sowing.

Second spray: Mix 10 litres of filtered *Jeevamrit* with 150 litres of water per acre and spray it 21 days after first spraying.

Third spray: Mix 5 litres of sour buttermilk or lassi with 200 litres of water per acre and spray it 21 days after the second spraying.

Fourth spray: Mix 20 litres of filtered *Jeevamrit* with 200 litres of water per acre and spray it 21 days after the third spraying.

Fifth spray: Mix 20 litres of filtered *Jeevamrit* with 200 litres of water per acre and spray it 21 days after the fourth spraying.

Last spraying: Mix 20 litres of filtered *Jeevamrit* with 200 litres of water per acre and spray it at the milking stage of growing or fruit initiation stage of the crop

Spraying *Jeevamrit* on Sugarcane, Banana, and Papaya

Spray as per the method directed above for the first five months after sowing or planting these crops. Then, every 15 days, spray a mixture of 20 litres of *Jeevamrit* with 200 litres of water per acre in sugarcane, banana and papaya.

Spraying of *Jeevamrit* on fruit crops

Spray *Jeevamrit* twice a month on the fruiting trees (of any age). Spray the fruit crops with a mixture of 20-30 litres of filtered *Jeevamrit* with 200 litres of water.

Spray a mixture of coconut water and 2 litres of water 2 months before fruit maturity. After 15 days, spray a mixture of sour buttermilk with 200 litres of water.

Preparation of *Ghanjeevamrit*

To prepare *Ghanjeevamrit*, start by making a bed of dried cow dung, one and a half feet high and three feet wide, with the length as per requirement. Using a stick with a 1.5-inch diameter, create holes one foot above the bed in a zigzag pattern, spaced one foot apart.

Fill these holes with fully prepared *Jeevamrit*, allowing it to spread into the bed naturally. Leave the bed undisturbed for seven days.

After seven days, shovel the mixture and reshape it into a bed in the same manner. Again, create holes at one-foot intervals and fill them with *Jeevamrit*. Let this new bed ripen for another seven days. At the end of this process, the *Ghanjeevamrit* will be ready for use.

Ensure the bed is kept away from direct sunlight and rainwater during preparation. If you plan to store *Ghanjeevamrit*, dry it thoroughly in strong sunlight and then store it in a dry place using a jute bag.

Soil Application of *Ghanjeevamrit*

For soil application, *Ghanjeevamrit* should be applied at the rate of **1000 kg per acre**. This should be done **before every plantation or at the beginning of each cropping season**.

2. Crop residues vary in composition and decomposition rates based on their **Carbon-to-Nitrogen (C: N) ratio**.

High C: N ratio residues, like maize and wheat, decompose slowly, providing long-term soil organic matter and reducing erosion.

Low C: N ratio residues, like soybean and legume crops, decompose quickly, enriching the soil with nitrogen and boosting microbial activity.

Oilseed residues moderately improve soil texture and fertility, while soft residues from vegetables offer rapid nutrient cycling.

Apply a **thick layer of mulch around plants but not more than 4-6 inches**. Ensure proper coverage to prevent moisture loss and weed growth.

3. **Cover crops or perennial and green manure crops** keep the soil covered during the summer and other periods of time reducing the risk of erosion, compaction and Recapture excess nutrients. The biomass produced by cover crops is usually returned to the soil, enhancing organic matter levels.

Cover Crops Grown for Specific Purposes

Organic matter (high C:N)	Sorghum-Sudan grass, cereal rye, annual ryegrass, triticale, oats, wheat, spelt, barley
Nitrogen (low C:N):	Cowpea, winter pea, red clover, sweet clover, alfalfa, soybeans, mung beans
Reduce soil compaction (deep rooted)	Sorghum-sudangrass, annual ryegrass, oilseed or tillage radish, sweet clover-deep taproot, cereal rye, oats
Forage or can be grazed	Oats, forage radishes, turnips, cereal rye, annual ryegrass, teff for dry fields, sorghum-sudangrass, barley
Prevent soil erosion	Grasses have fibrous root systems to bind soil, and the best grass cover crops include cereal rye, annual ryegrass, oats, wheat, and barley. Other cover crops include buckwheat with a shallow fibrous root system, cowpea, winter pea
Recapture excess nutrients (nitrogen, phosphorus)	Oilseed or tillage radish, turnips, annual ryegrass, cereal rye, oats, wheat, sorghum-sudangrass, and buckwheat, sweet clover, winter pea, cowpea, red clover, hairy vetch
Tolerate wet soils	Sweet clover, red clover, annual ryegrass, cereal rye, wheat, and oats
Tolerate heat and drought	Cowpea, mung beans, sweet clover, sorghum sudan grass, buckwheat, barley
Attract beneficial insects	Buckwheat, sweet clover, red clover
Cold tolerant	Cereal rye, wheat, spelt, triticale, winter pea, and sweet clover
Natural herbicides or allelopathic effects for weed suppression	Cereal rye, oilseed or tillage radish, mustard, oats, barley, Buck wheat, and sorghum-sudangrass. Annual ryegrass, cereal rye, sorghum-sudangrass may be used for controlling soybean cyst nematodes.

Pre-season Dry Sowing

To maintain year-round green cover (365DGC) and active microbial activity by cultivating diversified crops during the off-season while reducing dependence on chemical inputs.

Pre-season Dry Sowing Implementation Process

a. Seed Composition and Preparation:

- Use a seed kit containing 30-35 types of seeds (Cereals, Pulses, Millets, Oilseeds, Spices, Vegetables, etc.), 12-13 kg per acre. Pelletize seeds before sowing to improve germination.
- Seed compositions are customized for each district based on seed availability.

b. Time of Sowing: Broadcast seeds one week before the harvest of the preceding crop.

- Use **residual moisture** from:
 - Rainfed black soil (one week before harvest).
 - Irrigated systems (one week before harvest).

- c. **Sowing Method:**
 - Broadcast seeds evenly over the field during the harvest of the main crop
 - Provide a 1-2-inch layer of mulch using groundnut shells, crop residues, or paddy straw (less preferable).
 - d. **Management Post-Sowing:**
 - Offseason rains support seed germination and growth. If plants survive beyond 25 days, microbial benefits accrue. Provide life-saving irrigation where possible.
 - e. **Incorporation into Soil:**
 - Incorporate dry sowing crop foliage directly into the soil 7-10 days before sowing the main crop.
 - f. **Utilization as Fodder:**
 - Use PMDS crop foliage as cattle fodder, leaving roots to enrich the soil.
 - g. **Soil and Crop Health:**
 - Enhanced root-to-shoot ratio (PMDS: 0.312 vs. Chemical: 0.242).
 - Improved soil carbon content and microbial activity.
 - Resistance to lodging in heavy rains.
4. **Crop diversification** is essential to enhance ecological balance, improve resource utilization, and support sustainable farming practices. Follow the guidelines below to implement effective crop diversification:
- i. If the main crop is a **monocot**, the mixed/allied crop should be a **dicot**.
 - ii. If the main crop is **deep rooted**, then **shallow rooted** mixed/allied crop should be selected.
 - iii. The growing duration of a mixed/allied crop should be **less than half or one-third of the main crop**. That means, it should be ready in less time than the main crop.
 - iv. The shadow of mixed/allied crops should not fall on the leaves of main crop.
 - v. Mixed/allied crops should proliferate and **cover the soil** thoroughly like cucumber, water melon etc.
 - vi. If the leaves of the main crop can **tolerate intense sunlight**, then mixed/allied crops that **do not require intense sunlight should be planted**.
 - vii. If the primary crop grows **quickly**, select a mixed/allied crop that **grows slower than the main crop**.

B. Nutrient management practices

i. Name of the practice:

Plants meet their nutrient requirements from different sources. Natural biomass can also serve as a source of nutrient for crop plants and can be used after appropriate treatment or conversion, and in some cases even in raw form.

1. *Jeevamrit* and *Ghanjeevamrit*
2. *Saptadhanyaankur ark*
3. Residue Mulching
4. Permanent (365 Days) Soil Cover/ Pre-Season Dry Sowing/Manuring
5. Crop Diversification

ii. Brief description of the practice:

1. *Jeevamrit* and *Ghanjeevamrit* are fundamental to natural farming, **enhancing soil health and nutrient availability through microbial enrichment and organic matter decomposition**. These inputs introduce beneficial microorganisms that **convert complex organic compounds into plant-available nutrients, fostering natural nutrient cycling**. By improving soil structure, aeration, and water retention, they support root development and efficient nutrient uptake. **The organic acids and enzymes produced by microbes enhance the solubility of essential nutrients like phosphorus, zinc, and iron, often locked in inaccessible forms.**
2. **Mulching** plays a critical role in nutrient management by enhancing soil health and **ensuring the efficient cycling of nutrients**. By covering the soil with organic materials like **crop residues, straw, or leaves, mulching creates a microclimate that promotes microbial activity and decomposition**. This process **releases nutrients such as nitrogen, phosphorus, and potassium into the soil in plant-available forms**. Mulching also **minimizes nutrient loss** and facilitate better nutrient uptake by roots. Additionally, it suppresses weeds, reducing competition for nutrients, and improves soil organic carbon, which enhances nutrient retention and availability over time. This natural farming practice aligns with sustainable nutrient management, fostering a self-reliant and fertile soil ecosystem.

The **combined use of bio-inputs such as *Jeevamrit* and *Ghanjeevamrit*, along with mulching, significantly enhances the activity of earthworms in the surface soil.** Earthworm activity is stimulated by the favourable environment created by bio-inputs, which promote microbial growth, and by mulching, which maintains moisture and provides organic matter as food. **Earthworm faeces, or castings, contain nutrient concentrations that are five times higher in nitrogen, nine times higher in phosphorus, and eleven times higher in potassium compared to the surrounding soil.**

3. Growing and incorporation of tender biomass of legume crops in soil is known as ‘**green manuring**’. These crop plants fix atmospheric nitrogen, and after decomposition and incorporation, release this nitrogen into the soil, which helps improve its physical and biological properties. (Permanent Soil Cover/ Pre-Season Dry Sowing/Crop Diversification as per above practice A)
4. *Saptadhanyaankur* (Seven-Grain Sprouts) refers to a traditional preparation in Indian agriculture and natural farming practices that involves sprouting seeds of seven types of grains or cereals. These grains are often chosen for their nutrient content, potential to promote soil health, and cultural significance. The sprouted grains are used as a bio-activator in natural farming, especially for soil enrichment

iii. Benefits of the practices (impacts)

1. (*Jeevamrit*, *ghanjeevamrit*, Residue Mulching, Permanent (365 Days) Soil Cover/ Pre-Season Dry Sowing and Crop Diversification as per above Practice A)
2. *Saptadhanyaankur* provide plant with nutrients & other growth promoters. Spraying will increase the shine on fruits, vegetables and grains.

iv. Protocols to be followed:

1. (*Jeevamrit*, *ghanjeevamrit*, Residue Mulching, Permanent (365 Days) Soil Cover/ Pre-Season Dry Sowing and Crop Diversification as per above Practice A)
2. **Preparation of *Saptadhanyaankur***

Sr. No.	Components	Quantity
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1	Sesame seeds	100 g
2	Mung bean seeds	100 g
3	Urdbean seeds	100 g
4	Cowpea (Lobia) seeds	100 g
5	Moth seeds	100 g
6	Chickpeas (Chana) seeds	100 g
7	wheat seeds	100 g

In a small bowl, take 100 grams of sesame seeds and add enough water to soak all the seeds and keep it in the shade. After two days, take a large bowl and put 100 grams of whole grain seeds of green gram (mung bean) 100 grams of black gram (urad), 100 grams of cowpeas beans, 100 grams of moth beans, 100 grams of chickpeas, and 100 grams of wheat grains.

Mix all these well and add enough water to soak them well. Then keep it indoors. After three days take out all the seeds, tie them in a cloth bag and keep them inside the house for germination. Also cover the water in which the grains were soaked. Open the bag when 1 cm long plumules (ankur) come out, then make its paste. Mix the paste in 200 litres of water by hand and also mix the previously steeped water. Leave this prepared mixture for 2 hours. After two hours, shake the mixture and strain it through a cloth.

Saptadhanyaankur ark Usage:

The solution should be sprayed on the plants as a foliar spray at the rate of 200 L/Acre. This mixture should be sprayed within 48 hours of preparation without mixing with the water.

- Spray on the vegetable crops (kharif or rabi) when the vegetables are in milking stage.
- Spray on fruit trees when the fruits are small, medium or before they mature.
- When the fruit pods are small.
- In flower crops when the flowers are in budding stage.

A. Plant protection practices

i) Name of the practice:

1. *Beejamrit*
2. *Dasparni Ark*
3. *Neemastra*
4. *Brahmastra*
5. *Agniastra*
6. *Sour Buttermilk*
7. *Mixed cropping (Push Pull System):*
8. Trap crops:

ii. Brief description of the practice:

1. Majority of the diseases and insect-pest infestation in the plants are seed and soil-borne. ***Beejamrit*** is a natural formulation to treat seeds, seedlings or other planting material before sowing to prevent seed and soil-borne diseases and insect-pest infestation.
2. ***Dasparni Ark*** is a botanical-based bio-formulation used in natural farming made from the leaves of ten different plant species, that is why it is called Dashparni Ark. It helps control pests, specifically all kinds of sucking pests and caterpillars. This formulation is made from neem leaves, karanj leaves, castor leaves, belpatra leaves, mango leaves, dhatura leaves, basil leaves, guava leaves, desi bitter gourd leaves, papaya leaves, turmeric leaves, ginger leaves, acacia leaves, custard apple leaves, along with ginger powder, turmeric powder, ginger paste, asafetida powder, chewing tobacco powder, spicy chili paste and garlic paste.
3. ***Neemastra*** is a natural pesticide used in natural farming practices, made primarily from Neem leaves. It is prepared by soaking Neem leaves in water for a specific period, sometimes combined with other ingredients like garlic or chili paste, to enhance its pest-repelling properties. *Neemastra* is highly effective in controlling a wide range of pests, especially insects like aphids, whiteflies, ants and caterpillars, while being safe for beneficial organisms and the environment. It works by disrupting the pests' feeding and reproductive cycles, making it a sustainable alternative to chemical pesticides.

4. ***Brahmastra*** is a natural bio-pesticide used in natural farming practices, to control large caterpillars and many other types of pest. It is prepared using a combination of medicinal plants such as neem, karanj, cilantro, bael, castor and dhatura leaves along with other ingredients like cow urine and water, all known for their antimicrobial and insecticidal properties.
5. ***Agniastra*** is a natural bio-pesticide used in natural farming systems, to control harmful pests and diseases. It is prepared by soaking a mixture of ingredients such as spicy green chili paste, garlic, crushed neem leaves and chewing tobacco often combined with cow urine to enhance its potency. *Agniastra* is effective in repelling and controlling a wide range of pests, including large sized insect pests boring holes on leaves and living inside fruits. It works by acting as a strong irritant to pests, deterring them from feeding on plants, while being safe for beneficial organisms.
6. ***Sour buttermilk*** is used in natural farming formulations, particularly for disease control, due to its beneficial properties. It acts as a natural fungicide and helps in reducing the spread of harmful pathogens, making it an effective and eco-friendly alternative to chemical treatments in natural farming systems.
7. **Mixed cropping** systems are principal strategies among cultural practices to be followed for ecological pest management. It is mainly based on the principle of agro ecosystem diversity for pest and disease management, wherein two or more crops are raised Simultaneously. In the push-pull mix cropping systems, specifically identified companion crop plants are raised in the field.
8. **Trap crops:** Trap crops are plants raised to entice insects away from the main crop.

ii Benefits of the practices (impacts):

1. Most of the diseases, insect-pest infestation and other disorders in plants are seed and soil-borne. So, it is important to treat seeds, seedlings or other planting material with ***Beejamrit*** before sowing to prevent seed and soil-borne diseases and insect-pest infestation in plants
 - a. Use of *Beejamrit* increase germination capacity of seeds.
 - b. It leads to uniform growth of seedlings and faster development of roots.
 - c. Plants remain free from seed and soil borne diseases, insect-pests and other disorders.

2. ***Dasparni Ark*** is a powerful botanical bio-formulation used in natural farming. It effectively controls a wide range of pests, especially sucking pests like aphids, whiteflies, and mealybugs, as well as caterpillars. Safe for humans, animals, and beneficial organisms, Dasparni Ark is a non-toxic, sustainable solution for pest management in natural farming.
3. ***Neemastra*** is widely used in natural farming for pest control, particularly effective against a range of insect pests like aphids, whiteflies, and caterpillars. Made primarily from Neem leaves, it works by disrupting the pests feeding and reproductive cycles, acting as both a repellent and a growth regulator. *Neemastra* is eco-friendly, non-toxic to humans, animals, and beneficial insects. It enhances plant health by reducing pest damage.
4. ***Brahmastra*** is effectively controlling large caterpillars and many other types of pests. It is an eco-friendly alternative to chemical pesticides, it is safe for humans, animals, and beneficial organisms, supporting biodiversity and reducing environmental harm.
5. ***Agniastra*** is a natural bio-pesticide made from ingredients like chili, garlic, and ginger, known for their strong pest-repelling properties. It effectively controls a wide range of pests, especially sucking insects like aphids and whiteflies, as well as fungal infections. As an eco-friendly solution, *Agniastra* is safe for humans, animals, and beneficial insects. It helps reduce pest damage. It controls the large sized insect pests boring holes on leaves and living inside fruits.
6. ***Sour buttermilk*** offers several benefits in natural farming. It contains lactic acid bacteria (LAB) that possess antifungal and antibacterial properties, making it effective in controlling plant diseases like fungal infections and bacterial blight. The organic acids, enzymes, and beneficial microorganisms found in sour buttermilk promote soil health, enhance microbial activity, and stimulate plant growth. As a natural fungicide, it helps reduce the spread of harmful pathogens.
7. ***Mixed cropping (Push Pull System)***: In the push-pull intercropping systems, specifically identified companion crop plants are raised in between the main crop, which acts as a push component, and the pull components are plated as borders. The 'push' component in this system repels the insects (using semiochemicals), whereas the 'pull' component, i.e., the trap-crop, attracts the pest species away from the main crop.

8. **Trap crops** are plants raised to entice insects away from the main crop. While selecting a trap crop, care must be taken that the trap crop species must be more attractive to the insect pests than the main crop and in no case, the pest should migrate from the trap crop to the main crop. Hence, effective utilization of trap crop technology is a challenging strategy that needs a thorough understanding of the food preference behaviour of pests

(iv) Protocols to be followed:

1. Preparation of Beejamrit

Sr No.	Ingredients	Quantity
1	Cow urine	05 litres
2	Cow dung	05 kg
3	Lime	50 g
4	Soil collected from area under the large canopy trees added as a native microflora inoculant	500 g
5	Water	20 Liter

Mix above ingredient and keep it for overnight. Treat the seed or planting material before sowing with 200 ml *Beejamrit* per kg seed. Spread the seeds for overnight to dry. In case of vegetative propagating crops, put tubers/rhizomes/sets/grafts of selected crop in a bamboo basket and dip the basket in a tub containing *Beejamrit* for 15-20 seconds for treatment. Dry the seeds in shade and use for sowing the next day.

2. Preparation of Dasparni Ark

Sr	Ingredients	Qty.	Sr	Ingredients	Qty.
1	Water	200 liters	13	Belpatra leaves	2 kg
2	Cow urine	10 liters	14	Mango leaves	2 kg
3	Cow dung	2 kg	15	Dhatura leaves	2 kg
4	Turmeric powder	500 g	16	Basil leaves	2 kg
5	Ginger paste	500g	17	Guava leaves	2 kg
6	Asafetida powder	10 g	18	Desi bitter gourd leaves	2 kg
7	Chewing tobacco powder	1 kg	19	Papaya leaves	2 kg
8	Spicy chili paste	1 kg	20	Turmeric leaves	2 kg

9	Garlic paste	500 g	21	Ginger leaves	2 kg
10	Neem leaves	2 kg	22	Acacia leaves	2 kg
11	Karanj leaves	2 kg	23	Custard apple leaves	2 kg
12	Castor leaves	2 kg	24	Ginger powder	200 g

Note: Any 10 of the above-mentioned plant parts can be added. The first five are important.

Put all the above-mentioned plant parts, dung and urine in a barrel and stir clockwise with a stick twice a day i.e., morning and evening. Keep this mixture in shade and protect it from water as well as sunlight. Leave the mixture for 40 days to be ready and then filter it with a cloth. Filtered *Dasparni Ark* can be stored for 6 months.

Spray: Mix 500-600 mL of *Dasparni Ark* with 15 liters of water and sprayed during pest infestation

3. Preparation of Neemastra

Sr	Ingredients	Quantity
1	Neem leaves and crushed seed kernel	5 kg
2	Cow urine	5 litre
3	Cow dung	1 kg
4	Water	100 litres

Mix above mentioned ingredients in a barrel and keep for 48 hours. Stir with a stick thrice a day and filter through a cloth before spraying.

Spray 200 L *Neemastra*/ Acre during insect infestation

4. Preparation of Brahmastra

Sr	Ingredients	Quantity
1	Cow urine	10 litres
2	Neem leaves	3 kg
3	Karanj leaves	2 kg
4	Cilantro leaves	2 kg
5	Bael leaves	2 kg

6	Castor leaves	2 kg
7	Dhatura leaves	2 kg

To make *Brahmastra*, select any of the above five ingredients. Grind all the leaves together. Put this mixture in an earthen pot and boil it with 20 liters of water. When four boils come, take it off and keep it in the shade for two to three days to cool down. After this, mix cow urine in it and filter it with a cloth and keep it in an earthen pot to store. It is important to protect it from the direct sunlight

Spray: Mix 500 mL *Brahmastra* with 15 L of water and sprayed during insect infestation

5. Preparation of Agniastra

Sr	Ingredients	Quantity
1	Cow urine	10 litres
2	Spicy green chilli paste	500 g
3	Garlic paste	500 g
4	Crushed neem leaves	5 kg
5	Chewing tobacco	1 kg

Mix above mentioned ingredients and boil it until four boils than cool the mixture for 48 hours. Mix 2-3 litres in a day and after then filter it with a cloth which can store for 3 Months.

Spray: Mix 500 mL *Agniastra* with 15 L of water and sprayed during insect infestation

6. Preparation of Sour buttermilk

Sr	Ingredients	Quantity
1	Water	100 litres
2	Sour buttermilk	3 litres

Mix both together in the above-mentioned ratio and spray as effective fungicide.

D. Weed Management in Natural Farming

i. Name of Practices

1. Mulching
2. Mixed cropping
3. Crop diversification
4. Cover crop
5. Water management
6. Use of Weed Free Seeds

ii. Brief Description of the practice

1. **Mulching:** Mulching or covering the soil surface can prevent weed seed germination by blocking light transmission preventing seed germination. Mulches physically suppress weed seeds emergence. There are many forms of mulches available. The following two are suitable practices of mulching in natural farming:

Living mulch/Organic mulch: Living mulch is usually a plant species that grows densely and low to the ground such as clover. Living mulches can be planted before or after a crop is established. It is important to kill and till in, or manage living mulch so that it does not compete with the actual crop. Often, the primary purpose of living mulch is to improve soil structure, aid fertility or reduce pest problems and weed suppression may be merely an added benefit.

Soil mulch: Mulching with soil involves creating a protective layer of soil around plants or over the soil surface to mimic natural processes and provide benefits similar to conventional mulching methods. This practice is particularly suited to regions with limited access to organic mulch materials or in areas where soil conservation is essential.

Plant residue mulching (Mulching with plant/crop residue): Plant residue mulching involves using leftover parts of crops or vegetation (e.g., straw, stalks, leaves, or husks) as a protective layer on the soil surface. It is a highly sustainable and cost-effective practice in natural farming that enhances soil fertility and reduces reliance on external inputs.

2. **Mixed Cropping:** Mixed cropping is an age-old agricultural practice of cultivating two or more crops simultaneously on the same piece of land. In natural farming, this method

aligns with the principles of ecological balance, biodiversity conservation, and low-input farming. By leveraging the natural symbiotic relationships between crops, mixed cropping ensures sustainable agricultural productivity without relying on synthetic inputs.

3. **Crop diversification:** Crop diversification in natural farming is a sustainable approach that fosters ecosystem health, optimizes resource use, and improves farm resilience. It enhances productivity while reducing the environmental footprint, making it a key practice in achieving long-term agricultural sustainability.
4. **Cover crop:** Plants grown primarily to cover the soil rather than for harvest. In natural farming, cover crops are an essential practice for enhancing soil health, managing weeds, and promoting sustainable agricultural systems. They are typically planted during the off-season or between primary crop cycles to provide multiple ecological and agronomic benefits.
5. **Water management:** Effective water management is a crucial component of weed control in natural farming. Water influences weed growth directly by affecting seed germination, plant establishment, and competition between weeds and crops. By managing water efficiently, farmers can reduce weed populations and create a more favorable environment for desired crops.
6. **Use weed free seeds:** The use of weed-free seeds is an important practice in natural farming to prevent the introduction and spread of weed species on a farm. By selecting seeds that are free from weed contamination, farmers can reduce the need for chemical herbicides and minimize the risk of invasive weed species that could damage crops or decrease yields.

iii. Benefits of the practice (impact):

1. **Mulch creates a physical barrier** by covering the soil surface. This will block sunlight from reaching weed seeds. This inhibits germination and growth, reducing weed emergence.

Thick mulch layer will **cover small or emerging weeds, suffocating** them and preventing further growth of weed. Organic mulches like plant residues or grass clippings **fill available gaps in the soil, leaving less space for weeds to establish.**

Mulching **minimizing weed seed dispersion** by preventing the weed seeds from being blown onto the soil or carried by water, reducing the weed seed bank in the field.

Consistent moisture under mulch discourages certain weeds that thrive in dry conditions. Mulching stabilizes soil temperature, which can prevent the germination of weeds adapted to temperature fluctuations.

2. **Mixed cropping systems combine crops with different growth patterns** (e.g., tall and spreading) to create a dense canopy that shades the soil. This limits the sunlight available for weed germination and growth. *Example:* Maize and cowpea, where maize provides vertical growth and cowpea spreads across the ground. Mixed crops efficiently utilize nutrients, leaving fewer resources available for weeds.

Crops with varied rooting depths exploit water resources differently, depriving weeds of moisture. The combined root systems of mixed crops occupy soil niches, preventing weeds from establishing. Some crops release allelochemicals (natural compounds) through roots or residues that inhibit weed seed germination and growth. *Example:* Mustard and sorghum have allelopathic properties that suppress weeds.

Crop residues left after harvest in a mixed cropping system can act as a natural mulch, suppressing weeds for subsequent crops. The diversity of crops disrupts the lifecycle of specific weed species by altering soil conditions and reducing their preferred growing environment. Growing different family crops in mixed cropping will break the cycle of infestation and change the microclimate which will affect the germination and viability of the weed seed.

3. **Diverse crops use different amounts of sunlight, water, and nutrients, which limits the availability of these resources for weeds.** For instance, tall crops like maize may shade the soil, while ground-hugging crops like legumes spread to cover the soil and block weed growth. *Example:* Intercropping maize with beans or cowpeas, where beans help cover the ground, leaving less space for weeds to grow.

Diverse crops planted together create a dense canopy that shades the soil and prevents sunlight from reaching weed seeds. This inhibits germination and growth of weeds that rely on sunlight to thrive. *Example:* In intercropping systems like maize and

pigeon peas, maize forms a tall canopy, and the peas spread across the ground, blocking sunlight for weeds.

Some crops naturally release chemicals (allelochemicals) through their roots, leaves, or decaying residues that inhibit weed seed germination or growth. This natural herbicidal action can help control weeds in diversified systems. Example: Mustard, sorghum, and sunflower are known for their allelopathic properties, which can suppress weeds when used in intercropping or rotation.

By alternating different crops each season, farmers can break the lifecycle of weeds that are adapted to a particular crop. This reduces the chance of weeds becoming resistant to a particular crop or cultivation practice. Example: Rotating between wheat, legumes, and vegetables disrupts the growth cycles of weeds that are common in monoculture systems.

Different crops have varying root depths, types, and structures.

This diversity means that crops use different parts of the soil, leaving less room for weeds to establish themselves. Shallow-rooted crops may help suppress deep-rooted weeds, while deeper-rooted crops can outcompete weeds that grow near the surface. Example: Planting legumes (shallow roots) with deep-rooted crops like maize or sunflowers can improve soil structure and reduce weed competition.

In diversified systems, especially with perennials and cover crops, the soil is less disturbed, creating conditions where weed seeds may not be exposed to light or favorable conditions for germination. Example: Cover crops in a rotation system can suppress weeds by maintaining continuous ground cover, reducing the need for tillage that brings weed seeds to the surface.

4. Weed Suppression can be achieved as cover crops grow rapidly and form dense canopies that shade the soil, preventing sunlight from reaching weed seeds, thus inhibiting their germination and growth.

Cover crops, like legumes, grasses, or broadleaves, compete with weeds for essential resources such as water, nutrients, and space. This competition weakens the weeds and limits their growth potential.

Some cover crops, such as mustard or sunflower, release natural chemicals (allelopathic compounds) into the soil that can inhibit weed seed germination and growth. These chemicals can affect the growth of nearby plants, including weeds.

The roots of cover crops, especially deep-rooted ones like **radish**, can physically disturb the soil, making it harder for weeds to establish their root systems. Additionally, cover crops help to break up soil compaction, allowing for better root penetration by desirable plants.

When cover crops are mowed or left to decompose on the soil surface, they act as a mulch, creating a thick layer that prevents weeds from emerging. The decaying matter also adds organic matter to the soil, enriching it and enhancing soil health.

By rotating cover crops regularly (e.g., planting different species each season), farmers disrupt the growth patterns of weeds that are adapted to specific crops. This rotation prevents weeds from becoming accustomed to a particular cropping system. Continuous cropping with the same species can promote specific weed species; cover crops in rotation break this cycle, reducing weed pressure over time.

5. **Irrigating at the right time can prevent weeds from germinating.** For example, delaying irrigation until after the crop has emerged can help reduce weed seed germination, as weeds often need water to sprout. Conversely, avoiding over-irrigation reduces the excess moisture that promotes weed growth.

Drip or localized irrigation minimizes the moisture available to weeds that grow between crops, while promoting healthy growth for crops that need water. This system reduces weed germination between the rows or in areas where crops are not planted.

By maintaining the right moisture level, crops can develop deeper, stronger root systems, making them more competitive with weeds. Conversely, weeds are often shallow-rooted and sensitive to fluctuations in moisture. Maintaining optimal soil moisture encourages crops to grow vigorously while limiting the growth of weeds.

In natural farming a **dry period can prevent many weeds from germinating by inhibiting their growth cycle.** Wetting and drying cycles can be managed by alternating

irrigation and avoiding unnecessary moisture in between crop growing seasons, particularly during the early stages of weed germination.

By watering during the **early stages of crop growth and after weed emergence, crops establish strong root systems and reduce the chance of weeds becoming established.** Conversely, avoiding irrigation right before or during peak weed germination can limit weed growth.

Combining water management with mulching techniques can provide additional weed control benefits. Mulch reduces soil evaporation, moderates temperature, and provides a physical barrier for weed seeds.

In some natural farming systems, **controlled wetland or shallow water systems can be used to control invasive weeds.** Weeds that thrive in wet conditions, such as **water hyacinth** or **duckweed**, can be controlled by regulating water levels. Creating an environment that supports crops while limiting weeds' ability to grow can improve both yield and weed control.

6. Weeds often spread through contaminated seeds. By using weed-free seeds, farmers reduce the risk of introducing new weed species to their fields, which can otherwise become a major source of weed pressure and competition for resources.

The seed bank in the soil can be replenished with weed seeds, which may persist for years, making weed management harder in subsequent seasons.

When weed-free seeds are used, crops have a better chance to grow without competing for water, light, and nutrients with unwanted weeds. This can lead to improved crop yields and healthier plants, as they do not have to struggle with weed competition.

By consistently using weed-free seeds, farmers can prevent the establishment of new weed populations in their fields, ultimately reducing the weed seed bank in the soil and decreasing the need for frequent weeding.

iv. Protocol to be followed:

1. **Remove large weeds manually or through shallow tillage to prepare the field.** Crop residues like straw, husks, or leaves should ideally be by-products to minimize costs. **The**

material should decompose naturally, enriching the soil with organic matter over time. Examples: Legumes (e.g., cowpea, mung bean), cereal residues (e.g., wheat or rice straw). Crops with dense residues that effectively block sunlight and inhibit weed growth. Example: Maize stalks or sugarcane leaves.

Residues should not harbor pests or pathogens that could infect crops. **Avoid using diseased crop residues. Ensure the selected crop residues do not release toxic compounds** that could hinder the growth of the main crop. Example: Avoid walnut or eucalyptus leaves known for allelopathy.

Crops producing substantial biomass are ideal for thick and uniform mulch layers. Examples: Sunflower stalks, sorghum, and pigeon pea residues. The crop residues should suit the local soil type and climatic conditions. Example: Grass clippings work well in tropical regions, while straw is effective in temperate climates. **Residues from nutrient-rich crops can enhance soil fertility as they decompose.** Example: Legume crops like groundnut or chickpea. Select crops that offer additional benefits, such as erosion control, moisture conservation, or fostering beneficial microorganisms. Example: Cover crops like clover or alfalfa.

Apply a 5–10 cm thick layer of mulch for effective weed suppression and moisture conservation. Adjust thickness based on the material; finer materials require a thicker layer. Leave a small gap (3–5 cm) around plant stems to prevent rotting and pest attacks. Spread mulch evenly across the field to avoid patchy coverage.

2. Ensure crops have **complementary growth habits** and resource requirements. Use appropriate row ratios (e.g., 2:1 or 4:1) to maximize coverage. **Synchronize sowing of mixed crops to establish a competitive advantage over weeds.** Observe fields during the early stages to address weed emergence promptly. By growing crops with varying root depths, canopy structures, and growth habits, weeds have less opportunity to establish and compete for space, light, and nutrients. **Growing a variety of crops attracts different pollinators, beneficial insects, and wildlife, which contribute to the health and balance of the farm ecosystem.**
3. **Understand soil type, climate, water availability,** and pest pressure in the region. Select crops with complementary growth patterns (e.g., tall and short, deep-rooted and shallow-

rooted). **Ensure crops have different nutrient, water, and sunlight needs to reduce direct competition among them and with weeds.**

Choose crops that have natural weed-suppressing properties (e.g., mustard, sorghum, and sunflower) for their allelopathic effects. Design a crop rotation system that alternates between legumes (e.g., chickpeas, lentils) and cereals (e.g., wheat, rice) to disrupt weed cycles and reduce weed seed buildup.

Use appropriate crop density (e.g., 2:1 or 4:1) to ensure optimal canopy formation and soil cover. Choose crops with staggered maturity periods to ensure that the soil remains covered with crops at different stages, limiting weed opportunities.

4. Choose cover crops that **suit the local climate, soil conditions, and specific weed control needs**. The cover crop can belong to the class of: **Legumes** (e.g., clover, vetch) are ideal for adding nitrogen to the soil while suppressing weeds through canopy growth. **Grasses** (e.g., rye, oats) are great for quick ground coverage and root competition, preventing weed growth. **Brassicas** (e.g., mustard, radish) can help suppress weeds via allelopathy (chemical release). **Broadleaf plants** (e.g., buckwheat) provide quick growth and smother weeds through dense foliage.

Ensure the cover crop is **planted at the right time** to maximize weed suppression. Ensure cover crops are **dense enough** to effectively suppress weeds. Manage the growth of cover crops to maximize their weed-suppressing potential. **Rotate different cover crop species to avoid weed species becoming adapted to a particular cover crop.**

5. **Use drip irrigation systems** to directly apply water to the root zone of crops, reducing the water supply to surrounding areas where weeds may grow.

Water crops according to their specific needs at different growth stages, ensuring that the soil is not overly saturated, which can encourage weed germination.

Use soil moisture sensors or a simple **feel test to determine the moisture content in the soil** and adjust irrigation accordingly. Maintaining optimal soil moisture helps crops while minimizing conditions favorable for weed germination.

Apply water in small, frequent amounts to maintain consistent moisture in the crop root zone without promoting excessive moisture on the soil surface, which can favor weed seed germination.

Ensure that the soil drains well and doesn't become waterlogged, as stagnant water encourages weed growth and may harm crops.

Schedule irrigation to limit moisture availability for weeds, especially during their germination phase.

Avoid irrigating immediately after weeding or tilling, as this could provide ideal conditions for weed seeds to germinate. Allow the soil to dry slightly before irrigation to prevent a flush of new weed growth.

Weeds generally grow most vigorously during warm and moist conditions. Avoid excessive irrigation during this period to limit water availability to weeds.

If possible, use rainwater harvesting systems to supplement irrigation, reducing the need for water that may contribute to runoff and weed spread.

Before irrigating, manually remove or mulch weeds, as their competition for water will be higher once irrigation takes place. This ensures the crop benefits from water without excessive weed growth.

6. Choose **good quality indigenous seeds for sowing**. Before planting, carefully inspect the seeds for visible weed seeds or contaminants. If needed, **clean the seeds manually using techniques** such as:

Sieving: Use fine sieves to remove small weed seeds and debris.

Air Screening: Blow air over seeds to remove lighter weed seeds and chaff.

Washing: Wash the seeds to remove dirt and contaminants.

Store seeds in clean, dry, and cool conditions to avoid re-contamination. Ensure that storage containers are tightly sealed to prevent exposure to external contaminants.

Store seeds in **airtight containers to prevent contamination** from external sources. **Keep seeds in a cool, dry, and dark place to maintain seed quality** and reduce the risk

of contamination by mold, pests, or weeds. If storing large quantities, ensure that seed storage areas are cleaned regularly to avoid the buildup of debris or potential weed seeds.

Ensure that the **soil and growing environment are free from weed seeds**. Remove any visible weeds in the soil using manual weeding or mechanical methods (such as tilling or hoeing) before planting.

By ensuring that seeds are free from weeds, practicing proper soil management, and using complementary techniques like mulching and crop rotation, farmers can effectively manage weed populations

E. Seed Selection, Treatment and sowing practices

i. Name of Practice:

1. Use indigenous seeds
2. Selection of healthy plants
3. Mature seed harvesting
4. Lunar calendar practices

ii. Brief Description of the practice:

1. Indigenous seeds refer to locally adapted, traditional varieties of seeds that have been passed down through generations and are well-suited to the local environmental conditions, soil types, and farming practices. In natural farming, the use of indigenous seeds is a fundamental practice for promoting sustainability, biodiversity, and resilience in agricultural ecosystems. These seeds are typically non-GMO, open-pollinated, and exhibit strong natural traits such as resistance to local pests, diseases, and climate stressors.
2. Seeds were collected from the healthiest and most productive plants. Farmers believed that seeds from robust plants would ensure a better yield.
3. Seeds were harvested only when fully mature to ensure high germination rates. The moisture content should be optimum will storage and it will affect the viability and storage quality of the seed.
4. The **Lunar Calendar** has been traditionally used in various cultures around the world to guide agricultural practices, including seed selection, planting, and harvesting. This system is based on the phases of the moon, which influence various natural processes such as soil moisture, plant growth, and pest behavior. In natural farming, lunar calendar practices are often used to determine the most auspicious times for seed selection, sowing, and other important farming activities to promote healthy crop growth and improve yields.

iii. Benefits of the practices:

1. The use of indigenous seeds is a cornerstone practice in natural farming that emphasizes sustainability, biodiversity, and resilience. Indigenous seeds, often referred to as traditional or native seeds, are those that have been cultivated and adapted to a particular region over generations. They are well-suited to local environmental conditions and play a critical role in ensuring ecological balance and reducing dependency on commercial seed systems.
2. In natural farming, **selecting seeds from healthy plants** is a fundamental practice that ensures the quality, viability, and resilience of the seeds used in subsequent planting cycles. **This approach emphasizes the importance of choosing plants that have demonstrated robust growth, resistance to pests and diseases, and adaptation to the local environment.** By prioritizing healthy plants, farmers can maintain biodiversity, improve crop productivity, and align with the principles of sustainability in natural farming.

Plants that exhibit strong, uniform growth with sturdy stems, healthy foliage, and an absence of deformities.

Selecting seeds from high-yielding plants enhances productivity in subsequent planting cycles.

Local adaptation ensures that the seeds are more resilient to environmental stressors like drought, heat, or flooding.

3. Harvesting mature seeds is a crucial practice in natural farming that ensures the selection of high-quality, viable seeds for the next planting cycle. These seeds are better suited for the local environment and help build a more sustainable farming system.

Mature seeds have a higher germination rate compared to immature seeds. This is because they are fully developed and stored the necessary nutrients required for successful sprouting and initial growth.

Seeds harvested at the right maturity stage tend to produce healthier, more robust plants. These plants are better equipped to withstand environmental stresses like drought, pests, and diseases.

Harvesting seeds from plants that have thrived in the local environment ensures that the seeds are well-adapted to the soil, climate, and growing conditions

By selecting seeds from a wide variety of plants and avoiding hybrid varieties, mature seed harvesting helps preserve genetic diversity within the farming system.

Mature seeds are less prone to spoilage and have a longer shelf life than immature seeds. They are also more stable in terms of moisture content, making them easier to store.

4. During the new moon phase:

It is believed that the energy of the moon is inward and focused, which can promote seed germination and root growth. This phase is considered ideal for selecting and preparing seeds for sowing. Seeds are often soaked or treated during this time to enhance their viability. Some farmers believe that seeds planted at this time will grow stronger, as the moon's influence on the soil encourages deep root development.

The waxing moon occurs after the new moon:

The moon increases in size toward the full moon. The waxing moon is associated with increasing light, which is **believed to stimulate growth and development of above-ground plant parts, such as stems, leaves, and flowers.**

This phase is considered the **best time for planting seeds that produce above-ground crops**, such as leafy vegetables and fruits. **It is believed that planting during this phase helps the plants establish stronger above-ground growth. Additionally, seeds that are more sensitive to light or heat may be chosen for planting during this time.**

The full moon occurs when the moon is fully visible.

The full moon phase is associated with a peak in energy and light. It is believed that the full moon **encourages plant maturation and seed setting.**

This phase is considered ideal for harvesting mature crops that are ready for seed saving. Seeds collected during the full moon are believed to be more potent and have better germination rates. Additionally, this phase is associated with the collection of medicinal herbs and crops that need to be harvested at their peak vitality.

The waning moon occurs after the full moon, as the moon decreases in size.

Impact on Seed Selection: The waning moon is **believed to slow down plant growth and encourage the focus to shift below ground, improving root health and soil fertility**. This is a good time for seed selection that requires less above-ground growth, such as root vegetables (carrots, potatoes, etc.) or perennials. It is also a phase for maintaining the soil, applying compost, and ensuring that the land is in good condition for future planting. The waning moon is not typically considered an ideal time for planting seeds, but it's used for other important agricultural activities like weeding, soil preparation, and harvesting crops that are meant to be stored.

iv. Protocol to be followed:

- 1. Obtain seeds from community seed banks, traditional farmers, or indigenous knowledge holders. Use seeds saved from previous harvests** of traditional varieties grown on the farm. Prevent cross-pollination by isolating indigenous varieties from other crops or hybrids during flowering. Regularly monitor fields to rogue out plants that do not conform to the desired characteristics. **Dry seeds thoroughly after harvest to reduce moisture content** and prevent spoilage. **Store seeds in cool, dark, and pest-free conditions, using breathable containers such as cloth bags or clay pots.** Conduct germination tests before planting to ensure seed viability. This can be done by sprouting a small sample in moist conditions and observing the germination rate.
- 2. Choose fields practicing natural farming** methods to ensure the seeds are free from synthetic chemical contamination. **Ensure the crop is grown under optimal natural farming practices**, such as using Jeevamrut, mulching, and crop diversity. **Choose plants that exhibit robust growth, strong stems, and healthy foliage. Avoid plants with signs of stunted growth or pest and disease damage.** Choose plants with the highest yield potential. Reject plants showing symptoms of diseases, pest infestations, or physical injuries. **Ensure that selected plants have not required external pest control measures**, highlighting their natural resistance. Collect seeds at full maturity when the plant has dried or ripened naturally. Use clean and dry tools for harvesting and processing. **Dry seeds thoroughly before storage to reduce moisture content and prevent spoilage.** Store seeds in a cool, dry place using natural storage methods, such as neem leaves or ash, to prevent pest infestations.

3. **Choose fields managed under natural farming practices**, free from chemical inputs, and with healthy, pest-free plants. **Prefer indigenous, locally adapted varieties** that align with the agro-climatic zone. **Reduced moisture content in seeds** (generally below 20%). Drying of the surrounding plant parts like leaves and pods. **Harvest during dry weather to minimize moisture absorption. Morning hours are preferred when seeds are not exposed to excessive heat but have started drying.** Use sickles or clippers to cut mature plants or pods. **Avoid shaking or rough handling to prevent seed shattering.** Hand-pick the best seeds from robust, high-yielding plants to ensure quality. **Spread seeds thinly on a clean surface (e.g., jute mats or bamboo trays) under shade to avoid direct sunlight**, which can reduce seed viability. **Ensure seeds reach the ideal moisture content for storage** (8-12% for most crops). **Remove chaff, dirt, and immature or damaged seeds using sieves or hand sorting.** **Discard damaged or discolored seeds to prevent contamination.** Use neem leaves, ash, or turmeric powder to deter pests and fungal growth.
4. **Refer to a reliable lunar calendar to identify key phases. Align seed selection activities with specific phases based on crop type (e.g., above-ground crops during waxing/full moon, root crops during waning moon).** Observe the crop during the chosen lunar phase for maturity indicators. Harvest seeds on days considered auspicious in the lunar calendar, avoiding days marked as inauspicious (e.g., eclipse periods). Dry seeds during the waning moon for optimal moisture reduction. **Avoid direct sunlight during drying, especially during the full moon, to maintain seed vitality.** Store seeds in natural materials like clay pots, neem-treated cloth bags, or bamboo containers. **Label seeds with the harvest date and lunar phase for future reference.** Test seed germination and vigor periodically to ensure quality.

Waxing Moon (Shukla Paksha): Select seeds from plants showing robust growth and productivity. Avoid seed selection during eclipses or adverse astrological timings.

F. Seed Treatment Practice

Name of Practice:

1. *Beejamrit*
2. Cow Urine (Gomutra)
3. Neem Leaf Extract
4. Turmeric Powder
5. Wood Ash Coating
6. Lime Treatment
7. Butter Milk
8. Salt Water Treatment

Brief description of the practice:

1. *Beejamrit* is a liquid formulation made from indigenous cow dung, cow urine, lime or quicklime, water and handful of arable field soil. It is one of the main components of natural farming used for the seed treatment.
2. Cow urine is an important component used in natural farming.
3. Neem is having high herbal and medicinal value in our ancient heritage of the India. Different products made from the neem leaves are used for making bio-formulation like Neemastra. When it comes to seed treatment, neem leaf powder or extract can also be used for treating the seed in natural farming practice. Seeds are treated with neem leaf paste or decoction for protection from soil borne pathogens.
4. Turmeric powder or paste have high herbal and medicinal value in our ancient heritage of the India. Different products made from the turmeric are used for giving seed treatment. Coating the seed with turmeric helps in protection from soil borne pathogens.
5. The ash collected from Agnihotra or simple ash for protecting the seed from the disease-causing microbes in the soil.
6. Lime (calcium hydroxide or calcium carbonate) is used for seed treatment in natural farming due to its beneficial properties that enhance seed health, germination, and early

seedling vigor. It is used to treat the seed to prevent it from seed borne disease. Lime is also a constituent of *Beejamrit* which is also used for seed treatment.

7. Buttermilk (a fermented dairy product) is a traditional and eco-friendly input used in natural farming for seed treatment. It is valued for its beneficial microbial content, nutrients, and ability to promote healthy seed germination and early plant growth.
8. Salt water is a simple, traditional method used in natural farming for seed treatment. It primarily serves as a **seed quality test** and as a **pre-sowing treatment** to enhance seed germination and protect against pests and diseases.

Benefits of the practices (impact)

1. Seed treatment with Beejamrit germinate quickly and abundantly. Roots grow rapidly. Moreover, the plant can thrive well against soil-borne diseases and flourishes very well. Beejamrit protects the crop from soil-borne and seed-borne pathogens and it improves seed germination also. Cow urine has anti-fungal properties and is also a good source of plant nutrients.
2. Cow urine has strong **antimicrobial properties, effectively eliminating seed-borne pathogens such as fungi, bacteria, and viruses. Reduces the risk of early-stage diseases and promotes healthier crop establishment.** The nutrients and enzymes in cow urine **stimulate seed germination**, leading to quicker and more uniform sprouting. Bioactive compounds in cow urine enhance the seeds' natural defense mechanisms, making plants more resistant to pests and diseases throughout their lifecycle. Cow urine encourages the growth of beneficial microorganisms in the seed's immediate environment. **These microbes help improve nutrient availability, such as nitrogen fixation and phosphorus solubilization, which support plant growth.**
3. **Neem leaves contain compounds like azadirachtin, nimbin, and nimbidin, which have strong antifungal and antibacterial properties.** These compounds effectively combat seed-borne pathogens, reducing the incidence of fungal and bacterial infections. Neem leaves repel pests like nematodes, termites, and other insects that attack seeds during germination. **The bitter taste and smell of neem deter insects, ensuring pest-free germination. Neem also prevents fungal growth during seed storage.**

4. **Turmeric contains curcumin, a bioactive compound with potent antifungal, antibacterial, and antiviral properties. It prevents seed-borne infections and reduces the risk of fungal diseases such as damping-off and root rot. Turmeric's strong aroma and bitter taste repel seed and soil-borne pests, preventing damage during germination and early growth.**
5. **Wood ash contains potassium salts and other alkaline compounds that act as a natural repellent against seed and soil-borne pests, such as ants, termites, and weevils. The ash coating creates a barrier that deters pests from damaging the seeds. The alkaline nature of wood ash inhibits the growth of fungi and bacteria on seeds, protecting them from seed-borne diseases like damping-off and root rot. The mineral content of wood ash, especially potassium, calcium, and trace elements, stimulates seed germination and improves early root development. Potassium is essential for seed vigor and germination efficiency.**
6. **Lime has strong antifungal and antibacterial properties, which help protect seeds from seed-borne pathogens. It reduces the incidence of diseases like seed rot, damping-off, and root rot. Lime is alkaline in nature and helps neutralize acidity in the seed's immediate environment. It creates favorable conditions for germination by reducing the effect of soil acidity, especially in acidic soils. Lime absorbs excess moisture from seeds, preventing fungal growth and spoilage during storage. This extends seed shelf life, especially in humid conditions.**
7. **Buttermilk is a natural source of lactic acid bacteria (LAB), which suppress harmful seed-borne pathogens such as fungi and bacteria. These beneficial microbes colonize the seed surface and act as a biological barrier against diseases like damping-off and root rot. The lactic acid in buttermilk creates an acidic environment, inhibiting the growth of harmful pathogens on the seed surface. This reduces the incidence of seed-borne fungal and bacterial diseases. Buttermilk contains essential nutrients like calcium, proteins, and vitamins that nourish the seed and promote better germination.**
8. **Salt water is used to separate viable seeds (good quality) from non-viable or damaged seeds. Viable seeds sink to the bottom, while non-viable, hollow, or infested seeds float due to their lower density. Salt water helps eliminate some seed-borne fungal spores, bacterial pathogens, and pests present on the seed surface. The treatment ensures that only**

healthy, viable seeds are selected for sowing, leading to improved germination rates and better crop establishment.

Protocol:

1. Mix 5 kg of indigenous cow dung, 5 liters of indigenous cow urine, 50 g lime or quicklime and one handful of arable field soil mix them in 20 liters of water. Mix all of these ingredients in water and keep them aside for 24 hours. Twice a day, stir the mixture with a wooden stick. After that, the seeds are treated with this mixture and dried in the shade before being planted.

Dipping normal seeds for 6 – 7 hours and special seeds like bitter melon (Karela) and Ivy melon (Tinda) for 12 – 14 hours in Beejamrit helps with the process. After that, strain Beejamrit and dry seeds in shade and use them for planting.

2. Seeds are soaked in cow urine for 12-24 hours. Acts as a natural disinfectant and enhances seed vigor.
3. Take Fresh neem leaves: 200–500 grams (for 1 kg of seeds) and water as required to make a paste. Collect fresh neem leaves and wash them thoroughly to remove dirt. Crush or grind the leaves into a fine paste using a mortar and pestle or a grinder. Mix the paste with a small amount of water to achieve a spreadable consistency. Coat the seeds evenly with the neem paste, ensuring all seeds are covered. Allow the treated seeds to dry in shade for 2–3 hours before sowing.
4. Mix turmeric powder with seeds at a rate of 5–10 grams per kilogram of seeds. Coat the seeds evenly and allow them to dry in the shade before sowing. Prepare a paste using turmeric powder and a small amount of water. Apply the paste to the seeds and let them dry in the shade. Mix turmeric powder with crushed neem leaves or neem powder for dual-action protection against pathogens and pests. Dissolve 10 grams of turmeric powder in 1 liter of water. Soak the seeds for 2–3 hours, then dry them in the shade before planting.
5. Slightly moisten the seeds by sprinkling water, cow urine, or Beejamrit over them. Add fine wood ash to the moistened seeds. Mix thoroughly until the seeds are uniformly

coated with ash. Spread the treated seeds in a single layer and air-dry them in the shade for 2–3 hours. Once dry, sow the seeds or store them in a moisture-free environment.

6. Place the seeds in a clean, dry container. Sprinkle lime powder evenly over the seeds. Mix thoroughly until all seeds are uniformly coated. Allow the treated seeds to dry in shade before sowing.

For wet coating: Lime powder: 10–20 grams per kilogram of seeds. Water: Enough to moisten the seeds lightly. Slightly moisten the seeds using water or a natural adhesive like cow urine. Sprinkle lime powder over the moistened seeds and mix thoroughly. Dry the seeds in shade for 2–3 hours before planting.

7. Lightly coat the seeds with buttermilk to create a moist surface. Sprinkle wood ash, lime powder, or neem powder over the seeds for additional protection. Mix thoroughly and dry the seeds in shade before planting.
8. Mix 100 grams of salt per liter of water in a clean container. Stir well to dissolve the salt completely. Add the seeds to the prepared salt water solution. Stir gently to ensure all seeds are submerged. Allow the seeds to settle for 2–5 minutes: Viable Seeds: Sink to the bottom. Non-Viable Seeds: Float on the surface. Skim off the floating seeds and discard them. Collect the viable seeds from the bottom. Wash the selected seeds thoroughly with fresh water to remove excess salt. Air-dry the seeds in shade for 2–3 hours before sowing.

A. Cropping system practices

ii) Name of the practice:

A) Parallel Multiple cropping

- 1) Multilayer cropping
- 2) Mixed cropping
- 3) Intercropping
- 4) Relay cropping
- 5) Alley cropping

B) Sequential Multiple cropping

- 1) Sequential cropping
- 2) Double cropping
- 3) Triple cropping
- 4) Quadruple cropping

ii. Brief description of the practice:

A) Parallel multiple cropping

1. **Multilayer cropping** is a modern integrated cropping system where multiple crops are grown on a single field without relying on a single crop. In this system, symbiotic crops are intercropped or mixed between the main crops, considering factors such as **plant height, sunlight intensity, root area, and the symbiosis between different crops. This approach allows the ground to be fully covered, and when several crops are grown together, one crop benefits from the nutrients provided by the others.** Additionally, the inclusion of various layers and trees in the field creates a soil biome that enhances **carbon sequestration, improves water retention, increases nutrient availability, and boosts farmers' income year-round.** Trees also contribute to microclimate regulation, reducing temperature, and promoting rainfall.
2. **Mixed cropping** in natural farming is the practice of **growing two or more different crops simultaneously in the same field, without a fixed pattern, to enhance biodiversity and optimize land use.** This system leverages complementary relationships between crops, such as varying root depths, growth habits, or nutrient requirements, to improve soil fertility, reduce pest and disease pressure, and increase overall resilience. By diversifying crop species, mixed cropping helps in weed suppression, promotes efficient nutrient cycling, and reduces the dependency on synthetic inputs, all while supporting ecological balance and sustainability in natural farming.
3. **Intercropping** enhances productivity by planting **two or more crops together on the same plot, using row patterns like 1:1 or 1:2, where the primary crop is in the first row and subsidiary crops in subsequent rows.** It is especially useful for small farmers

relying on rain. The crops, though having different nutrient needs, are chosen to optimize resource use and prevent pest and disease spread. The subsidiary crop should grow faster and be harvested earlier, complementing the main crop's growth. Crops should have different root depths to avoid competition and agronomic practices for both should align. Erect crops can be paired with cover crops like pulses to reduce soil erosion, control weeds, and **minimize water loss**. Avoid crops with similar pest or disease and ensures planting and management is simple, efficient, and profitable for wider adoption.

4. **Relay cropping** is a unique cropping practice where a second crop is planted before the previous one has been harvested, allowing both crops to share a portion of the growing season. Examples of relay cropping include planting **rice (or wheat) alongside Black gram, onions, lady's fingers and maize simultaneously**. This approach reduces risks associated with relying solely on a single crop, as farmers can simultaneously grow multiple crops in the same field.
5. **Alley cropping** is an agroforestry practice where **trees or shrubs and agricultural crops are grown in alternating rows**. Trees are often pruned to minimize shading of crops. This method **supports nutrient cycling and erosion control**. Alley cropping replicates a savanna structure, with multiple canopy layers: an over story of nut trees, a mid-layer of fruiting trees and shrubs, and a groundcover of annual crops or perennial grasses. *e.g.* Growing of pineapple, sweet potato, black pepper, tapioca, turmeric, ginger etc. in coconut or arecanut.

B) Sequential Multiple cropping

1. **Sequence cropping** in this cropping system two or more crops are grown in sequence one after another on the same piece of land in a year.
2. **Double cropping** It is multiple cropping system in which two crops are grown in sequence on the same piece of land in a year. *e.g.* Black gram-Jowar, Black Gram-Wheat, Rice-Gram, Groundnut-Wheat *etc.*
3. **Triple cropping** It is the multiple cropping system in which three crops are grown in sequence on a same land in a year. It is possible when irrigation facilities are available throughout the year. *e.g.* Groundnut-Wheat-Okra, Rice-Wheat-G.nut, Jowar-Potato-Green gram, Soybean-Wheat-G.nut *etc.*

4. **Quadruple cropping** It is the multiple cropping system in which four crops are grown in a sequence on the same land in a year. It is possible under irrigated conditions throughout the year. *e.g.* Groundnut-Coriander-Wheat-Green gram, Soybean-Methi-Wheat-Green gram *etc.*

ii) Benefits of the practices (impacts):

1. The multilayer cropping system offers several benefits, including reduced **farming costs** and consistent income throughout the year. It allows for **higher production per unit area, optimizing land, water, and resource use**. The system also mitigates the risk of crop damage from climate change, **improves soil characteristics**, and **enhances soil moisture retention, which boosts nutrient utilization efficiency**. By maintaining environmental balance, the system reduces water evaporation, **saving up to 70% of water**. Additionally, the presence of more friendly insects decreases the impact of pests and diseases. As the land is **fully covered with crops, weed growth is minimized, leading to increased productivity**.
2. Mixed cropping the crops included in mixed cultivation system are inherently in tune with the local ecological and climatic system. The different crops are supportive of each other rather than competitive as the creepers of legumes use stems of grain plants as a natural support and replenishing nitrogen, while the grain roots grip the soil firmly, preventing soil erosion. The diverse root system holds the soil tightly and much efficiently along with **prevention of soil erosion, water loss and maintains the water holding capacity of soil particles with increasing nutrient quality of soil**. In these traditional cropping system, Simultaneous cultivation of multiple crops prevents not only the losses by pests but also shows effective for weed control.
3. Intercropping offers several advantages, including increased profit, as secondary crops can provide **higher yields and guarantee income even if the primary crop fails**. It ensures more efficient land use compared to monocropping by planting species between rows, reducing unused soil space. Intercropping also protects cash crops by deterring pests, attracting beneficial insects, and providing shade from excessive sunlight or wind, which reduces the need for chemical pesticides and lowers costs. Additionally, the roots of plants in alley intercropping help prevent soil erosion and crusting. Leguminous plants contribute **nitrogen to the soil and enhancing soil fertility**. This method also allows for

more efficient use of natural resources such as water and solar energy, benefiting secondary crops. Weed management is improved as beneficial plants occupy the spaces between rows, and the overall biodiversity and ecological stability are enhanced by the increased variety of agricultural species.

4. Relay cropping offers numerous benefits for natural farming, including improved soil quality, as it enhances soil fertility and structure by incorporating different crops that contribute unique nutrients. It also aids in pest and weed control by disrupting pest cycles and suppressing weed growth. By overlapping crop cycles, relay cropping ensures more efficient use of resources like water and nutrients. This method saves time and money by optimizing labor and reducing the need for additional inputs. It helps prevent soil degradation by maintaining ground cover and promoting soil health. Additionally, relay cropping can increase farm profitability by optimizing land use and boosting crop yields, addressing issues like resource inefficiency, timing conflicts, and soil degradation.
5. Alley cropping, which involves planting trees or shrubs alongside crops, offers several benefits. It improves soil structure and fertility by adding organic matter from the trees and shrubs, and helps reduce soil erosion, particularly in sloping areas, through the roots of these plants. This technique also enhances water quality by minimizing nitrogen leaching and controlling erosion. Alley cropping creates a favorable microclimate, increasing shade and reducing wind, which can boost crop performance. Additionally, it enhances soil fertility and controlling weeds. **The practice promotes biodiversity by providing habitats for wildlife, diversifies farm income through short-term crop sales and long-term tree products, and contributes to climate change adaptation and mitigation by improving environmental resilience.**

(iv) Protocols to be followed:

1. Multilayer cropping

The multilayer cropping system involves the classification of crops based on their sunlight intensity requirements in a multilayer cropping system.

- **Group 1** Crops belonging to this group can withstand the most intense heat of the solar energy **8000 to 12000 candle feet**, and during this whole time they keep the leaf openings open to make food so that more production can be taken. Fruit crops like **tamarind, mango, sapota, jamun, cashew, coconut and monocots crop like**

sugarcane, paddy, maize, millet and fodder crops are also included. Crops belonging to this group are long-term investments with high economic returns and contribute to carbon sequestration.

- **Group 2** Under this group **6,000 to 8,000 candle feet** of solar energy keep the leaves open all the time and produce more food through photosynthesis. Fruits like **ber, guava, bitter gourd, etc.**, and all types of pulse crops are not exposed to severe heat but thrive in the sun with light shade. Cultivation and pruning are necessary to manage light to the lower levels and to maintain the desired shape and density of the tree. By reducing the intensity of sunlight, the lower level plants get moderate shade which is useful for their growth. Regular income can be earned by harvesting fruits in this level.
- **Group 3** Under this group, plants produce food with **5000 to 6000 candle feet** of solar energy. This group includes **medium-sized fruit trees and succulent crops**, which thrive in environments with moderate light intensity, benefiting from indirect sunlight to support their growth and productivity.
- **Group 4** Under this group **3700 to 5000 candle feet** of solar energy keep the leaves open all the time and allowing for increased food production through photosynthesis. *e.g. All tuber crops include turmeric, ginger, chillies and alavi etc.*
- **Group 5** In this group, plants in less than **3700 candle feet** of light keep their stomata open and make food. This includes **herb plants, cucurbitaceous crops**.

Thus, through this multilayer cropping system, the amount of sunlight required for photosynthesis is adequate for production. Therefore, optimizing the management practices outlined above will enhance natural farming. In this way, by combining the multilayer cropping system, agricultural production can be enhanced more efficiently. This method helps to make agriculture more efficient by making the best use of natural resources.

2. Mixed cropping

Crop selection: Choose a variety of crops with complementary growth habits, nutrient needs, and root depths. Ensure that these crops can co-exist without competing for the

same resources, thereby supporting biodiversity. Crop selection should follow the criteria below:

- If the main crop is a **monocot**, the mixed/allied crop should be a **dicot**.
- If the main crop is **deep rooted**, then **shallow rooted** mixed/allied crop should be selected.
- The growing duration of a mixed/allied crop should be **less than half or one-third of the main crop**. That means, it should be ready in less time than the main crop.
- The shadow of mixed/allied crops should not fall on the leaves of main crop.
- Mixed/allied crops should proliferate and **cover the soil** thoroughly like cucumber, water melon etc.
- If the leaves of the main crop can **tolerate intense sunlight**, then mixed/allied crops that **do not require intense sunlight should be planted**.
- If the primary crop grows **quickly**, select a mixed/allied crop that **grows slower than the main crop**.

By following these protocols, mixed cropping can enhance ecological balance, support sustainable farming, and reduce reliance on synthetic inputs, as outlined in the paragraph.

3. **Intercropping:** crop selection should focus on choosing varieties with complementary **growth habits, nutrient requirements, and root depths to ensure that the crops can coexist without competing for resources, thus supporting biodiversity**. The selection should follow these criteria: (i) If the main crop is a monocot, the mixed or allied crop should be a dicot. (ii) If the main crop is deep-rooted, a shallow-rooted mixed/allied crop should be chosen. (iii) The growing duration of the mixed/allied crop should be less than half or one-third of the main crop's growing period, ensuring it matures sooner. (iv) The shadow of the mixed/allied crops should not fall on the main crop's leaves. (v) Mixed/allied crops should spread and cover the soil effectively, like cucumbers or watermelons. (vi) If the main crop can tolerate intense sunlight, plant mixed/allied crops that do not require strong sunlight. (vii) If the main crop grows quickly, select a slower-growing mixed/allied crop. By adhering to these guidelines, intercropping can enhance ecological balance, promote sustainable farming practices, and reduce reliance on synthetic inputs.
4. **Relay cropping** in relay cropping system should ensure optimal crop growth, minimal competition, and improved soil health. Start by selecting compatible crops with different

growth durations and ecological needs, so one crop can establish while the other is still growing. The main crop is planted first, followed by the relay crop, seeded before the main crop is fully harvested. The relay crop should complement the main crop's growth habits, such as pairing deep-rooted crops with shallow-rooted ones, to optimize nutrient and water use. Timing is crucial to avoid shading and competition for sunlight, with proper spacing for airflow and root expansion. Select relay crops that enrich the soil, like nitrogen-fixers, and diversify crop species to maintain biodiversity and reduce pest pressure. Monitor soil fertility and implement practices like mulching, composting, and crop rotation to enhance soil health. Regular monitoring of crop growth is essential to adjust the system for successful and sustainable relay cropping.

5. **Alley cropping** In alley cropping system should aim to optimize plant growth, **maintain soil health, and minimize resource competition**. This system involves **planting rows of trees or shrubs, often nitrogen-fixing, alongside crops, creating alleys for crop growth**. The trees provide **shade, windbreaks, and organic matter, while reducing soil erosion and improving the microclimate**. Choose tree species that complement the crops and are suited to the local environment, ensuring minimal competition for sunlight, water, and nutrients. Proper spacing of trees is crucial to allow crops to thrive. Pruning or thinning must be timed to prevent excessive shading and ensure adequate sunlight for crops. Leguminous crop should be used to enhance soil fertility through nitrogen fixation. Regular monitoring of plant health, soil fertility, and moisture levels is essential to ensure long-term productivity and ecological balance.

Water management in natural farming

I. Name of practices

1. Drip Irrigation
2. Micro-Sprinklers
3. Moisture Sensors

II. Brief description of the practices:

1. Drip irrigation is a water-efficient farming technique that delivers water directly to plant roots through a system of tubes, pipes and emitters. It minimizes water wastage by reducing evaporation and runoff, promoting healthier plants and conserving water. Drip irrigation in natural farming helps create a VAPSA condition by efficiently delivering water directly to plant roots, promoting healthy soil structure and moisture retention. It minimizes water wastage, supports microbial life and reduces stress on plants, fostering a balanced and thriving ecosystem in natural farming.
2. A micro-sprinkler is a small irrigation device that distributes water in fine droplets or a gentle spray over a targeted area, similar to natural rainfall. It is designed to cover a small to medium-sized area and is typically used in gardens, orchards or greenhouses. Micro-sprinklers provide efficient water distribution, reduce evaporation and ensure uniform irrigation, promoting healthy plant growth while conserving water. Micro-sprinklers in natural farming help create a VAPSA condition by evenly distributing water over the soil surface, fostering healthy plant growth and soil moisture retention. This gentle irrigation method promotes soil life, reduces water wastage and supports a balanced, thriving ecosystem, essential for sustainable farming.
3. Moisture sensors in natural farming help create a VAPSA condition by providing real-time data on soil moisture levels. This allows for precise irrigation, preventing overwatering or underwatering and maintaining optimal moisture for plant health. The sensors promote healthy soil structure, support microbial life and conserve water, enhancing overall farm sustainability.

III. Benefits of the practices

1. **VAPSA Condition:** Water management is indeed crucial in natural farming for creating a favourable microclimate and ensuring the health of crops, soil and the surrounding

ecosystem. **VAPSA** (Vapour-Aided Plant Soil Air) conditions is a sophisticated concept that works with the principles of capillary action to ensure plants receive adequate moisture without over-relying on irrigation systems.

2. **Capillary action** refers to the ability of water to move through the soil or any porous medium due to the forces of adhesion and cohesion. Water naturally rises through the small pores in the soil, moving from wetter areas to drier areas. This is especially important for plants, as capillary action ensures that the root zone remains moist and plants get water even when there are no immediate sources of irrigation.

Principles of VAPSA Condition in Water Management

In natural farming, **VAPSA** (Vapor-Aided Plant Soil Air) water management leverages capillary action to ensure plants get sufficient moisture while minimizing water wastage. Here's how it works:

- **Vapor (Humidity):** High humidity reduces evaporation, keeping moisture in the soil, which supports capillary rise and water retention.
- **Air (Soil Aeration):** Proper aeration prevents soil compaction, allowing water to move through pores and reach plant roots via capillary action.
- **Plant (Root System):** Healthy, deep-rooted plants access water from deeper soil layers, optimizing capillary water uptake.
- **Soil (Water-Holding Capacity):** Well-structured, organic-rich soil retains moisture, aiding capillary rise and reducing water loss.
- **Atmosphere (Climate Control):** Temperature and humidity management (through shade, windbreaks or mulching) prevent excessive evaporation and stabilize moisture levels.

Water Management Techniques to Support VAPSA Conditions

To enhance capillary action and ensure plants get enough water under **VAPSA** conditions, the following natural farming techniques can be applied: **Mulching**, using organic materials like straw, leaves and grass clippings, helps retain soil moisture by reducing evaporation and improving the soil's ability to hold water, which supports

capillary movement. **Cover cropping**, with plants such as legumes, clovers and grasses, prevents soil erosion, enhances organic matter content and maintains soil moisture, creating favourable conditions for capillary action. **Contour farming techniques** capture rainwater and slow runoff, allowing it to infiltrate the soil for capillary action. Swales also promote localized microclimates that aid in moisture retention. **Rainwater harvesting** captures water for irrigation during dry periods, ensuring consistent moisture levels for capillary rise. Encouraging **deep-rooted plants** enables them to access moisture from deeper soil layers, helping them withstand drought and utilize water through capillary action. **Agroforestry**, through planting trees and shrubs, stabilizes the microclimate by providing shade, reducing wind and increasing humidity, which improves water retention in the soil. Finally, **soil aeration practices**, such as minimal tillage and adding organic matter, maintain soil porosity, allowing water to move efficiently through the soil via capillary action.

IV. Protocols to be followed

Irrigation practices and water conservation strategies play a crucial role in supporting soil health by maintaining moisture levels that promote a healthy, active and balanced soil food web. For instance, effective water management can help farmers prevent soil degradation issues like waterlogging, salinization and surface crusting, which can occur due to the impact of overhead irrigation on bare soil. In natural farming, micro irrigation protocols focus on promoting sustainability and soil health. First, install micro-irrigation systems, like drip or micro-sprinklers, to ensure efficient water delivery directly to plant roots. Use organic mulches to minimize evaporation. Maintain proper soil structure and avoid over-irrigating, which can disrupt soil microbes. Regularly monitor moisture levels with sensors to adjust water usage based on plant needs. Ensure the water quality is free from chemicals and pollutants. Lastly, ensure the system is regularly maintained to prevent clogging and ensure long-term efficiency.

1. **Name of the State:** Himachal Pradesh

2. **Name of Crop:** Apple

3. **Natural Farming Practices:**

a. **Soil Health Management Practices:**

- i. **Name of the practice:** Application of *Ghanjeevamrit*, soil and foliar application of *jeevamrit*, intercropping, mulching.
- ii. **Brief description of the practice:**

✓ **Ghanjeevamrit application at the time of field/ basin preparation:**

- a. In rootstock plants: 200gm/plant
- b. In seedling plant: 400gm/plant
- c. Application at time of intercropping @10 q/ha

✓ **Jeevamrit drenching at an interval of 21 days:**

- d. In rootstock plants: 3L/plant
- e. In seedling plant: 5L/plant

✓ **Jeevamrit spraying at an interval of 15 days: 10L in 100L water**

Application of Ghanjeevamrit and jeevamrit in soil and foliar application of jeevamrit @ 10 % at 15 days interval provide congenial conditions/ culture for multiplication of beneficial microbes which make nutrient available to the plants. Using intercrop with atleast one leguminous crop fix atmospheric nitrogen. Use of mulch (dry/ live) conserve soil moisture and make congenial conditions for microbial and earthworm activities.

- iii. **Benefits of practices (Impacts):** All these practices improve nutrient availability, improve soil organic carbon and humus as well as soil microbial activities and soil enzymatic activities.
- iv. **Protocols to be followed:** Application of *Ghanjeevamrit* @ 10 q/ha, soil and foliar application of *jeevamrit* @ 10 % at 15 days interval, using intercrop with atleast one leguminous crop, use of mulch (dry/ Live).

b. **Nutrient Management Practices:**

- i. **Name of the practice:** Application of *Ghanjeevamrit*, soil and foliar application of *jeevamrit*, intercropping and mulching.
- ii. **Brief description of the practice:** Application of *Ghanjeevamrit* @ 10 q/ha and soil and foliar application of *jeevamrit* @ 10 % at 15 days interval provide congenial conditions/ culture for multiplication of beneficial microbes which make nutrient available to the plants. Using intercrop with atleast one leguminous crop fix atmospheric nitrogen. Use of mulch (dry/ live) conserve moisture and make congenial conditions for microbial and earthworm activities

- iii. **Benefits of practices (Impacts):** All these practices improve nutrient availability, improve soil organic carbon and humus as well as soil microbial activities and soil enzymatic activities.
 - iv. **Protocols to be followed:** Application of *Ghanjeevamrit* @ 10 q/ha, soil and foliar application of *jeevamrit* @ 10 % at 15 days interval, using intercrop with atleast one leguminous crop, use of mulch (dry/ live).
- c. **Pest and Disease Management Practices:**
- i. **Name of the practice:** Application of *beejamrit*, *agnister*, *brahmasteer*, *Neemasteer*/*Drakeaster*, *Dashparni ark*, 3-5 days butter milk, *saunthaster*.
 - ii. **Brief description of the practice:** Seed treatment of different intercrops with freshly prepared *beejamrit* @ 10-20 %.
- Foliar application of *agnister*, *brahmasteer*, *Neemasteer*, *Drakeaster*, *Dashparni ark* @ 3 %, 3-5 days butter milk, *saunthaster* @ 3-5% at different plant stages for the management of diseases and insect pests of apple:
- iii. **Benefits of practices (Impacts):** All these practices act as pest repellent and pathogen protectants. Other natural farming practices like intercropping act barrier for spread of pests and pathogens. Improvement in population of beneficial microbes result bicontrol of the pathogens. Mulching and bio-inoculants like Jeevamrit and Ghanjeevamrit create a protective barrier and suppress soil-borne pathogens, promoting soil health and plant growth.
 - iv. **Protocols to be followed:** Seed treatment with freshly prepared *beejamrit* @ 10-20 %. Foliar application of *agnister*, *brahmasteer*, *Neemasteer*, *Drakeaster*, *Dashparni ark* @ 3%, butter milk, *saunthaster* @ 3-5% at different plant stage as per detail given below for the management of diseases and insect pests of apple.

Sprayed natural preparations at 15-day intervals/ as and when required for management of diseases and insect pests

- ✓ **Sonthastra, khatti lassi, ramban rognashi, and jeevaamrit for fungal foliar diseases**
- ✓ **Darekastr, bhramastr, and agniast for insect pest management**
- ✓ **Dashparni ark for woolly apple woolly aphid control**

Foliar application

Jeevamrit :10L in 100L water

Khatti lassi (Butter Milk) :5L in 100L water

Ramban: 7L jeevamrit+ 3L lassi in 100L water

Sonthastr: No dilution

Darekastr: No dilution

Agniast: 5L in 100L water

Brahmastr: 5L in 100L water

Dashparni ark: 5L in 100L water

Spray schedule for the control of insect of apple

Tree Stage	Pests Targeted	Keetnaasi Astra (NF) sprayed for management of pests under NF	Keetnaasi Astr
Half-inch/ Green tip	Sanjose scale/mites		Neem oil (5lts)
Pink bud	Thrips		-
Petal fall	Sanjose scale/ Mite	Sanjose scale/ woolly apple aphid	Agniast/ Bhramastr/ Dashparni ark
Fruit Development (Walnut size)	Mite	Woolly apple aphid	Darekast/ Bhramastr/ Agniast/ Dashparni ark
Fruit development (20-25 days after 3 rd spray)	Mite	Woolly apple aphid	Bhramastr /Agniast
Fruit development (20 days after 4 th spray)	Mite	Defoliating beetles Woolly apple aphid	Darekast/ Dashparni ark
Pre harvest	Sanjose scale/ Aphid	-	-
Post harvest	Woolly apple aphid	Woolly apple aphid	Agniast

Spray schedule for the management of diseases of apple under NF

Tree Stage	Diseases Targeted	Rognashi Dawa for management of diseases
Green tip	Powdery mildew	Khatti lassi (Butter milk)
Pink bud	Powdery mildew	-
Petal fall/ Pea Stage	Powdery mildew, Alternaria leaf spot	Sonthastr / Khatti lassi/ Ramban rognashi dawa
Fruit Development (Walnut size)	Alternaria leaf spot/ pre mature leaf fall	Khatti lassi / Ramban rognashi dawa/ Sonthastr
Fruit development (20-25 days after 4 th spray)	Pre mature leaf fall/ Alternaria leaf spot	Ramban rognashi dawa / Kandiastr
Fruit development (After 5 th spray)	Alternaria leaf spot/ premature leaf fall	Khatti lassi / Ramban rognashi dawa
Pre harvest	Sooty blotch/ fly speck/ Alternaria leaf spot/ Premature leaf fall	Kandiastr /Sonthastr
Post harvest	Cankers	Jeevamrit

d. Weed Management Practices:

- Name of Practice:** Mulching/ *Achhadan*, live mulch, intercropping, *ghanjeevamrit*.
- Brief description of the practice:** Mulching/ *Achhadan* with dry grass / crop residues in the tree basin, Sowing of different intercrops in between the plants as well as in between the rows, application of *ghanjeevamrit* @ 10q/ha.

- iii. **Benefits of practices (Impacts):** Soil surface remain covered during crop season and not allowed weed to germinate and etiolate. *Ghanjeevamrit* remain free from weed seed.
- iv. **Protocols to be followed:** Mulching/ *Achhadan* with dry grass / crop residues in the tree basin 2-3 times in a year, Sowing of different intercrops in between the plants as well as in between the rows, *ghanjeevamrit* @ 10q/ha.

e. Water Management Practices:

- i. **Name of Practice:** Mulching/ *Achhadan*, live mulch, intercropping, *Waaphasa*
- ii. **Brief description of the practice:** Mulching/ *Achhadan* with dry grass / crop residues just after crop emergence, Sowing of live mulch before transplanting, intercropping, Creation of *waaphasa* channels (1.5' × 1.5') outside the basin area.
- iii. **Benefits of practices (Impacts):** Soil surface remain covered during crop season and reduce evaporation and conserve soil moisture. *Waaphasa* channels creates congenial conditions / maintaining moisture at field capacity.
- iv. **Protocols to be followed:** Mulching/ *Achhadan* with dry grass / crop residues in the basin and *Waaphasa* channels, Sowing of live mulch in between the plants and plant rows, intercropping, Creation of *waaphasa* channels (1.5' × 1.5') outside the basin area of plants.

f. Any other practices being followed:

It is important to note that since Natural Farming is holistic farming system hence, use of all the components such as Beejamrit, ghanjeevamrit, jeevamrit, achhadan, waaphasa and intercropping along with concoctions for plant protections provide maximum benefits when applied in toto in a field.

Case studies of practicing natural farming farmers implementing these practices:

Subhash Shadru's case study

Subhash Shadru's journey is a testament to the transformative power of natural farming. Initially, transitioning to this sustainable approach presented challenges, but his dedication to understanding and implementing natural farming principles has yielded remarkable results. Since 2019, Shadru has diligently applied techniques like Beejamrit, Ghanjeevamrit, Jeevamrit, mulching, and waaphasa channels to his apple orchard. The outcome is evident: juicier apples with a richer flavor. Moreover, his expenses have significantly reduced, and the premium price for his naturally-grown apples has substantially boosted his overall profit. Shadru's success story has inspired fellow apple growers in Shimla to embrace natural farming. This shift towards sustainable agriculture not only benefits individual farmers but also contributes to a healthier environment and a more resilient food system.

1. **Name of the State** : Himachal Pradesh

The following are the recommendations of package of practices for vegetable crops under Natural Farming:

1. Tomato+ Frenchbean + Brinjal
2. Cucumber + Frenchbean + Okra
3. Pea + Spinach + Coriander
4. Cabbage + Fenugreek + Coriander

2. **Natural Farming Practices:**

a. **Soil Health Management Practices:**

- i. **Name of the practice:** Application of *Ghanjeevamrit*, soil and foliar application of *jeevamrit*, intercropping, mulching.
- ii. **Brief description of the practice:**
Application of *Ghanjeevamrit* @ 10 q/ha and soil and foliar application of *jeevamrit* @ 10 % at 15 days interval provide congenial conditions/ culture for multiplication of beneficial microbes which make nutrient available to the plants. Using intercrop with at least one leguminous crop fix atmospheric nitrogen. Use of mulch (dry/ live) conserve soil moisture and make congenial conditions for microbial and earthworm activities.
- iii. **Benefits of practices (Impacts):** All these practices improve nutrient availability, improve soil organic carbon and humus as well as soil microbial activities and soil enzymatic activities.
- iv. **Protocols to be followed:** Application of *Ghanjeevamrit* @ 10 q/ha, soil and foliar application of *jeevamrit* @ 10 % at 15 days interval, using intercrop with at least one leguminous crop, use of mulch (dry/ Live).

b. **Nutrient Management Practices:**

- i. **Name of the practice:** Application of *Ghanjeevamrit*, soil and foliar application of *jeevamrit*, intercropping and mulching.
- ii. **Brief description of the practice:** Application of *Ghanjeevamrit* @ 10 q/ha and soil and foliar application of *jeevamrit* @ 10 % at 15 days interval provide congenial conditions/

culture for multiplication of beneficial microbes which make nutrient available to the plants. Using intercrop with at least one leguminous crop fix atmospheric nitrogen. Use of mulch (dry/ live) conserve moisture and make congenial conditions for microbial and earthworm activities

- iii. **Benefits of practices (Impacts):** All these practices improve nutrient availability, improve soil organic carbon and humus as well as soil microbial activities and soil enzymatic activities.
- iv. **Protocols to be followed:** Application of *Ghanjeevamrit* @ 10 q/ha, soil and foliar application of *jeevamrit* @ 10 % at 15 days interval, using intercrop with atleast one leguminous crop, use of mulch (dry/ live).

c. **Pest and Disease Management Practices:**

- i. **Name of the practice:** Application of *beejamrit*, *agniaaster*, *brahmaster*, *Neemaster*/*Drakeaster*, *Dashparni ark*, sour butter milk, *saunthaster*.
- ii. **Brief description of the practice:** Seed treatment with freshly prepared *beejamrit* @ 10-20 %. Foliar application of *agnister*, *brahmaster*, *Neemaster*, *Drakeaster*, *Dashparni ark* @ 3 %, sour butter milk, *saunthaster* @ 3-5% alternatively at weekly intervals, just after 15 days of transplanting. In case of cucumber start spray of different asters at cotyledon stage.
- iii. **Benefits of practices (Impacts):** All these practices act as pest repellent and pathogen protectants. Other natural farming practices like intercropping act barrier for spread of pests and pathogens. Improvement in population of beneficial microbes result biocontrol of the pathogens. Mulching also creates barrier for soil borne pathogens.
- iv. **Protocols to be followed:** Seed treatment with freshly prepared *beejamrit* @ 10-20 %. Foliar application of *Agniaster*, *Brahmaster*, *Neemaster*, *Drakeaster*, *Dashparni ark* @ 3 %, sour butter milk, *saunthaster* @ 3-5% alternatively at weekly intervals, just after 15 days of transplanting. In cucumber at cotyledon stage. Incorporation of antagonistic crops should be done while selecting the intercrop.

d. **Weed Management Practices:**

- i. **Name of Practice:** Mulching/ *Achhadan*, live mulch, intercropping, *ghanjeevamrit*.

- ii. **Brief description of the practice:** Mulching/ *Achhadan* with dry grass / crop residues just after crop emergence, Sowing of live mulch before transplanting, intercropping, application of ghanjeevamrit @ 10q/ha.
 - iii. **Benefits of practices (Impacts):** Soil surface remain covered during crop season and not allowed weed to germinate and etiolate. *Ghanjeevamrit* remain free from weed seed.
 - iv. **Protocols to be followed:** Mulching/ *Achhadan* with dry grass / crop residues just after crop emergence, Sowing of live mulch before transplanting, intercropping, *ghanjeevamrit* @ 10q/ha.
- e. **Seed Selection, treatment and sowing practice:**
- i. **Name of Practice:** Use of Open Pollinated varieties in natural farming.
 - ii. **Brief description of the practice:** Seeds of OP varieties of the vegetables released and recommended by the University for the State are used for growing the vegetables under natural farming system.
 - iii. **Benefits of practices (Impacts):** All the varieties of vegetables are well adapted to this region and are less prone to various diseases and insect pests.
 - iv. **Protocols to be followed:** The seeds of all the vegetables are treated with freshly prepared *beejamrit* before sowing, dried in shade for 1-2 hours and sown in nursery or directly sown in the field at proper spacing as per the crops and cropping system.
- f. **Cropping Systems:**
- i. **Name of Practice:** Multiple/ intensive cropping system.
 - ii. **Brief description of the practice:** Two or more crops are grown on the same piece of land in succession within one calendar year. During the *Kharif* season the main crops are tomato, cucumber and capsicum and intercrops are French bean, brinjal and okra while in *Rabi* season the main crops are pea, cabbage and intercrops are fenugreek, coriander spinach or radish/turnip.
 - iii. **Benefits of practices (Impacts):** In multiple cropping systems we get maximum production of crops per unit area per unit time. There is also multiple use of resource. We can grow more number of crops within a year & more number of crops on the same piece of land at any given period. By growing the diverse crops there are less attack of diseases and insect pests.

- iv. **Protocols to be followed:** Main crops are sown/ transplanted in the rows and intercrops are sown in between the main crop rows by following the proper spacing as per the crops. All the crops are raised by following the natural farming practices.

g. **Water Management Practices:**

- i. **Name of Practice:** Mulching/ *Achhadan* , live mulch, intercropping, *Waaphasa*
- ii. **Brief description of the practice:** Mulching/ *Achhadan* with dry grass / crop residues just after crop emergence, Sowing of live mulch before transplanting, intercropping, Creation of *waaphasa* channels (1.5' × 1.5') at 1.2 m distance.
- iii. **Benefits of practices (Impacts):** Soil surface remain covered during crop season and reduce evaporation and conserve soil moisture. *Waaphasa* channels creates congenial conditions / maintaining moisture at field capacity.
- iv. **Protocols to be followed:** Mulching/ *Achhadan* with dry grass / crop residues just after crop emergence, Sowing of live mulch before transplanting, intercropping, Creation of *waaphasa* channels (1.5' × 1.5') at 1.2 m distance.

h. **Any other practices being followed:**

It is important to note that since Natural Farming is holistic farming system hence, use of all the components such as Beejamrit, ghanjeevamrit, jeevamrit, achhadan, waaphasa and intercropping along with concoctions for plant protections provide maximum benefits when applied in toto in a field.

3. **Case studies of practicing natural farming farmers implementing these practices:**

- Ms. Shushma Chauhan, a progressive natural farming farmer from Shimla District in Himachal Pradesh, has turned her 0.64-hectare plot into a natural farming system. She cultivates high-value crops such as Apple, Cherry, Pea, Plum, and Beans, all of her produce known for their superior natural farming quality and enjoys strong demand in the market. With her commitment to sustainable agriculture, she has earned a prestigious 3-star certification, highlighting the excellence of her farming practices.

Ms. Chauhan achieves an impressive annual income of approximately ₹3,32,720, while keeping her expenses as low as ₹22,123. Her success demonstrates how natural farming can offer both economic viability and environmental sustainability, serving as a powerful example for other farmers.

Name	Sushma Chauhan	 Multi-tier cropping at Sushama's farm
Agro Ecological Zone	High Hill, Temperate wet	
Major Crops	Apple, Cherry, Pea, Beans, Plum	
Area	8 Bigha = 0.64 hectares	
CETARA Rating	3 Star	
Cost of Cultivation	20,000 INR (Input cost only)	
Profit	3,00,000 INR	

- Mr. Kartar Singh, a progressive farmer from Bilaspur District in Himachal Pradesh, has embraced sustainable agricultural practices on his 1.2-hectare farm. He cultivates a diverse range of crops, including Wheat, Mango, Gram, Turmeric, Ginger, and Legumes, which are renowned for their superior quality and enjoy strong demand in the market. Recognized for his commitment to sustainability, Mr. Singh has earned a 3-star certification under a natural farming rating system. By employing eco-friendly methods, Mr. Singh achieves an impressive annual income of approximately ₹5,32,352, while maintaining minimal expenses of just ₹28,779. His success showcases the economic and environmental benefits of sustainable farming and serves as a model for others to follow.

Name	Kartar Singh	 Quality seed of local varieties
Agro Ecological Zone	Mid Hill, Sub Humid	
Major Crops	Wheat, Mango, Gram, Turmeric, Ginger, Taro root, Legumes	
Area	15 Bigha= 1.2 hectares	
CETARA Rating	3 Star	
Cost of Cultivation	26,000 INR (Input cost only)	
Profit	480000 INR	

- Mr. Vijay Singh, a progressive farmer from Kullu District in Himachal Pradesh, has revolutionized agriculture by adopting natural farming techniques. Managing a 2-hectare farm, he cultivates high-quality crops such as Apples, Maize, Kidney Beans, Pears, and Finger Millet. His superior produce has earned significant recognition and enjoys strong demand in the market. His commitment to

sustainable agriculture is further validated by a 3-star certification under the natural farming rating system, highlighting the excellence of his methods.

Mr. Singh achieves an impressive annual income of approximately ₹8,31,800, while keeping his expenses as low as ₹7,725, proving the financial viability of natural farming. His success serves as an inspiration for other farmers seeking to transition toward environmentally friendly and profitable agricultural practices.

Name	Vijay Singh	
Agro Ecological Zone	High Hill, Temperate wet	
Major Crops	Apple, Maize, Kidneybean, Pear, Finger Millet	
Area	25 Bigha = 2 hectare	
CETARA Rating	3 Star	
Cost of Cultivation	7,000 INR (Input cost only)	
Profit	7,50,000 INR	
		Crop intensification in Vijay's farm

- Mr. Jeet Negi, a farmer from Kinnaur District in Himachal Pradesh, has set an example in sustainable agriculture by adopting natural farming practices. On his 1.6-hectare farm, he cultivates high-quality crops, including apples, potatoes, peas, and maize, which are highly sought after in the market. His produce, known for its superior quality, has earned him a 3-star certification under a natural farming rating system, further validating his commitment to eco-friendly agriculture. Mr. Negi's dedication to natural farming has resulted in an impressive annual income of approximately ₹16,63,600, with operational expenses as low as ₹16,636, demonstrating how cost-effective and profitable this approach can be. His achievements inspire other farmers to adopt sustainable practices, contributing to ecological balance and better livelihoods.

Name	Jeet Negi	
Agro Ecological Zone	High Hill, Temperate Dry	
Major Crops	Apple, Potato, Pea, Maize,	
Area	20 Bigha= 1.6 hectare	
CETARA Rating	3 Star	
Cost of Cultivation	15,000 INR (Input cost only)	
Profit	15,00,000 INR	

Jeet Singh in his natural farm

Also, some research revealed that average soil organic content analyzed was found to be within the range of 1.5-2.6% which is higher from conventional farming (SOC 0.9-1.84 %). As a result, there is a significant increase in the presence of beneficial insects, pollinators, and other wildlife, contributing to ecosystem balance and resilience. NF leads to saving 46 % irrigation water. 23-27 % increase in farmers income. Reduce the cost in Apple (56.5%, n=110), Wheat (28.1%, n=254), Cereals and pulses (14.3%, n=325), Cereals, pulses & vegetables (26.46%, n=60) Fruit, pulses & vegetables: 45.5%, Vegetables and pulses: 20.4%. Increase in net returns of Apple (27.4%), Wheat (63.6%), Cereals and pulses (16.1%), Cereals, Pulses & vegetables (18.8%), Fruit, pulses & vegetables (21.5%), Vegetables and pulses (11.8%). Incidence of Scab and *Marssonina blotch* in Apple and Yellow rust in wheat was lower vis-a-vis conventional practices (n=110). Large crop diversity as revealed from case study different agro ecological zones is capable of fostering the various crops in the State. By avoiding the use of synthetic chemicals, natural farming reduces the presence of harmful residues in food products, promoting consumer health and well-being.

Template for sharing the natural farming package of practices

1. Name of the State: Himachal Pradesh, CSK HPKV, Palampur

2. Natural Farming practices: The natural farming practices and its concept are based upon the four principles/ components *i.e.* *Beejamrit*, *Jeevamrit*, *Achadana* (Mulching) and *Whapsa* (Moisture). These four components need to be applied in totality, and skipping any leads to considerable reduction in the crop yield.

a. Soil health management practices		
Name of the practice	:	Application of <i>Jeevamrit</i>, <i>Achadana</i> and <i>Whapsa</i>
Brief description of the practice	:	<i>Jeevamrit</i> is the efficient application of consortium of beneficial organisms and nutrients in growing crops to improve productivity. It is top dressed @ 10% at 14-days interval, till the grain filling stage. <i>Achadana</i> (Mulching) is one of the important components of natural farming. In this practice, the surface around the crop is covered with natural mulch mainly straw, dry grass, crop residues <i>etc.</i> It is applied @ 10 t/ha on dry weight basis. <i>Whapsa</i> (Moisture) is a practice of growing crops on raised bed not more than 1.5-2.0 m width and forming a channel of 10-15 cm deep, facilitates the availability of air and moisture to the roots as well as microorganisms living in the soil.
Benefits of the practice (Impacts)	:	Application of <i>Jeevamrit</i> increases soil fertility, activity of beneficial micro-organisms in the soil and crop yields. These microorganisms have antagonistic effects on pathogenic fungus, bacteria in the roots as well as foliage and also provide nutrition to the crop. <i>Achadana</i> (Mulching) is helpful in suppressing the weeds, better management of upper soil surface, addition of organic matter, moderation soil temperature & moisture and protect the crop against soil borne diseases. It creates the micro-climate, darkness and warmth in the soil under which micro-organisms can develop and proliferate. <i>Whapsa</i> (Moisture) favours better roots growth and ultimately efficient utilization of resources resulting in better crop growth & yield.
Protocols to be followed	:	<i>Jeevamrit</i> should be used within 7-8 days of its preparation and must be prepared from cow dung and cow urine. It should be sprayed at 14 days interval.
b. Nutrient management practices		
Name of the practice	:	Application of <i>Ghanjeevamrit</i> and <i>Jeevamrit</i>
Brief description of the practice	:	It is necessary to balance the amount of nutrients in the soil depending upon the crop growth stage. It is applied @ 500 kg/ha at sowing of crops. For meeting out the nutrient requirement of more exhaustive crops, <i>Ghanjeevamrit</i> must be applied @ 1000 kg/ha. Whereas, <i>Jeevamrit</i> should be sprayed at fortnightly interval.
Benefits of the practice (Impacts)	:	<i>Ghanjeevamrit</i> makes the soil fertile and increases the crop yield.
Protocols to be followed	:	It should not be older than six months and must be prepared from cow dung and cow urine.

c. Pest and Disease management practices		
Name of the practice	:	Application of Botanicals and Bio formulations
Brief description of the practice	:	To bar the menace of destructive insect-pests and diseases in different crops under natural farming system, the prophylactic sprays with the extracts of potential insecticidal plants and cow urine & dung based bio formulations at 10 days interval like <i>Fermented butter milk</i> , <i>Ram baan</i> , <i>Neemastra</i> , <i>Darekastra</i> , <i>Dashparni</i> etc. is mandatory
Benefits of the practice (Impacts)	:	These extracts and bio formulations are locally available, cost effective, residue free and eco-friendly
Protocols to be followed	:	These must be sprayed @ 200 l/acre as 10% prophylactic and in rotation right from the 15 days old crop till maturity
d. Weed management practices		
Name of the practice	:	Application of Achadana (Mulching) and Intercropping
Brief description of the practice	:	The surface around the crop is covered with natural mulch mainly straw, dry grass, crop residues etc. It is applied @ 10 t/ha on dry weight basis. On the other hand, the inclusion of legume crops as intercrop acts as complementary with SPNF inputs, in comparison to sole crop.
Benefits of the practice (Impacts)	:	It is helpful in suppressing the weeds, better management of upper soil surface, addition of organic matter, moderation soil temperature & moisture and protect the crop against soil borne diseases. It creates the micro-climate, darkness and warmth in the soil under which micro-organisms can develop and proliferate.
Protocols to be followed	:	Perform manual weeding at early stages of crop i.e. 15-30 days after sowing. Whereas, incorporation of legumes as intercrop suppress the weeds, improves the soil health and yield of the main crop.
e. Seed selection, treatment and sowing practices		
Name of the practice	:	Application of Beejamrit
Brief description of the practice	:	<i>Beejamrit</i> is an ancient amazing formulation & very common practice under modern natural farming. It is used for seed treatment, seedling and other planting material. Treat the dry seeds thoroughly with freshly prepared <i>Beejamrit</i> (24 hrs required for preparation) not older than 2-3 days applied @ 100-200 ml/kg seed. Treated seeds are air dried in the shade for 10 to 15 minutes.
Benefits of the practice (Impacts)	:	It protects the seed from seed/soil borne pathogens and soil borne insect pests, provides the strength to the root system and improves the seed germination viability. It cost very less and require little time for preparation.
Protocols to be followed	:	<i>Beejamrit</i> should be prepared fresh & need based and can be mixed with biofertilizers before use. It must be used within 2 days of its preparation.
f. Cropping systems		
Name of the practice	:	Intercropping and Crop rotation
Brief description of the practice	:	The inclusion of legume crops as intercrop acts as complementary with SPNF inputs, in comparison to sole crop.

		Crop rotation is the practice of growing different crops sequentially on the same plot to improve soil health, optimize nutrients in the soil, and combat pest and weed pressure.
Benefits of the practice (Impacts)	:	Intercropping improves the crop equivalent yield (CEY), fertility status and microbial properties of the soil. Whereas, crop rotation improves water use efficiency by increasing the amount of organic matter in the soil, which improve soil structure and water-holding capacity. It also helps in preserving moisture in deep soil layers, which plants can use during droughts.
Protocols to be followed	:	Suitable legumes should be planted following the additive series or replacement series.
g. Water management practices		
Name of the practice	:	Whapsa (Moisture)
Brief description of the practice	:	Growing crops on raised bed not more than 1.5-2.0 m width and forming a channel of 10-15 cm deep, facilitates the availability of air and moisture to the roots as well as microorganisms living in the soil. Besides, crop rotation also improves the water use efficiency by increasing the amount of organic matter in the soil, which improve soil structure and water-holding capacity.
Benefits of the practice (Impacts)	:	This condition favours better roots' growth and ultimately efficient utilization of resources resulting in better crop growth and yield.
Protocols to be followed	:	Care must be taken to drain out the excess water through constructed drains around raised bed in <i>kharif</i> season as plenty of rainfall is there.
h. Any other practices being followed Instead of single crop, suitable cropping system should be followed to increase the production per unit area and to improve the soil health.		

3. Case studies of practicing natural farming farmers implementing these practices: N.A

Natural Farming Package of Practices

University of Agricultural Sciences, GKVK, Bangalore

1.Name of the State : Karnataka

1a. Name of the University : University of Agricultural Sciences, GKVK, Bangalore

2. Natural Farming Practices :

a. Soil health management practices

i. Name of the Practice: Adoption of natural farming practices like zero tillage, mulching with crop residues, green manuring, green leaf manuring, intercropping and multilayer cropping (Five layer model)

ii.Brief Description of the Practice: Zero-tillage is practiced which avoids compaction of soil. In between the two rows of crops mulching is adopted with use of straw, crop residues or weeds. Mulching is practiced using 4 (four) tonnes of straw / crop residues / weeds per hectare. *In-situ* green manuring with sunhemp, sesbania, dhaincha, velvet bean, cowpea can be practiced using 10 kg seeds per acre. Green leaf manuring with use of leaves of Glyricidia, Simarouba, neem, pongamia etc., can be practiced. Growing of short duration pulse / legume crops like green gram, black gram, cowpea, field bean, French bean, peas as intercrops in cereals, oilseeds, vegetable and plantation crops like mango, pomegranate, guava, sapota, coconut, arecanut is practiced. Five layer model can be established with timber / fruit / field crops like Teak, Mahagani, Bamboo, Coconut, arecanut, Drum stick, banana, redgram, castor, field bean, cowpea.

This practice involves the use of natural farming systems, integrating traditional farming methods with scientific insights to maintain and enhance soil health. These approaches aim to enhance the organic carbon content, microbial population, and enzymatic activity in the soil while avoiding chemical inputs. The practice also incorporates sustainable techniques like intercropping, crop residue mulching, and minimizing deep ploughing to conserve soil structure and fertility.



Fig.1. *In-situ* green manuring with sunhemp, dhaincha in Mango orchards



Fig.2. *In-situ* green manuring with Velvet bean in coconutgardens

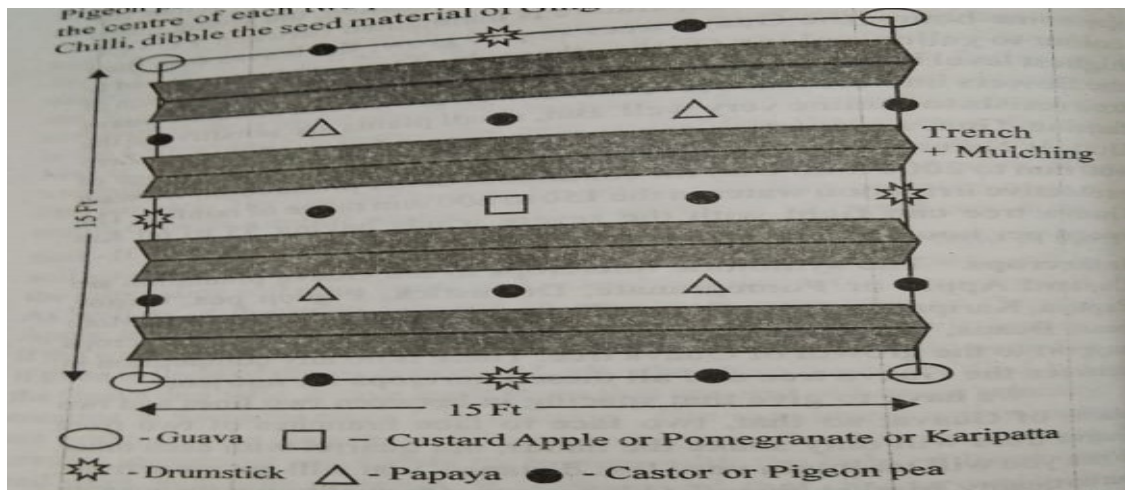


Fig.3. Five layer model in natural farming

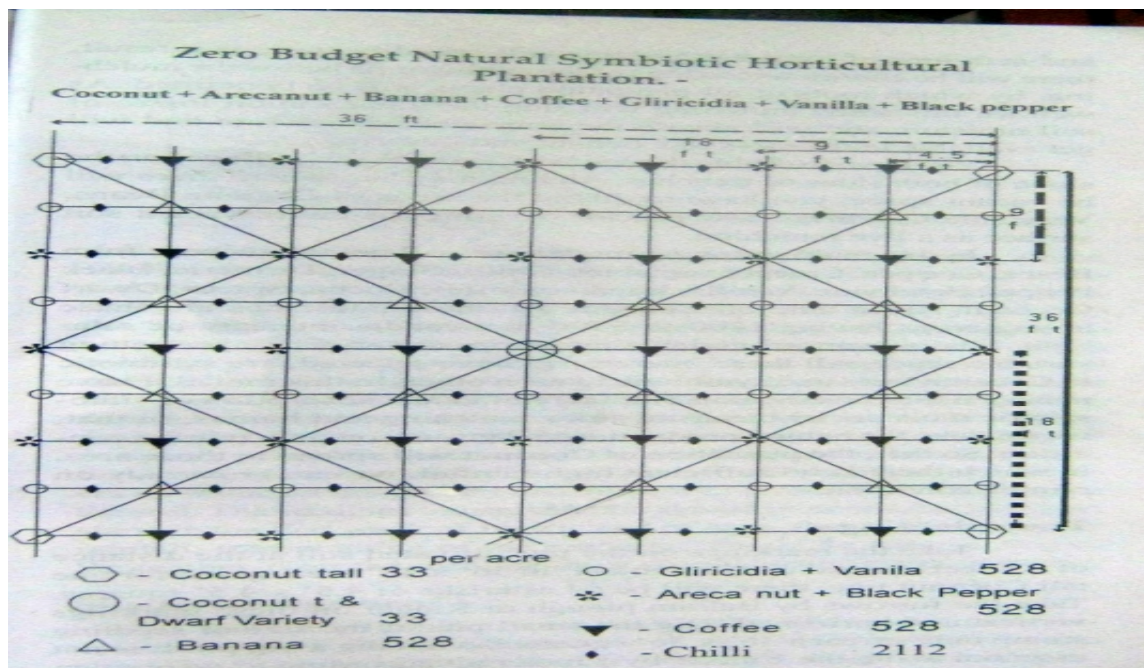


Fig.4. Five layer model in natural farming

iii. Benefits of the Practice (Impacts):

- Enhanced soil organic carbon content
- Improved nutrient availability
- Boosts soil microbial and enzymatic activity:
- Sustainability
- The practice minimizes environmental harm by avoidance of synthetic fertilizers and pesticides.

iv. Protocols to be followed:

1. **Soil testing:** Baseline soil testing need to be conducted to identify organic carbon levels, pH and nutrient deficiencies.
2. **Crop management:** Intercropping with legumes and other cover crops to enhance nitrogen fixation and reduce erosion.
3. **Residue management:** crop residue mulching to conserve moisture and increase soil organic matter.
4. **Monitoring and Evaluation:** Regularly monitoring of soil health parameters and microbial activity to assess the effectiveness of the practices.

This holistic approach ensures that soil health is maintained and enhanced, contributing to long-term agricultural sustainability and resilience against climate change

b. Nutrient management practices

i. Name of the practice : Jeevamrutha, Ghanajeevamrutha and intercropping practices

ii. Brief description of the practice : Jeevamrutha is a fermented microbial culture developed to enhance soil health by increasing the population of beneficial microorganisms. Regular application of Jeevamrutha promotes nutrient mineralization, ensuring a steady supply of nutrients to crops. Similarly, Ghanajeevamrutha not only enriches the soil with essential nutrients but also fosters the growth of soil microflora, improving overall soil fertility. Intercropping with pulses / legumes results in biological nitrogen fixation. Enormous amount of leaf litter or crop residues are added to the soils at the time of harvest of the intercrops which adds organic matter to the soil thereby improves organic carbon content in the soil.

iii. Benefits of the practices (Impacts) :

- The application of Jeevamrutha enhances soil microflora and improves nutrient availability to crops.
- Ghanajeevamrutha enriches the soil with nutrients and boosts soil health by enhancing both soil microflora and nutrient status.
- It aids in the decomposition of straw mulch used in natural farming, converting it into valuable nutrients.
- Jeevamrutha improves the mobilization and mineralization of nutrients in the soil.
- It promotes soil health by fostering the growth of both soil microflora and macroflora.
- Intercropping with pulses / legumes adds organic matter to the soil and improves the organic carbon content in the soil.

iv. Protocols to be followed :

Jeevamrutha: Add 500 liters of water in a barrel, add 25 kg fresh desi/local cow dung and 25 liters desi/local cow urine. Add 5 kg chemical free jaggery, 5 kg pulse flour and a two – three hand full of soil from the farm. Mix the ingredients well in shade with gunny bag cover, stir in morning, afternoon and evening using wooden pole in clock-wise direction for 7 days and apply Jeevamrutha to crops in between 5th to 11th day of preparation. Dosage of Jeevamrutha

prescribed is 500 litres of Jeevamrutha for one hectare of field crops once in every 15 days until harvest. For plantation crops like coconut / arecanut / mango / papaya / banana 500 litres of Jeevamrutha per hectare once in 30 days continuously.



Fig.5. Application of Jeevamrutha to crops

Ghanajeevamrutha: Mix 1000 kg of desi/local cow dung with 20-25 liters of desi/local cow urine, 5 kg of chemical-free jaggery, 5 kg of pulse flour, and a tow to three handful of soil from the farm. After thoroughly mixing, cover the mixture with a gunny bag and leave it overnight. The next day, shape the mixture into dung cakes or small balls and allow them to dry in the sun. Before application, grind the dried cakes into a powder and apply 1000 kg Ghanajeevamruth per hectare to the field 15 -20 days prior to sowing of field crops. For perennial plants, apply 1-2 kg per plant once in every six months. Once Ghanajeevamrutha is prepared, it can be stored for up to six months.

Intercropping : The practice of growing two or more crops at same time with definite or indefinite row proportion is called as intercropping. For intercropping short duration pulses / legumes like green gram, black gram, cowpea, field bean, horsegram, groundnut, soybean, French beans etc are sown inbetween two rows of main crops so that space inbetween the rows of main crop is very efficiently utilized.



Fig.6. Muching with pulse / legume crops for better nutrient fixation

C. Pest and disease management practices

i. Name of the practice: Pest and disease management by using agni astra, neemasthra, shunti astra, brahmastra and sour butter milk

ii. Brief description of the practice :

Seed treatment: Seed / seedling / set treatment with beejamrutha at 1 lit/ 10 kg of seeds.

Spraying: shuntiastra, neemasthra, agniasthra and brahmastraas prophylactic measures in anticipation of pest and disease incidence.

Cultural practices : Timely sowing, inter cropping, crop rotation, mulching, use of pest and disease free planting materials, use of pest and disease resistant varieties.

Biological practices : Use of light traps, yellow sticky traps, boarder crops, bird perches, owls, natural predators / enemies

iii. Benefits of the practices (Impacts) :

Helps in managing the pests and diseases, ecological approach, natural predators are encouraged, biological life chain is restored, access to chemical free food.

iv. Protocols to be followed :

Crop rotation: It involves growing different crops in the same area over multiple growing seasons. It's an effective way to control pests with a limited host range, long life cycle, and limited ability to move between fields.

Mixed cropping: Multiple cropping, also known as mixed cropping, is a method of growing different crops at the same time to help control pests and diseases

Use of disease free seed materials: Using disease-free seed or plant material can help eliminate some pathogens

Use of pest and disease resistant varieties to avoid the incidence

Regular monitoring for pest and disease incidence.

Preparation and use of astras: On farm preparation and use of need based astras viz., shunti astra, neemasthra, agniasthra, brahmastra and sour buttermilk.

Shunti astra:

Take 2 litres of water, add 200 grams of ginger powder and mix it and cover with a lid. Now boil it till it reduces to half of the solution. Keep this solution for cooling. Take 2 litre desi cow milk in another container and boil it slowly on low flame. After boiling milk, allow it to cool down,

remove the cream from the milk. Now take 200 litres of water, add a solution of ginger powder and milk without cream. Mix it properly and cover this solution with gunny bags for two hours. During this process ion exchange will occur, filter it with muslin cloth and spray this solution within 48 hours.

Neemastra (broad spectrum botanical pesticide)

Take neem leaves - 5 kg, desi cow urine - 5 litres, desi cow dung - 5kg, water - 100 litres. Take five kg of green leaves of neem or take five kg of dried fruits of neem and keep the leaves or fruits crushed. Add this crushed neem or fruit powder in 100 liters of water. Put 5 liters of cow urine in it and mix with five kg of cow dung. Stir it with wood and keep it covered for 48 hours. Dissolve thrice a day and after 48 hours filter the solution with a cloth. Now spray on the crops as prophylactic sprays.

Agniastra

Take 20 liters capacity earthen pot, add 10 litres of desi cow urine, add five kg of crushed green leaves of neem + 0.5 kg crushed green chilli + 0.5 kg crushed garlic. Now dissolve all this mixture in cow urine and boil it. After 3-4 boils, take it down from the fire. Let it cool for 48 hours and then filter the solution with a cloth. Now the solution is ready to spray on the crop. For insects living in tree trunks or stalks, all types of large bollworms and caterpillars' four litres of agniastra has to be dissolved in 200 litres of water and has to be sprayed at 20 ml agniastra per litre of water.

Brahmastra (broad spectrum botanical pesticide) for management of leaf roller, boll worm, pod borer and sucking pests :

Take 20 liters capacity earthen pot add 10 liters of cow urine + 3 kg of crushed green neem leaves + 2 kg crushed karanj leaves + 2 kg crushed custard apple leaves + 2 kg crushed papaya leaves + 2 kg crushed pomegranate leaves + 2 kg crushed guava leaves + 2 kg crushed *Lantana camera* weed leaves + 2 kg crushed *Datura stramonium* weed leaves . Now dissolve all this mixture in cow urine and boil it. After 3-4 boils, take it down from the fire. Let it cool for 48 hours and then filter the solution with a cloth. Now the solution is ready to spray on the crop. Four litres of brahmastra has to be dissolved in 200 litres of water and has to be sprayed at 20 ml brahmastra per litre of water.

d. Weed management practices

i. Name of the practice: Mulching (live mulching and straw mulching), passing cycle weeder and hand weeding

ii. Brief description of the practice : Mulching is an important practice in natural farming it involves covering the soil surface with decomposable materials or spreading type of leguminous crops as live mulch.

iii. Benefits of the practices (Impacts) :

- a. Straw mulch plays a crucial role in weed management in natural farming by preventing weed seed germination. Additionally, it reduces soil moisture evaporation and decomposes over time to enhance soil fertility.

- b. The spreading type of legumes effectively suppresses weed growth, while also contributing leaf litter that enriches the soil and boosts fertility.

iv. Protocols to be followed :

- a. Apply 4 tonnes of well-dried straw per hectare between the rows of field crops, 25–30 days after sowing. For perennial trees, use approximately 5 kg of straw as mulch around the base of each tree.
- b. Sow 7–10 kg per hectare of spreading leguminous crops such as cowpea, horse gram, or soybean between the rows of the main crop and around the base of trees. These crops produce abundant foliage, helping to suppress weeds and improve soil fertility.
- c. In light textured soils cycle weeder can be passed. Now – a days power operated / solar operated cycle weeders are available in the market and can be utilized for effective weed management followed by better aeration to crops.
- d. Age-old practice of hand weeding can be practiced for weed management and the uprooted weeds can be again used for mulching in between crop rows.



Fig.7. Mulching in basins to avoid weeds.



Fig.8. Live - mulching with pulses to avoid weeds.

e. Seed selection, treatment and sowing process

i. Name of the practice : Selection of local / desi varieties, Beejamrutha, use of seed drill for sowing, manual seed dibbling / transplanting

ii. Brief description of the practice : Farmers can select the bold, good filled, disease and pest-resistant, drought resistant seeds with their own expertise and experience. This helps in getting virulent and vigorous seedlings. Beejamrutha can be prepared at farmers level using desi cowdung and lime. Sowing is done using seed drill / dibbling of seeds / transplanting of seedlings.

iii. Benefits of the practices (Impacts):

- Selection of desi / local varieties ensures better response to applied jeevamrutha, ghanajeevamrutha.
- Seed treatment with beejamrutha ensures better germination, disease free, vigorous seedlings
- Manual dibbling of seeds / transplanting of seedlings ensures better germination, optimum plant population and optimum spacing for further growth and development of crops.
- Avoiding tractor for sowing avoids compaction of soil thereby soil microbes and structure of the soil is maintained.

iv. Protocols to be followed :

Beejamrutha : 5kg of desi cowdung is tied in cotton cloth and immersed in a bucket containing 20 litres of water with the help of stick. 50 grams of lime is dissolved in a mug of water. Next day in the morning the cowdung is squeezed along with cloth and is taken out. Water dissolved with lime is added to this and 5 litres of desi cowurine is added to this solution. One hand full of soil taken from field bunds is added to this solution. Now beejamrutha is ready. Seeds before sowing have to be immersed for one minute and then shade dried. Seedlings also has to be immersed for one minute before transplanting in the main field. In case of potato, sugarcane etc, the cut tubers can be immersed for one minute in beejamrutha and used for sowing.



Fig. 9. Beejamrutha preparation



Fig.10. Methodology of seed treatment with beejamrutha

f. Cropping systems

i. Name of the practice : Cereals - pulses, highly nutrient exhaustive crops – soil fertility restorative crops

Finger millet-field bean, Paddy –soybean, Pigeon pea+finger millet , Finger millet – cowpea, Paddy – green gram, sugarcane + soybean, cotton + field bean

ii. Brief description of the practice : Adopting a cereal-legume cropping system is highly recommended for enhancing soil health, optimizing nutrient utilization, and reducing the occurrence of weeds, pests and diseases. This system not only boosts economic returns but also promotes the growth of beneficial soil microflora, natural predators contributing to sustainable agricultural practices.

iii. Benefits of the practices (Impacts):

- Since legumes have the unique ability to fix atmospheric nitrogen through their symbiotic relationship with Rhizobium bacteria in their root nodules. This natural nitrogen enrichment reduces the dependence on chemical fertilizers and improves soil fertility for the subsequent cereal crop.
- The complementary nature of cereals and legumes often results in higher combined yields compared to cereal-cereal cropping system.
- Inclusion of pulses like cowpea, green gram, black gram, field bean in cropping system encourages the growth of natural enemies like lady bird beetles resulting in biological pest management.

- Growing of yellow flower crops like marigold, sesamum all along the borders will reduce the pest incidence on main crop, reduce the nematode problem in the soil.

iv. Protocols to be followed :

Cultivation of leguminous crop in the early monsoon seasons (May-June) and then sowing of cereal or millets in the monsoon season.

g. Water management practices

i. Name of the practice : Rainfed crop, trenches, mulching

ii. Brief description of the practice : Rainfed cropping, trench cultivation and mulching are integral practices in natural farming, aimed at ensuring sustainable agricultural productivity in water-scarce conditions. Rainfed cropping relies on rainfall, making it essential to adopt water-conservation methods to optimize moisture use. Protective irrigation is a strategic approach that involves providing minimal water during critical crop growth stages to prevent yield loss during dry spells. Mulching complements these practices by covering the soil with organic materials such as straw or live mulching, reducing moisture evaporation, suppressing weeds, and enhancing soil fertility. Together, these techniques create a resilient farming system that conserves resources, improves soil health and supports sustainable crop production.

iii. Benefits of the practices (Impacts) :

- Rainfed cropping maximizes the use of natural rainfall, reducing dependency on external water sources.
- Protective irrigation ensures minimal but critical water application during key growth stages, safeguarding crops during dry spells.
- Mulching suppresses weed growth and reduces soil evaporation, maintaining consistent soil moisture.

iv. Protocols to be followed : Sowing of the crop as a rainfed crop or sowing immediately after rainfall, followed by mulching which reduces the loss of soil moisture and protective irrigation to be provided during the dry spells or during critical stages of crop growth.

3. Case studies of practicing natural farming farmers implementing these practices

Age is just a number- innovative farmer adopts natural farming

Farmer: Somappa

Address: S/O Parvatappa Mandanahalli, Jayapura Taluk, Mysore, Karnataka

Contact no: 9900629858

Crops: Banana, vegetables: brinjal, tomato and long cowpea

Livestock-Desi cow-2



Mr. Somappa and his family depending on agriculture as their occupation. In his 4 acre ancestral land property he was growing finger millet and maize as major crops and adopted organic farming during 2009. He faced scarcity of capital, high chemical input cost, lack of storage facilities, scarcity of irrigation water, labour scarcity and high labour costs, unavailability of quality seeds, soil and water pollution due to fertilizers application. Realizing the ill-effects caused due to indiscriminate and unscientific use of pesticides he thought of shifting to natural farming. During 2019 he was selected as demonstrating farmer for natural farming project, UAS Bangalore. He undergone frequent training by university research staff on preparation of natural farming inputs under natural farming project. He got aware of natural farming and its importance for the sustainable agriculture, with the help of his son Mr. Swamy he started natural farming in his farm with a small piece of land and after knowing its importance he converted his land to grow banana, tomato and other vegetable under natural farming. He used to prepare natural farming inputs viz., beejamruta, ghanajeevamruta, jeevamruta and pest and disease management by preparing astras by his own. He sells his produce to big basket outlet which is situated in Mandanahalli, Jayapura, Mysore. His way of farming was impressed by his neighbouring farmers and they also started to follow natural farming by taking advice from Mr. Somappa.

Salient feature

- Providing information on natural farming and natural farming input preparation to neighbour farmers
- Master in conducting trainings and demonstrations on natural farming at village level for ZBNF farmers.

Major Interventions

1. Crop diversity banana, vegetables and pulses.
2. Cultivation of yelakki banana under natural farming practice
3. Inter crop with green manuring crops and *in-situ* mulching.
4. Adopted multiple cropping systems to generate additional income.
5. Natural inputs viz., beejamrutha, ghanajeevamrutha and jeevamrutha were used as nutrient supplements.
6. Adapted minimum farm mechanization
7. Astras viz., neemastra, shuntiastra and agniastra were used for management of pest and diseases

Practical utility:

1. Reduction in the labour requirement
2. Conservation of soil moisture, avoid erosion and weed management through mulching.
3. Limited intercultural operation
4. Reduced pesticide residue.
5. Input cost reduction by utilizing on farm resources.
6. Many farmers consulted for natural farming practice



Economics

Parameters	Natural farming (Ha ¹)	Traditional farming (Ha ¹)
Cost of cultivation (Rs.)	96732	172480
Production (qt)	116	160
Gross return (Rs.)	267121	369423
Net returns (Rs.)	170389	196943

Conclusion (after 4 years):

1. Reduction in cost of cultivation
2. Higher benefit cost ratio
3. Soil health is improved
4. Quality of the product is improved. Chemical free produce obtained.
5. Eco-friendly management of pest and diseases

Natural Farming -As Gracious As Jaggery

Farmer: Manju. M

Address: Hebbakavadi, Mandya, Karnataka

Crop: Sugarcane, paddy, banana and vegetables

Livestock: 2 Hallikar cows

Contact: 9008691194



Mr. Manju, M., basically came from farming family, his father and forefather were practicing farming as their occupation. His grandfather was growing crops only by using FYM and using green leaf *in - situ* mulching. But after the introduction of high yielding varieties they have also migrated to chemical farming. Initially they have got very good yield but Mr. Manju could see that there is depletion of nutrients and organic carbon after soil testing. He also faced lot of challenges while doing chemical farming viz., high chemical input cost for nutrient and weed management problems, labour scarcity and high labour cost and lack of knowledge about innovative practices.

Mr. Manju neighbour farmer Mr. Nandish, a software engineer was following Shri. Subash Palekar ZBNF practices and he forced Mr. Manju to switch over to natural farming. Earlier he refused to switch over to natural farming. During 2015 he attended one workshop on natural farming hosted by Shri. Subhash Palekar at Amritabhoomi, Chamarajanagar and



Karnataka. After the programme he got confidence to start natural farming with leafy vegetables in 100 sqmeter he got good harvest. He extended his land to grow sugarcane in 0.2ha with spacing of 8×8ft and with mixed cropping of vegetables and pulses in between the rows but he could not able to

continue ratoon crop. Later he planted sugarcane variety COVC 517 received from research station, Mandya. He got 18 quintals of jaggery from that harvest. He has also grown banana in 1 acre with spacing of 6×6ft in mixed cropping of vegetables and rotated by pulses. He is also growing paddy variety Siddasanna in 20 guntas. They have made a group along with other

farmers following natural farming and developed their marketing channel for selling their produce.

Practical utility of natural farming:

1. Mulching-reduces weeds and increases soil moisture availability
2. Dependence on chemical fertilizers and pesticides was reduced
3. Input cost reduction
4. Saving in labour cost
5. Effective use of locally available resources
6. Produce is free from chemical pollutants
7. Increased farm income

Description of innovative technology

1. Cultivation of high yielding sugarcane varieties
2. Proved natural farming yield with quality was equivalent to inorganic farming in sugarcane crop with modification in crop geometry.
3. Application of ghanajeevamrutha, jeevamrutha and growing *in - situ* green manuring crops for nutrient management.
4. Adopted cultivation of different crops (mixed cropping) in sugarcane for additional income.
5. Spraying of neemastra and sour butter milk for disease management.

Methodologies for transfer of technology

1. Using ICT mechanism (WhatsApp Group)
2. Participated in training programs and exposure visits
3. Participated in exhibition and discussion forums

Impact of natural farming

1. Reduced cost of cultivation
2. Higher farmers net profit
3. Improvement in the yield levels of natural farming
4. Gradual improvement in soil health status.
5. Eco friendly management of pest and disease

Performance of sugarcane under natural farming

Parameters	Natural farming (ha.)	Traditional (ha.)
Cost of cultivation (Rs.)	70000	80000
Production (t.)	90	125
Gross return (Rs.)	540000	472264

Integrated Farming System nourishes Natural Farming

Farmer: Chikkaswamy

Address: S/O Lt.Mallappa, Muddalli, Nanjangudu, Mysuru,
Karnataka

Contact no: 8722859420

Crops: Coconut, areca nut, banana, finger millet, guava,
coffee, nut mug, sugarcane, lemon, butter fruit etc.

Livestock-Desi cow, bandoor sheep, buffallo.



Awards;

- Shrestakrushika prashasti-2019-20 (ATMA)
- Taluk level shrestakrushikaprashasti (republic day)-2022-23
- District level pragathipararaitha prashasti-2023 (UAS Bangalore)

Mr.Chikkaswamy is from farming family, after his basic education SSLC in Vidyavardhaka High school, Nanjangudu, he started farming with his father. They were growing mono-cropping with traditional crops maize, Finger millet, jowar, groundnut under rainfed condition. To overcome this, his father digs an open well and started mulberry. He found it was also laborious and the uji fly on silkworm infestation for was severe. He decided to start Integrated farming system he introduced crop diversification with Coconut, Areca nut and Banana initially. During 2019 he was selected for Natural farming project, UAS Bangalore. He started to convert his farm into natural farming, He got trained by university research staff on preparation of natural farming inputs under Natural farming project.

Challenges faced

1. Uneven rain fall pattern
2. High chemical input cost
3. Degraded soil quality
4. Lack of knowledge on innovative practices
5. Labour scarcity and high labour cost.

6. Unavailability of quality seedlings.
7. Pest resurgence, pesticide resistance and pesticide residue problem.
8. Lack of storage and high fluctuation in marketing price

Practical utility:

1. Many farmers consulted on Natural farming practice.
2. Multiple cropping system generates additional income
3. Savings in labour requirements
4. Improved soil physical and biological properties
5. Input cost reduction

Salient feature

1. Innovative and implementation of new technologies to farm.
2. Master in conducting trainings and demonstrations on Natural farming and organic farming practices at village level for ZBNF farmers.
3. Awarded shrestakrushika award in taluk and district level by KSDA.

Description of Interventions

1. Adopted tissue culture banana cultivation through natural farming practice.
2. Inter cultivation with green manuring crops.
3. Adopted integrated farming system Bee keeping, 5 layer model etc
4. Natural inputs viz., Ghana Jeevamrutha and jeevamrutha in bio digester were used as nutrient supplements
5. Ashtras viz., Neemashtra, Shuntiashta and Agniashtra were used for management of pest and diseases



Methodologies for transfer of technologies

1. Participation in group discussions, meetings and method demonstration on Natural farming
2. Participation in training programmes, exhibition, and exposure visits on Natural farming
3. Beneficiary of natural farming project, UAS, Bangalore.
4. Using ICT mechanism (WhatsApp group, Television and Facebook)

Parameters	Natural farming (Ha¹)	Traditional farming (Ha¹)
Cost of cultivation (Rs.)	132099	185530
Production (qt)	128	153
Gross return (Rs.)	321237	383013
Net returns (Rs.)	189138	197484

Conclusion:

1. Reduction in cost of cultivation
2. Higher benefit cost ratio
3. Soil health in improved
4. Quality of the product is improved
5. Eco-friendly management of pest and diseases
6. Doubled the income by Natural farming

A Young Farmer Metamorphose Into Natural Farmer

Farmer: Shashikumar M

Address: Doddattuppur, Gundlupet, Chamarajanagara, Karnataka,

Crops: Finger millet, Vegetables and turmeric

Contact no: 9611491313

Land: 3 Acre



Mr.ShashiKumar was born and brought up from Doddaputtur village, which is located in Gundlupettaluk of Chamarajnagar district (Karnataka) is basically from a farming family. He completed his schooling from his village then he completed his 10+2 from Chamarajanagar. After completion of his graduation, he wanted to improve his farm. His father was doing conventional farming. After his father he took responsibility of the farm he was started integration of Organic farming in his farm. He found that organic farming require more labour, high input cost and require high cost of production, so he decided to shift towards other best farming system. After discussion with the family members and other friends he motivated since 2018 he has adopted natural farming and growing different crops. He owns 3 acres of land with one and a half acres of turmeric + small onion, 10 guntas of brinjal, 10 guntas of tomatoes, 10 guntas of fodder crops. He is using cow dung, cow urine, Jeevamruta, Neemastra, Gokripamruta water, sour buttermilk, coconut water as a source of nutrient for crops. He also set up a Farmer Produce Organization for direct marketing system of crops grown in small quantities. He also adopted multi-layer system of natural farming and adopted tree based agriculture in all parts of the farm (Growing 80 coconut, 30 sapota, 15 mango, 15 Guava, jammuneel, ramphal, jackfruit and other types of fruit plants) and he got an exposure through different training programmes through Organic Farming Training KVK, Nursery Worker Training, Dairy Training, Natural Farming Workshops held at AmritBhoomi, Value Based Products Training, CFTRI, Mysore. Many programs are organized and guided by KVK, Department of Agriculture, progressive farmers and he also participated in various awareness programmes under the ZBNF scheme, Karnataka and also he worked for oneyear at the taluk level for creating awareness among the farmers through agriculture department and he also giving suggestions to neighbouring farmers through different training programmes on disease and nutrient management of crops under

natural farming. He also serves as a resource person for the school children, departmental programs on adoption of natural farming and environmental protection programmes organised by Rural Education and Health Institute and he awarded taluk level "YuvaKrishika Award" at GKVK KrishiMela held in 2020



Practical utility:

1. Reduced cost of cultivation
2. Reduction on dependency of chemical fertilizers
3. Self-sustaining in availability of seed
4. Labour cost saving
5. Reduced pesticide residual effect

Salient features

1. Innovative and implementation of new technologies to farm.
2. Master in conducting trainings and demonstrations on Natural farming and organic farming practices at village level for ZBNF farmers.
3. Awarded shrestakrushika award in taluk and district level by KSDA.

Description of innovative technology

1. Adopted tissue culture banana cultivation through natural farming practice.
2. Inter cultivation with green manuring crops.
3. Adopted integrated farming system.
4. Natural inputs viz., Ghana

jeevamrutha, Jeevamrutha and Panchagavya were used as nutrient supplements



5. Ashtras viz., Neemashta, Shuntiashta and Agniashta were used for management of pest and diseases

Methodologies for transfer of technologies

1. Participation in group discussions, meetings and method demonstration on Natural farming
2. Participation in training programmes, exhibition, and exposure visits on Natural farming
3. Beneficiary of natural farming project.
4. Using ICT mechanism (WhatsApp group, Television and Facebook)

Impact

1. Protection and preservation of natural resource
2. Improved soil fertility by conserving soil moisture
3. Reduction in cost of cultivation.
4. Higher yield- Grain production and straw yield(
5. Increased income

Economics of Finger millet under different farming system

Parameters	Natural farming (acre)	Conventional farming (acre)
Cost of cultivation (Rs.)	10000	15000
Production (qt.)	10	13
Gross return (Rs.)	40000	37000
Net returns (Rs.)	30000	22000

Mushrooming sugarcane under natural farming

Farmer: Mr. Chandrashekar

Address: Chinnegowda, Sollepura, Mandya, Karnataka

Land: 3.5 acre

Crops/ Livestock: Paddy Sugarcane, Coconut, Banana and Vegetables

Livestock: 2-Amrutmahal, 2-Malnad gidda

Contact: 9844152845



Mr. Chandrashekar is from farming community and he completed his primary education in his native Sollepura and he completed his PUC from Govt. PU College Maddur, after completion of 10+2 he diverted himself to start working in his farm with his father and grandfather. He is having 3.5 acre of ancestral properties. Earlier he was growing local varieties Paddy Mysore mallige in his land. Once, he has got seedling from neighbour farmer and planted but even after 3 months there was no flowering took place, he visited to ZARS V C farm Mandya to discuss regarding the issue faced by him then he confirmed with scientist that it was seedlings of hybrid seeds of previous season from the neighbour farmer. He faced complete yield loss then he decided to stop using hybrid seeds and chemical fertilizers to his farm.

Earlier his father was using more chemical fertilizers and pesticides in his farm and the expenditure was high compare to income. In conventional agriculture high chemical input cost, Water scarcity during summer season, Labour scarcity and high labour cost. Degraded soil quality, Nutrient and weed management problems, Pest and disease management, Lack of knowledge about innovative practices. He also observed that after few years the soil fertility and organic carbon was getting depleted and the earth worm count was getting decreased. So he decided to shift from chemical farming to organic farming system. He started to read Natural farming books written by Padma Sri. Subhashpalekar after had discussion with friends he developed interest on Natural farming. During 2007 he attended training programme conducted by Shri Subhashpalekar on Zero Budget Natural Farming at Arsikere then he realised the importance of Natural farming in sustainable agriculture. He also attended another training programme conducted by SIT College Tumkur Karnataka on Natural farming by Subhashpalekar. He started to convert his farm into Natural farming. Meanwhile he found that the

soil organic carbon was started to recover. He purchases one desiHallikar cow he started to grow local paddy varieties viz., Rathnachooda, Rajamudi and siddasanna. He also planted coconut seedlings with 30*30 spacing and broad casted pulses seeds for nitrogen fixation and to avoid weeds. He makes trenches of 1.5 feet deep and he started fill up the trenches with crop debris and weeds which were in vegetative stage. Mulching was done by using weeds. He thought to expand his livestock and purchased desi cows such as Amritmahal, Malnadgidda and Gir cows. He also planted Banana with spacing of 8*8ft and between the rows he grown pulses, short duration vegetables and drum stick. He also growing Sugarcane varieties VSI8005 with spacing 10*4ft. annually he is producing 50-60 quintal of jaggery.

Major Interventions

- Multilayer farming system
- Application of Ghanajeevamritha, Jeevamrutha and Gokrupamrutha for nutrient management.
- Spraying of shuntiastra and sour butter milk for disease management.
- Maintenance of desi cow breeds.



Major Marketing developments.

- Developed own brand for Natural jaggery and Natural ghee.
- Supplying desi cow buttermilk, natural jaggery for other farmers.
- Mentoring natural farming and input preparation.

Natural farming - Sugarcane cultivation

Farmer: Mr. Ramesha

Address:Hebbakavadi,Mandya, Karnataka

Land: 3.5 acre

Crops/ Livestock:Paddy and Sugarcane

Livestock: 2-Amrutmahal

Contact: 9900365446



Major Interventions

1. Cultivation of high yielding local paddy varieties
2. Proved natural farming yield with quality equal to inorganic farming in paddy crop using naturally available resources
3. Application of Ghanajeevamrutha, jeevamrutha and growing in situ green manuring crops for nutrient management.
4. Adopted cultivation of different crops (mixed cropping) for additional income.
5. Spraying of jeevamrutha and sour butter milk for disease management.
6. Adopted application of azolla as organic manure.

Salient features

1. Mentor in natural farming and input preparation
2. Providing natural farming information to the farmers group
3. Providing natural farming information at panchayat level

Methodologies for transfer of technology

1. Using ICT mechanism (WhatsApp Group)
2. Participated in training programs and exposure visits
3. Participated in exhibition and forums

Challenges faced:

1. Lack of marketing facility for natural farming produce.
2. Nutrient and weed management problems.
3. High chemical input cost
4. Labour scarcity and high labour cost.
5. Lack of knowledge about innovative practices.

Practical utility:

1. Mulching-reduces weeds and increasing soil organic matter
2. Dependence on chemical fertilizers and pesticides was reduced
3. Input cost reduction
4. Saving in labour cost
5. Effective use of locally available resources
6. Free of chemical pollutants
7. Increased farm income

Performance of Sugarcane under natural farming

Parameters	Natural farming /ZBNF (ha.)	Traditional (ha.)
Cost of cultivation (Rs.)	70000	80000
Production (tonne)	90	125
Gross return (Rs.)	540000	472264
Net returns (Rs.)	470000	356250

Conclusions:

1. Reduced cost of cultivation
2. Higher farmers net profit
3. Improvement in the yield levels of natural farming
4. Gradual improvement in soil health status.
5. Eco friendly management of pest and disease

Natural Farming Package of Practices

Directorate of Extension Education, OUAT, Bhubaneswar

1. Name of the State : Odisha

2. Natural Farming practices

a. Soil health Management practices

- Name of the practices – Green manuring
 - Application of Jeevamrit
 - Crop rotation with legumes
- Brief description of the practices
 - Growing of green manure crops Dhaincha and incorporated in soil at 40 DAS.
 - Application of Jeevamrit @200 l per acre along with foliar spray @ 10 % solution
 - Cultivation of green gram after rice
- Benefits and detailed practices – It can improve the soil health by addition of organic matter to the soil. Application of jeevamrit can improve microbial activity in soil. Crop rotation with legumes restores the soil fertility through biological Nitrogen Fixation.

b. Nutrient Management practices

- Name of the practices
 - Application of FYM (On farm produce)
 - Application of organic matter through decomposition of organic mulch
 - Application of Jeevamrit
- Brief description of the practices
 - FYM is produced in the farm itself and applied @ 5t/ha during the final land preparation

- The organic mulching material like straw and leaves are added to the soil after decomposition and supplements to the soil nutrient pool
- Application of Jeevamrit @200 l per acre along with foliar spray @ 10 % solution
- Benefits and detailed practices – Organic source of nutrients is a very stable source and supply the nutrients to the plants as per its demand without any adverse effect on soil health.

c. Pest and Diseases Management practices

- Name of the practices
 - Application of Neemastra
 - Application of Brahmastra
 - Application of Sour butter milk
- Brief description of the practices
 - Application of Neemastra and Brahmastra @ 4% spray at 15 days interval as a preventive pest care against sucking pests, borer etc .
 - Sour butter milk is applied as foliar spray @ 3l/100 l water against fungal diseases
- Benefits – It is purely organic base . It has no adverse effect on environment. Low cost attributes to reduction in cost of cultivation.

d. Weed Management practices

- Name of the practices
 - Mulching
 - Manual Weeding
 - Cover crops
- Brief description of the practices-
 - Application of organic mulch suppress the weeds
 - Intercropping with short duration crop acts as cover crops and manage the weeds

- Need based manual weeding is done
- Benefits
 - Chemical free weed management practices
 - Low cost of weed management through cover crops /mulching

e. Seed selection and treatment

- Name of the practices
 - Quality seed
 - Seed treatment with Beejamrit
- Brief description of the practices
 - Quality seeds of preferably indigenous variety are opted
 - Seed treatment with Beejamrit @ 1 1/2 kg of seeds(Based on size of seeds, it is to ensure a thin coating of beejamirt over each seed)
- Benefits and detailed practices
 - Good germination
 - Better vigour and growth
 - Reduction in infestation of soil borne diseases

f. Cropping system followed

- Name of the practices
 - Kharif – Rice
 - Rabi- Sole greengram, Brinjal+ Cowpea, Cowpea+pointed gourd, Okra + Guar
- Benefits - Increased resource use efficiency with reduction in risk

g. Water Management practices

- Name of the practices
 - Alternate furrow irrigation
 - Sprinkler irrigation
 - Mulching
- Brief description of the practices with benefits

- Alternate furrow irrigation methods conserves irrigation water and maintains whapsa condition.
- Sprinkler irrigation methods also conserves water and energy
- Mulching helps in conservation of soil moisture and reduces the number of irrigation by increasing irrigation interval

h. Any other ITKs/practices followed

- For storing of seeds neem leaves and begunia leaves are kept along with the stored seeds
- Cow urine is sprayed on the leaves to avoid damage from monkey and bulls.
- Yellow coloured earthen pot smeared with castor oil is used as insect traps for sucking pests.

3. Case Study of practising natural farming farmer implementing these practices

1. Name of the Farmer: Mr. Parshuram Pradhan

2. Address with mobile number: S/o- Mr. Shibnarayan Pradhan,
At- Pateli (Gadataila), Po- Kumurisingha, Dist-Angul,
PIN- 759129. Mobile No-8637257950

3. Total Land area: 3.0 acre

4. Area under natural farming practice: 1.5 acre

5. Natural Farming practices

a. Soil health Management practices

Green manuring: Sowing of *Dhaincha* @ 30 kg/hain the rice field and incorporation at 6-7 weeks before transplanting rice crop which improves the soil health by enhancing organic matter content, soil structure and microbial activity. Application of 100 kg FYM mixed with 100 kg Ghanajeevamrit per acre before final ploughing helps plant growth by enriching the soil and providing nutrients.

b. Nutrient Management practices

Jeevamrit: Application of Jeevamrit @200 litre/acre in soil through irrigation water 3-4 times during the crop growth stages which is excellent source of natural carbon and biomass that contains macro and micro nutrients and enhances the nutrient availability.

c. Pest and Diseases Management practices

Neemastra & Agniastra: Spraying of Neemastra @ 5 litre/200 litre water and Agniastra @ 8 litre/200 litre water for controlling sucking pest and spraying of Sour butter milk @3 litre/100 litre water at weekly interval to check bacterial/fungal diseases.

d. Weed Management practices

Mulching: Crop residue mulches (dried leaves and straw etc) are used to cover soil for weed management which reduces weed growth, runoff, conserve moisture, prevent the soil against severe sunlight, cold, rain and saves seeds from birds, insects and animals and increases the soil microorganisms after incorporation in soil.

e. Seed selection and treatment

Beejamrit: Use of Local seeds and seed treatment with Beejamrit @2 litre/2-3 kg seeds to increase viability of seeds and prevents seed born diseases.

f. Cropping system followed

Intercropping: Intercropping with pulse improves soil fertility, ensures enough fodder stocks for cattle, enhances the income and biodiversity.

Rice - greengram/blackgram/groundnut, Maize + cowpea and Cowpea+marigold

g. Water Management practices

Ridge and furrow methods. Irrigation in Ridge and furrow method to develop soil microclimate on which soil organisms and roots depend for

most of their moisture and nutrients which Increases water availability, enhances water use efficiency and build resilience in plants against drought. To develop soil microclimate on which soil organisms and roots depend for most of their moisture and nutrients.

h. Any other ITKs/practices followed

Use of Handikhata for controlling various types of pest and diseases and Panchagavya 8 litre/200 litres of water at flowering stage for better growth and plant immunity

Natural Farming Package of Practices

Natural Farming (NF) is a chemical free farming, involving livestock integrated natural farming methods and diversified crop systems rooted in the Indian traditional knowledge. It is aimed towards improving soil health and reducing input cost to the farmer with greater climate resilience. NF recognises the natural interdependence of soil, microbiome, plants, animals, climate and human agricultural requirements.

Template for sharing the natural farming package of practices is given below.

1. Name of the State: Punjab

2. Natural Farming practices

a. Soil health management practices

i. Name of the practice: Natural farming involves four core practices: NFCs (*Beejamrit*, *Ghanjeevamrit* and *Jeevamrit*), crop residue mulching, intercropping and promoting soil aeration (*Whapasa*) are used for soil health management and pre-*kharif* green manuring with multiple crops.

ii. Brief description of the practice:

NFCs are microbial formulations used in natural farming to enhance soil health. *Beejamrit* treats seeds, while *Ghanjeevamrit* (1000 kg/ha) is applied as organic manure before sowing, and *Jeevamrit* (500 liters/ha) is applied with irrigation or as a foliar spray to boost microbial activity and nutrient availability.

Mulching with crop residues or paddy straw reduces erosion, moisture loss, and improves soil fertility. Intercropping with legumes enhances nitrogen fixation and supports crop growth.

Whapasa refers to maintaining the balance of air and water in soil pores, promoting healthy root development and nutrient uptake.

Pre-*kharif* green manuring with crops like pearl millet, sorghum and sunhemp improves soil fertility and structure through nitrogen fixation and organic matter addition.

Preparation procedure of NFCs

1. *Beejamrit*

Take 5 kg of indigenous cow dung (preferably fresh or not more than 3 days old) in a cloth and bound it by tape and hang this in the 20 liters of water for 12 hours. Take one liter water and add 50 g lime in it, keep for a night. Squeeze this bundle of the cow dung in water 3

times to extract material. Add the soil from undisturbed bunds or forest or under tree cover in the solution and stir it well. Add 5 liters of indigenous cow urine in the solution & add the lime water and stir it well. Beejamrit is ready to treat the seeds

2. *Jeevamrit*

In a plastic drum of 250 litres capacity, add 10 kg of fresh cow (indigenous) dung, 10 litres of indigenous cow urine, 2 kg of Jaggery, 2 kg of pulse flour (Basan, Chickpea flour) and 150 g of soil from undisturbed bunds or forest or under tree cover in 200 litres of water and mix thoroughly. Keep the drum in shade covering with gunny bag or cotton cloth or plastic mosquito net. Stir the mixture for 5-10 minutes for twice a day (morning and evening) with wooden stick. *Jeevamrit* is ready for application at 9th day and it can be applied up to 12th day from its preparation.

3. *Ghanjeevamrit*

Ghanjeevamrit was prepared by mixing 100 kg indigenous cow dung (4 to 5 days old), 1 kg chickpea flour, 1 kg jaggery, 3 litres cow urine and 250 g undisturbed virgin soil collected from canal bank. It was then kept for 10 days under shade away from direct sunlight and then applied in the field @ 1000 kg/ha before sowing.

iii. Benefits of the practices (Impacts):

Jeevamrit and *Ghanjeevamrit* enhance the availability of essential nutrients and support the growth of beneficial microorganisms, including bacteria, fungi and actinomycetes, in the soil. These microbial formulations improve soil fertility by increasing biological activity and mineralizing nutrients from unavailable to available forms. *Jeevamrit*, rich in plant growth hormones, stimulates microbial activity, leading to enhanced nutrient cycling and improved soil health when applied through irrigation water or foliar spray. *Ghanjeevamrit*, the solid form of *Jeevamrit*, provides similar benefits but in a more concentrated form, further promoting soil fertility and microbial diversity.

The retention of crop residues or the application of paddy straw as mulch offers several agronomic benefits. It helps protect the soil from erosion, reduces moisture loss through evaporation, and improves the organic matter content over time, contributing to better soil structure and enhanced fertility. This practice also fosters increased soil microbial activity, supporting long-term soil health.

Intercropping with legumes not only aids in nitrogen fixation, improving soil fertility, but also contributes to weed management by reducing weed competition. When legumes are

intercropped with cereal crops, nitrogen is directly transferred from the legume to the cereal, enhancing the nitrogen nutrition of the cereal crop. This synergistic relationship boosts crop yields and promotes sustainable farming practices.

Whapasa, the balance of air and water in the soil's pore space, is a key concept in sustainable agriculture. By optimizing this balance, crops benefit from better root respiration and nutrient uptake, reducing the risk of waterlogging and ensuring efficient water use.

Pre-kharif green manuring involves the use of a mixture of nine crops—pearl millet, sorghum, cowpea, greengram, marigold, clusterbean, sesame, sesbania and sunhemp. These crops are grown to enhance soil fertility through nitrogen fixation, organic matter addition, and the creation of a favorable soil environment for subsequent crops. Green manuring improves soil structure, microbial diversity, and nutrient availability, promoting sustainable agricultural practices.

iv. Protocols to be followed: Protocol prepared by ICAR-IIFSR, Modipuram, Meerut, UP under All India Network Programme on Organic Farming (AI-NPOF) for Natural Farming experiments.

b. Nutrient management practices

i. Name of the practice: Natural farming involves four core practices: NFCs (*Beejamrit*, *Ghanjeevamrit* and *Jeevamrit*) and intercropping are used for nutrient management in NF along with some other liquid organic manures like vermiwash, cowurine, cow dung cake extract etc.

ii. Brief description of the practice:

NFCs are microbial formulations primarily used in natural farming. *Ghanjeevamrit* is applied as an organic manure at a rate of 1000 kg/ha before sowing, while *Jeevamrit* is applied at 500 liters/ha either through irrigation water or as a foliar spray. Intercropping with legumes like cowpea enhance soil health through biological nitrogen fixation.

Vermiwash is a liquid organic manure derived from the excretory products and excess secretions of earthworms, enriched with micronutrients and organic molecules from the soil.

Cow dung cake extract used as foliar application for various agricultural purposes since ancient times, plays a crucial role in crop growth.

iii. Benefits of the practices (Impacts):

Jeevamrit and *Ghanjeevamrit* accelerate nutrient availability in the soil by supplying essential nutrients, plant growth hormones, and enhancing soil biological activity. *Ghanjeevamrit*, the solid form of *Jeevamrit*, provides similar benefits but in a concentrated, stable form. *Jeevamrit* promotes microbial activity, which helps in mineralizing nutrients from unavailable forms into available forms. When applied with irrigation water or through foliar spray, the consortium of beneficial microorganisms in *Jeevamrit* supports soil fertility improvement and nutrient mobilization.

Intercropping with legumes like cowpea enhance soil health through biological nitrogen fixation, which improves soil nutrient availability, reduces the need for synthetic fertilizers, and supports sustainable cropping systems. These integrated practices are effective components of conservation agriculture and agroecological farming systems, promoting both productivity and ecological balance.

Vermiwash contains water-soluble nutrients and provides immediate nutrient availability for crops. It also contains enzymes, earthworm secretions, soluble plant nutrients and organic acids that stimulate growth and improve yields. Crops treated with vermiwash exhibit enhanced resistance to insects, pests, and diseases, due to the presence of beneficial microorganisms that protect the plants from infestations and promote growth.

Cow dung cake extract has been used for centuries as a vital input in agriculture. It contains humic compounds, essential nutrients and bioelements that support crop growth. Cow dung cake extract is a rich source of bio-fertilizers and natural pest repellents, enhancing soil fertility and microbial populations, thereby improving soil health and boosting agricultural productivity.

iv. Protocols to be followed: Protocol prepared by ICAR-IIFSR, Modipuram, Meerut, UP under All India Network Programme on Organic Farming (AI-NPOF) for Natural Farming experiments.

c. Pest and Disease management practices

- i. Name of the practice:** *Beejamrit* protects seeds and seedlings from diseases. Biopesticides like *Neemastra*, *Agniastra*, *Brahmastra*, *Dashparni* and sour better milk used to manage pest and diseases. Okra and marigold are used as trap crops in natural farming.
- ii. Brief description of the practice:** *Beejamrit* is used to treat seeds, seedling roots, and other planting materials, protecting crops from soil- and seed-borne diseases that commonly

affect plant health. Biopesticides such as *Neemashtra*, *Agniastra*, *Brahmastra* and *Dashparni* are applied to manage insect pests effectively. Additionally, trap crops like okra and marigold are utilized in natural farming to attract pests away from the main crops, providing an integrated approach to pest management.

Preparations of Biopesticides

1. *Neemashtra*

Neemashtra is a neem and cow urine based liquid formulation. Take 5 kg of fresh neem leaves or 5 kg of neem seed kernels (3-8month old) crush the materials to make fine/small pieces. Mix the crushed leaves/kernels in 100 litres of water in a plastic drum. Mix 5 litres of indigenous cow urine and 1kg of indigenous cow dung. Mix the content thoroughly with the help of a wooden stick for 2-3 minutes. Cover the mouth of drum with a fine cotton cloth and incubate for 48 hours. Stir the content thrice a day for 2-3 minutes After 48 hours, filter the content with a fine mesh or cloth and spray on one acre crop.

2. *Agniastra*

Agniashttra is a botanical formulation prepared by using neem leaves, chilli fruits, garlic and cow urine. Take half kg each of green chilli and garlic and 5 kg of fresh neem leaves. Crush all the three materials to make a fine paste. Add crushed materials in 20 litres of indigenous cow urine and mix it thoroughly. Boil the content with intermittent stirring with wooden stick for about 20 minutes. Cool the content for about 48 hours. Filter the content with fine cotton cloth. Take 5-6 L of filtrate and dilute in 250 L of water for spraying on one ha crop.

3. *Brahmashttra*

It is a cow urine and botanicals-based formulation. Take 3kg of fresh neem leaves and 2 kg of karanj (*Pongamia pinnata*) leaves. If karanj leaves are not available, take 5 kg of neem leaves and crush then to fine pieces. Take 2 kg of custard apple leaves and 2 kg of Datura leaves and crush them to fine pieces. Mix all the above crushed leaves in 10 litres of indigenous cow urine. Boil the mixture for about 20-25 minutes. Cool the mixture for 48 hours. Filter the content with fine cotton cloth. Take 5-6 litres of filtrate and dilute in 250 litres of water for spraying on one ha.

4. *Dashparni*

Take 5kg of neem leaves and 2 kg leaves, each of any ten plant species. Take 10 litres of indigenous cow urine, 10kg indigenous cow dung, 500-gram turmeric powder, 500-gram garlic paste, 500 grams ginger paste, 1kg tobacco leaf powder, 1kg hot chilli paste. Crush

the leaves to fine pieces. Protocol for preparation of inputs for Natural Farming experiment under AI-NPOFMix all above ingredients in 200 litres water drum in shade. Shake the mixture thrice a day with the help of a wooden rod and ferment the content for 30-40 days. Filter the content with fine cotton cloth. Store the filtrate in containers and it can be used for 6 months.

iii. Benefits of the practices (Impacts):

Fresh *neemashtra* is sprayed on crops to control insect-pests like aphids, jassids, mealy bugs, thrips, whiteflies, small caterpillars and other sucking pests. For spraying, one hectare of crop, 250 litres of solution is required.

Agniashttra is used to manage stem borers, fruit borers and other different types of caterpillars of the crop. Five to six litres of *Agniashttra* is sufficient for spraying in one hectare of crop after dilution in 250 litres of water.

Brahmashttra is used to manage large size borers and caterpillars of the crops. Five to six litres of *brahmastra* is sufficient for spraying in one hectare of crop after dilution in 250 litres of water.

Dashparni extract is useful to manage all kind of insect-pests of crops and orchards. Take 5-6 litres of *dashparni* extract and dilute in 250 litres of water for spraying on one hectare of crop.

iv. Protocols to be followed: Protocol prepared by ICAR-IIFSR, Modipuram, Meerut, UP under All India Network Programme on Organic Farming (AI-NPOF) for Natural Farming experiments.

d. Weed management practices

i. Name of the practice: Intercropping, Mulching, Mechanical weeding

ii. Brief description of the practice:

The retention of crop residues on the soil surface or the use of paddy straw as mulch helps in weed suppression, protect the soil and conserve moisture. Additionally, intercropping with legumes such as cowpea, a cover crop, effectively suppresses weeds by shading the ground and blocking sunlight. Mechanical weed management by different field implements by reshaping of the beds.

iii. Benefits of the practices (Impacts):

Mulching acts as a physical barrier, significantly suppressing weed germination and growth by limiting their access to sunlight and reducing temperature fluctuations on the soil surface. Additionally, intercropping with legumes such as cowpea, a highly effective cover crop, further contributes to weed suppression. Cowpea forms a dense canopy that shades the ground, blocking sunlight and creating an environment unfavourable for weed germination. Furthermore, legumes like cowpea enhance soil health through biological nitrogen fixation, which improves soil nutrient availability, reduces the need for synthetic fertilizers, and supports sustainable cropping systems. These integrated practices are effective components of conservation agriculture and agroecological farming systems, promoting both productivity and ecological balance.

Mechanical weed management through the use of various field implements, combined with the reshaping of beds, is an effective strategy for reducing weed pressure in agricultural systems. Field implements such as rotary hoes, harrows, cultivators, and tine weeders are commonly used to physically disrupt weeds at various growth stages, particularly in the early stages of crop development. Overall, mechanical weed management, when integrated with bed reshaping and other cultural practices, supports sustainable weed control by reducing reliance on chemical herbicides and promoting healthier crop growth environments.

iv. Protocols to be followed: Protocol prepared by ICAR-IIFSR, Modipuram, Meerut, UP under All India Network Programme on Organic Farming (AI-NPOF) for Natural Farming experiments.

e. Seed selection, treatment and sowing practices

i. Name of the practice:

Traditional varietal seeds, seed treatment with *beejamrit*.

Sowing is done on furrows, raised beds, or broad beds to maintain *Whapasa*.

ii. Brief description of the practice:

Only traditional or indigenous varietal seeds are used, avoiding the use of genetically modified (GMO) or hybrid seeds to maintain biodiversity and ecological balance. Before sowing, the

seeds are treated with *beejamrit*, a natural seed treatment solution prepared from cow dung, cow urine, lime and water.

The concept of *Whapasa* in natural farming emphasizes maintaining an optimal balance of air and water in the soil's pore spaces. This is achieved through practices such as alternate furrow irrigation, where only the open furrows receive water while the ends of alternate furrows are closed to prevent over-irrigation.

iii. Benefits of the practices (Impacts):

Beejamrit is used for treatment of seed, seedling roots or other planting material. *Beejamrit* helps protect seeds from pathogens, enhances germination, and promotes the growth of beneficial soil microbes.

Whapasa, the optimal balance of air and water in soil pores, promoting healthy root growth and efficient irrigation. This technique reduces water wastage, improves root aeration, and minimizes the risk of waterlogging, thereby creating favorable conditions for plant growth. By combining these traditional practices, natural farming ensures sustainable and eco-friendly agricultural productivity.

iv. Protocols to be followed: Protocol prepared by ICAR-IIFSR, Modipuram, Meerut, UP under All India Network Programme on Organic Farming (AI-NPOF) for Natural Farming experiments.

e. Cropping systems

i. Name of the practice:Maize-Wheat

Rice- Wheat

Maize + Cowpea – Wheat+Chickpea

Baby corn – Chickpea – Summer moong

Sesame – Chickpea

ii. Brief description of the practice:In the maize-wheat system, maize is sown as a *kharif* crop and wheat as a *rabi* crop on furrows or raised beds to optimize growth. In the maize + cowpea and wheat + chickpea intercropping system, legumes enhance soil fertility, provide extra income and improve soil health.

iii. Benefits of the practices (Impacts):

In the maize-wheat cropping system, maize is cultivated as a *kharif* crop and wheat as a *rabi* crop, leveraging furrows, raised beds, or broad beds to optimize growth conditions. These methods improve soil aeration, enhance root penetration and minimize waterlogging, ensuring efficient nutrient uptake and moisture retention.

In the second cropping system, maize + cowpea during *kharif* and wheat + chickpea during *rabi* are implemented as intercropping systems. Intercropping cowpea and chickpea provide multiple benefits, including additional income, enhanced soil fertility through biological nitrogen fixation and improved soil structure due to the varied root systems. Raised beds and furrows further optimize water use efficiency, support better crop establishment, and reduce competition between crops.

These cropping systems enhance resource use efficiency, suppress weeds, and diversify income sources, making them economically and ecologically sustainable while promoting long-term soil health.

iv. Protocols to be followed: Protocol prepared by ICAR-IIFSR, Modipuram, Meerut, UP under All India Network Programme on Organic Farming (AI-NPOF) for Natural Farming experiments.

g. Water management practices

i. Name of the practice: *Whapasa*, the optimal balance of air and water in soil pores, promoting healthy root growth and efficient irrigation.

ii. Brief description of the practice: *Whapasa*, the optimal balance of air and water in soil pores, promoting healthy root growth and efficient irrigation. The concept of *Whapasa* in natural farming emphasizes maintaining an optimal balance of air and water in the soil's pore spaces. This is achieved through practices such as alternate furrow irrigation, where only the open furrows receive water while the ends of alternate furrows are closed to prevent over-irrigation.

iii. Benefits of the practices (Impacts):

This technique reduces water wastage, improves root aeration, and minimizes the risk of waterlogging, thereby creating favourable conditions for plant growth. By combining these traditional practices, natural farming ensures sustainable and eco-friendly agricultural productivity.

iv. Protocols to be followed: Protocol prepared by ICAR-IIFSR, Modipuram, Meerut, UP under All India Network Programme on Organic Farming (AI-NPOF) for Natural Farming experiments.

h. Any other practices being followed

Most of the farmers adopt natural farming and its component in with integration of organic farming.

3. Case studies of practising natural farming farmers implementing these

Practises

S. Surjit Singh Rai

S. Surjit Singh Rai, a progressive farmer from Langeri village, Banga block, district Shaheed Bhagat Singh Nagar, has been practicing organic and natural farming since 2010 and has become a role model for sustainable agriculture. He practiced all four pillar of natural farming with organic farming. He received the prestigious **National Award of Millionaire Farmer of India** in Delhi on 03-12-2024. He cultivates a variety of crops, including wheat, paddy, mustard, sugarcane, green fodder, turmeric and vegetables, using natural farming methods. On a demonstration plot under the ATMA Scheme, he grew organic wheat varieties such as Bansi, Black Wheat and Sharbati, achieving high-quality yields and preparing organic seeds. For soil health management, he uses homemade compost, vermicompost, bio-enzymes, and Jeevamrit, enriching the soil with organic matter and microbial activity. He conserves water by integrating natural farming techniques and avoids chemical inputs. Weeds are managed manually through crop rotation and mulching. Surjit Singh also processes sugarcane into jaggery, a highly valued product among consumers, and sells organic wheat and seeds at premium rates, ensuring higher profitability.

In addition to his staple crops, he grows medicinal plants like Neem, Tulsi, and Haar-Singar, along with fruit trees such as Guava, Papaya, Amla, Lemon, and Falsa, all organically. His achievements include a remarkable yield of 54 fruits per plant from his organic tomato crop, weighing about 5 quintals during May 2020. He avoids burning crop residues, turning them into compost to enhance soil fertility and sustainability. In 2021-22, he cultivated organic paddy on one acre using natural farming principles, producing 26 quintals of high-quality grain.

Surjit Singh regularly participates in farm schools, training camps, and exposure visits organized by the Department of Agriculture and Farmers Welfare under the ATMA Scheme to

promote organic farming. He shares his knowledge freely with students, farmers, and institutions, conducting training sessions on natural farming techniques. He has also been honoured by various organizations, including the Honourable Speaker of Punjab, S. Kultar Singh Sandhwa.

A passionate environmentalist and social worker, Surjit Singh Rai inspires others with his slogan: "*Ayo ral mil k jor lagayie, kheta whicho jehara bhajayie, shud bhojan sab nu khavayie, samaj nu tandrus bnayie, parali nu agg kdena layie, dharti hethla pani bachayie*" ("*Let us work together, drive poisons out of the fields, feed everyone pure food, make society healthy, never burn stubble, and save groundwater.*") His practices exemplify sustainable agriculture, balancing productivity with ecological stewardship.

Mahaveer Bisnoi

Mahaveer Bishnoi, a progressive farmer from V.P.O. Rukanpura Khuikheda, Tehsil Abohar, District Fazilka, has excelled in natural farming by integrating scientific and sustainable practices. He cultivates diverse crops such as wheat, basmati rice, sugarcane, potato and citrus orchards while adhering to all four components of natural farming principles. He grows wheat varieties like 1482, C-306, Sharbati, Bansian and sugarcane variety Co-85, prioritizing biodiversity and crop rotation for ecological balance. For soil health management, he incorporates green manure to enhance organic matter and fertility, applies *Ghanjeevamrit* before sowing to enrich microbial activity, and uses *Jeevamrit* as both a soil amendment and foliar spray to promote soil health and nutrient availability. Nutrient supplementation is achieved through the integration of farmyard manure (FYM) and natural farming concoctions (NFCs), ensuring balanced nutrient delivery and sustained soil productivity. He effectively controls pests and diseases using biopesticides, including *Neemashtra*, *Agniashttra*, *Brahmastra*, and sour buttermilk, ensuring crop health without harming beneficial organisms. By adopting the *Whapasa* technique—a component of natural farming—he reduces water usage through alternate furrow irrigation, maintaining soil moisture while conserving resources. Mahaveer also adds value to his crops by processing sugarcane into jaggery at his on-farm unit and producing cold-pressed wheat flour, retaining its nutritional quality. He directly sells all products at the farm gate, ensuring traceability and higher profit margins. His dedication and innovative practices earned him the Outstanding Farmer Award during the CIPHET-IFA, 2024, organized by KVK, Fazilka and ICAR-CIPHET Regional Station, Abohar. This recognition highlights his commitment to organic and natural farming, particularly his innovations in processing cane juice into jaggery and other products. His

practices serve as a model for sustainable agriculture, balancing productivity with ecological stewardship.

Natural Farming Package of Practices

1. Name of the State: Rajasthan

2. Natural Farming practices

Cropping systems: *Kharif*: Groundnut + Sesame intercropping (4:2)

Rabi: Fennel + Cabbage intercropping (2:1)

a. Soil health management practices

i. Name of the practice: Ghanjeevamrit + Jeevamrit and multiple crops in summer

ii. Brief description of the practice:

- Application of Ghanjeevamrit @250 kg/ha + Jeevamrit @ 500 lit/ha/ Twice a month (with irrigation or spray on soil as per rainfall availability).
- In summer season, multi-crops (Maize, blackgram, greengram, sorghum, clusterbean, cowpea, sesame) should be sown.

iii. Benefits of the practices (impacts): Application of Ghanjeevamrit @250 kg/ha + Jeevamrit @ 500 lit/ha/ Twice a month recorded higher groundnut equivalent yield (450 kg/ha) in *kharif* and fennel equivalent yield (2765 kg/ha) in *rabi* as compare to control in groundnut + sesame intercropping and fennel + cabbage intercropping, respectively.

iv. Protocols to be followed:

- Ghanjeevamrit: Take 100 kg indigenous cow dung and spread it on clean floor. Add 1 kg of jaggery + 1 kg pulse flour + 3 lit cow urine and 250 g soil from undisturbed bund or under tree cover to cow dung. After 10 days, ghanjeevamrit is ready for application.
- Jeevamrit: 200 litres of water was put in a barrel; 10 kg of fresh desi cow dung was added followed by 10 litres of aged cow's urine; To this 2 kg of jaggery, 2 kg of pulse flour (gram) and a handful of live soil from under canopy of the banyan tree (about 100 g) are added. Stir the solution well and let it ferment for 48 hours in the shade. Now jeevamrit is ready for application. 200 litres of jeevamrit is sufficient for one acre of land

b. Nutrient management practices

i. Name of the practice: Ghanjeevamrit + Jeevamrit

ii. Brief description of the practice: Application of Ghanjeevamrit @250 kg/ha + Jeevamrit @ 500 lit/ha/ Twice a month (with irrigation or spray on soil as per rainfall availability)

iii. Benefits of the practices (impacts): Application of Ghanjeevamrit @250 kg/ha + Jeevamrit @ 500 lit/ha/ Twice a month recorded higher groundnut equivalent yield (450 kg/ha) in *kharif* and fennel equivalent yield (2765 kg/ha) in *rabi* as compare to control in groundnut + sesame intercropping and fennel + cabbage intercropping, respectively.

iv. Protocols to be followed:

- Ghanjeevamrit @250 kg/ha should be applied before sowing
- Jeevamrit @ 500 lit/ha/ Twice a month (with irrigation or spray on soil as per rainfall availability) should be applied.

c. Pest and Disease management practices

i. Name of the practice: Neemaster, Dashparni ark, Brahmaster, Neem seed kernel extract, border crop and trap crop

ii. Brief description of the practice: Application of Neemaster (10%), Dashparni ark (10%), Brahmaster (5%), border crop and trap crop, (Marrigold and tulsi at space between replication and subabul at outside border)

iii. Benefits of the practices (impacts): Application of Neemaster (10%), Dashparni ark (10%) and Brahmaster (5%) recorded higher pest and disease control as compare to control in Groundnut + Sesame intercropping and Fennel + Cabbage intercropping.

iv. Protocols to be followed:

- **Neemaster:** Add the 5 Litres of Cow's Urine to the 100 Litres of Water. Add the 5 Kgs of Cow Dung to the water as well. Crush the 5 Kgs of Neem Leaves to the water. Allow this solution to ferment for 24 hours. Stir the solution twice a day by using a stick. Filter the solution using a cloth and use it.
- **Dashparni ark:** Crush neem leaves 5 kg, aak leaves 2 kg, babool leaves 2 kg, papaya leaves 2 kg, tulsi leaves 2 kg, Custard apple leaves 2 kg, Karanja leaves 2 kg, Castor leaves 2 kg, Nerium 2 kg, dhatura leaves 2 kg, Green chilli paste 0.5 kg, Garlic paste 0.5 kg, turmeric paste 0.5 kg, ginger paste 0.5 kg, tobacco leaves 1 kg, Cow dung 5 kg and cow urine 5 lit in 200 lit water and ferment for 40 days. Shake regularly three times a day. Extract after crushing and filtering. The extract can be stored up to 6 months and is sufficient for one acre.
- **Brahmaster:** Take 10 liters of desi cow urine. Add 3kg Neem leaves extract, 2 kg Custard apple leaves extract, 2 kg pongamia leaves extract, 2 kg of Beal leaves extract, 2 kg Castor leaves extract, 2 kg Datura leaves extract in it. Then boil this solution, keep the solution stable for 48 hrs after boiling.

Then filter the solution and store it. Spray prepared Brahmastra on the crops to control all the sucking pests, pod borer, fruit borer, etc. For spraying take 5 liters in 100 liters of water. Brahmastra solution can be stored for 6 months.

- Border crop and trap crop, (Marrigold and tulsi at space between replication and subabul at outside border) should be grown.

d. Weed management practices

- i. Name of the practice: Crop residue mulching
- ii. Brief description of the practice: Crop residue mulching (Fennel stover in *kharif* and Groundnut + sesame straw + cabbage stover/leaves in *rabi* @ 5 t/ha)
- iii. Benefits of the practices (impacts): Application of crop residue mulching reduce the weed intensity of different weeds such as *Portulaca*, *Amaranthus viridis*, *Digitaria sanguinalis*, *Echinochloa crus-galli*, *Chenopodium album*, *Chenopodium murale*, *Anagallis arvensis* etc.
- iv. Protocols to be followed:
 - Fennel stover in *kharif* @ 5t/ha should be applied
 - Groundnut + sesame straw + cabbage stover/leaves in *rabi* @ 5 t/ha should be applied

e. Seed selection, treatment and sowing practices

- i. Name of the practice: Beejamrit
- ii. Brief description of the practice: Beejamrit should be treated with beejamrit in such a way that each seed should be wetted
- iii. Benefits of the practices (impacts): Seed treatment with beejamrit gave higher yield as compare to control.
- iv. Protocols to be followed: Take 20 liter Water, 5 Kg Local Cow Dung, 5 liter Local Cow Urine, 50 Gram Lime & Handful soil from the bund of the farm. Take 5 Kg Local Cow Dung in a cloth and bound it by tape. Hang this in the 20 Liter water up to 12 hours. Take one liter water and add 50 gm lime in it, let it stable for a night. Then next morning, squeeze this bundle of the cow dung in that water thrice continuously, so that all essence of cow dung will accumulate in that water. Then add a handful of soil in that water solution and stir it well. Then add 5 liter Deshi cow urine or human urine in that solution & add the lime water and stir it well. Now Beejamrit is ready to use.

f. Cropping systems

Kharif: Groundnut + Sesame intercropping (4:2)

Rabi: Fennel + Cabbage intercropping (2:1)

- i. Name of the practice:
- ii. Brief description of the practice:
- iii. Benefits of the practices (impacts)
- iv. Protocols to be followed:

g. Water management practices

- i. Name of the practice: Whapasa
- ii. Brief description of the practice: Whapasa (irrigation in alternate furrows at noon)
- iii. Benefits of the practices (impacts): Water management
- iv. Protocols to be followed: Irrigation should be done in alternate furrows at noon

- h. Any other practices being followed: -

3. Case studies of practising natural farming farmers implementing these practises

1. Sh. Tejpal Jatiya s/o Sh Chena

Village	:	Badwai
Tehsil	:	Bhupalsagar
District	:	Chittorgarh
Contact No.	:	9928547412

Practices adopted

- Practicing natural farming for the last 6 years.
- Adopted indigenous practices for crop cultivation viz., jeevamrit, beejamrit, intercropping etc.
- Applied crop residues mulch, resulting in control over weeds infestation.
- Avoided insecticides, by growing different herbal plants like aak, dhatura, castor, neem, desi babul on the farm's boundary, which are used for preparation of different bio-pesticides.
- Perform Homa farming for early and enhanced flowering in lemon.
- Prepare lemon seedlings from seed for market purpose.
- Participated in kisanmela, goshti, exposure visits, trainings, exhibitions etc.

Comparison between Natural Farming and Conventional Farming

Parameters	Natural Farming (0.75 ha)	Conventional Farming (0.75 ha)
Crop	Lemon	Lemon
Cost of cultivation (₹)	38500	60000
Production (q)	105	100
Gross return (₹)	315000	300000
Net return (₹)	276500	240000
BC ratio	7.19	4.0

Benefits and achievements

- Well known natural farmer at district level. Consumers directly purchase citrus fruits as well as seedlings from farm.
- Improved soil organic matter and beneficial soil microorganisms.
- Utilized crop residue and farm wastage.
- No dependency on external inputs like agri-chemicals and agri-inputs. So, reduced cost of cultivation.



2. Sh. Bheru Lal S/o Sh. Teja Jatiya

Village	:	Daulatpura
Tehsil	:	Bhupalsagar
District	:	Chittorgarh
Contact No.	:	9929269773



Practices adopted

- Practicing natural farming for the last 6 years.
- Adopted indigenous practices for crop cultivation viz., jeevamrit, beejamrit, ghanjeevamrit, waste decomposer, mulching, intercropping, herbal ark etc.
- Use old papaya leaves for mulching in plant basin and also for biopesticide preparation for insect control.
- Perform Homa farming and use its ash for flowering and insect control.
- Participate and exhibit his farm produce and inputs in krishi mela of different Universities.
- Conduct training about natural farming for farmers at his own farm.

Comparison between Natural Farming and Conventional Farming

Parameters	Natural Farming (0.75 ha)	Conventional Farming (0.75 ha)
Crop	Papaya	Papaya
Cost of cultivation (₹)	41500	70000
Production (q)	160	155
Gross return (₹)	320000	310000
Net return (₹)	278500	240000
BC ratio	6.71	3.42

Benefits and achievements

- Well known natural farmer at district level.
- No dependency on external inputs like agri-chemicals and agri-inputs.
- Reduced cost of cultivation.
- Improved soil organic matter and beneficial soil microorganisms.
- Utilized crop residue and farm wastage.
- Awarded at block level by District Collector under ATMA Project.



PROTOCOL FOR NATURAL FARMING

Subject: Inclusion of farm practices in protocol for conducting Research and training activities in natural farming

Natural farming is a programme of national importance. Hon'ble Prime Minister and His Excellency, Governor of Gujarat, Acharya Devvrat have time and again emphasized the need of natural farming for self Reliant Bharat by providing self reliance to the farmers of the country. For realising the vision of our Prime Minister, different activities have been initiated at various ICAR Institutes and agricultural universities of the country. In these activities it has been observed that the topics and protocol regarding natural farming are quite variable at different Institutes like NCONF, Ghaziabad and some centres run by the Department of Horticulture and Department of Agriculture and farmers welfare, Haryana. As per my knowledge, the level and standard of trainings at these centres do not meet the requirement of different Agro climatic regions of the country. Like wise in research studies, partial protocol of farm practices is being followed. In research experiments, in most of the cases, protocol remains limited to the inclusion of Jeevamrit and Ghanjeevamrit alone.

The selection of topics of Extension trainings on natural farming and treatments in research experiments primarily remains based on four pillars (Jeevamrit, Ghanjeevamrit mulching and wapsaa) as suggested by Sh. Subhash Palekar who worked under rainfed conditions of Maharashtra by adopting intercrop options in all the crops. Both of these conditions (rainfed and intercrop) are not applicable under different agroclimatic conditions of the country particularly in the high productivity zone of North Western India, which is known as 'Grain Bowl' of the country. Hence, there is need to include and address the issues as per need of the farmers of the country.

We can define natural farming as the farming of microbes. In other words, it can be said, if we exclude chemical inputs from conservation agriculture and include natural inputs/bioformulations in it, it becomes natural farming as all the conservation practices help in preserving and multiplying soil microbiome.

Based on my vast experience on conducting trainings for different category of stakeholders and research on natural farming and needs of agriculture in different parts of the country, I want to submit the basic requirements/farm practices of natural farming required to be included in the training schedule and conducting research on natural farming (details attached in Annexure 1).

I request your good self to get approved a common protocol of natural farming by including all the crop/farm practices as explained in Annexure 1 for whole of the country with some area and location Specific variations / modifications.

Submitted for your kind consideration please.

(Dr Hari Om)
State Training Advisor (Natural Farming)
Natural Farming Training Institute
Gurukul Kurukshetra

Annexure 1

Farm practices required to be included in protocol for conducting trainings and research on natural farming

(A) All the farm practices which help in nurturing & multiplication of soil biota such as:

1. Desi/native cow: Preference should be given to cow dung and cow urine of native cow available in the area, however, comparative studies may be undertaken with other animal breeds for alternate options. Cows are hardly available on high altitude temperate areas like Ladakh and Jammu and Kashmir. In such cases, the experiments may be conducted on locally available animals like goats and sheep.

2. Zero/minimum tillage: Adoption of tillage practices particularly in hot summer months is one of the biggest threats to soil organic carbon. Efforts should be made to minimise the tillage practices.

3. Mulching: Emphasis should be laid on residue and live mulching.

4. Combination of natural farming, plastic mulching and drip irrigation: Mulching is one of the inevitable components of natural farming. Crop residues for mulching are hardly available in rainfed areas and dry regions of the country like Badhmer and Jodhpur areas of Rajasthan, Kutch and Bhuj areas of Gujarat, Bundelkhand areas, and Vidarbha areas of South India. Even the residues are not available for animals in these areas. In the initial evaluation trials at the farm of Gurukul Kurukshetra, use of plastic mulch with drip irrigation and application of Jeevamrit through drip irrigation has given very encouraging results. These results are showing that by the use of this technique there is a possibility of taking multi-crops for multi-seasons without ploughing the field and without incurring cost on the control of weeds.

5. Residue management: Preference should be given for retaining crop residues on the soil surface instead of their incorporation in the soil. No burning of crop residues.

6. Green manuring of Sesbania or any other legume crop before planting of the crop as it acts as a source of food for microbes as well as the crop planted.

7. Use of bio-formulations/natural inputs of natural farming as suggested by Shri Subhash Palekar.

8. Wapsaa: Wapsaa approximately emulates the condition of field capacity in which soil macropores are filled with air and micropores with moisture. Drip/sprinkler irrigation maintains wapsaa condition in more better way. Flood irrigation should be avoided.

9. Drip/sprinkler irrigation with the conjunctive use of Jeevamrit as given in **Point No. 3.**

10. Biodiversity: Combination of different crops (intercrops) grown simultaneously. Multilayer farming is the best model of biodiversity particularly under problematic, nutrient deficient soils and rainfed conditions.

11. Water harvesting in rainfed areas and having limited irrigation facilities.

(B) Besides the training of extension scientists, research and teaching scientists should also take training on natural farming for better understanding of different aspects of natural farming.

(C) For conducting experiments on natural farming one acre field should be separately earmarked for natural farming treatments as the results may be influenced in smaller plots which restrict the activities of earthworms. Application of chemical and organic inputs in smaller plots may also affect the results. For chemical and organic farming treatments, separate adjacent field should be selected.

(D) The trials conducted at KVK farm should also be replicated at farmers' fields who are already doing farming by adopting natural/organic farming practices and who have better understanding of chemical-free farming. The scientists of different disciplines including Microbiologist from research stream should be associated with the farmers who are doing On-farm trials.

(E) There should be frequent interaction and coordination of all the Experts associated with research experiments and extension activities related to natural farming.

(F) No experiment should be permitted in which only jeevamrit and Ghanjeevamrit treatments are included. Inclusion of maximum possible package of natural farming practices should be ensured.

(G) The farmers should be encouraged to initially adopt natural farming in smaller area (half to one acre) which may be increased after gaining proper experience of natural farming.